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Advances in Bioacoustics Sensing and Signal Processing Technologies

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Advances in Bioacoustics Sensing and Signal Processing Technologies

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Title

Advances in bioacoustic sensing and signal processing technologies

Chengyu Li^{1,2, §}, Wenbo Li^{1,2, §}, Jingyang Wu^{1,2, §}, Shuo Wang^{1,2, §}, Han Liao^{3, §}, Xiang Yu³, Li Cheng³ and Xiaoming Tao^{1,2,*}

¹ Research Institute for Intelligent Wearable Systems, The Hong Kong Polytechnic University, Hong Kong SAR, People’s Republic of China

² School of Fashion and Textiles, The Hong Kong Polytechnic University, Hong Kong SAR, People’s Republic of China

³ Department of Mechanical Engineering, The Hong Kong Polytechnic University, Hong Kong SAR, People’s Republic of China

§ These authors contributed equally to this work.

* Corresponding author email: xiao-ming.tao@polyu.edu.hk

Abstract

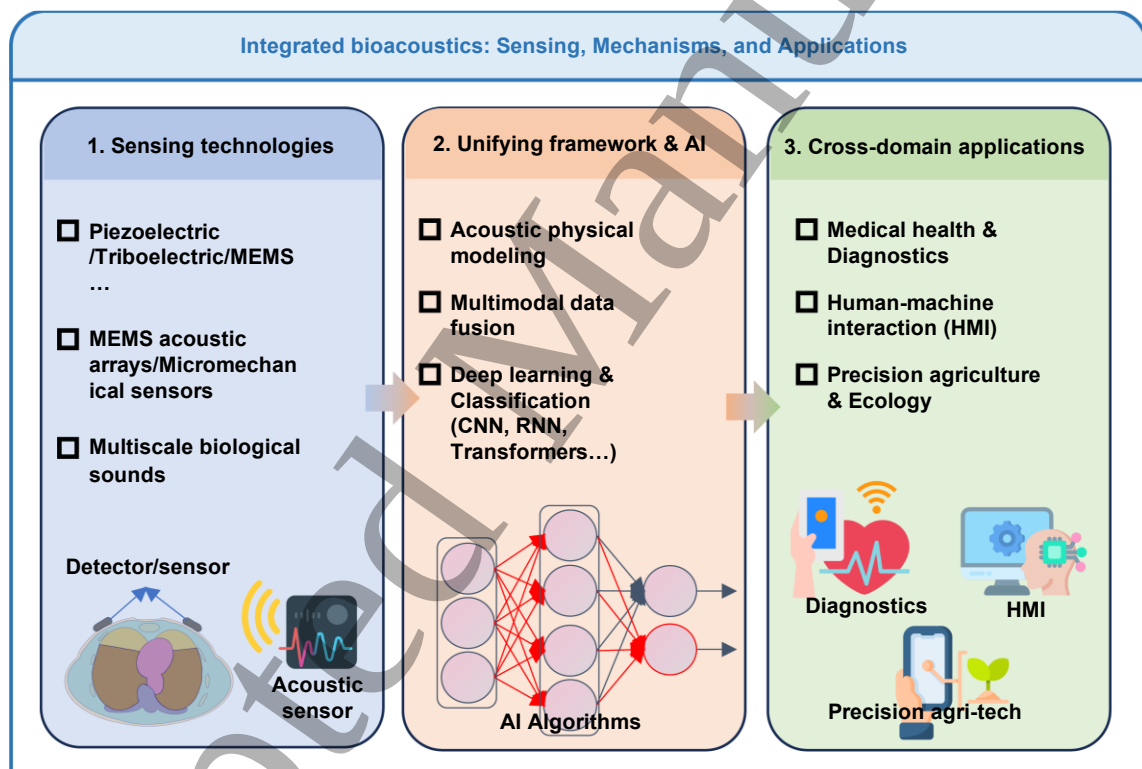
Driven by advances in flexible electronics, microfabrication, and artificial intelligence, bioacoustics is revolutionizing sensing and signal processing across health, ecology, and intelligent systems. This review systematically examines key progress in high-performance acoustic sensors (e.g., piezoelectric, triboelectric, MEMS) and acquisition schemes for multiscale biological sounds, ranging from heart tones to plant stress emissions. Furthermore, we delve into advanced methodologies for acoustic signal preprocessing, feature extraction, and classification, with a particular emphasis on the powerful analytical capabilities of multimodal data fusion and deep learning algorithms in complex real-world scenarios. Crucially, although the sensing requirements and biological targets vary drastically across medical, ecological, and agricultural domains, this review establishes a unifying framework centered on the shared physical and algorithmic challenges of bioacoustic information flow. By juxtaposing these diverse fields, the potential strategies for the interdisciplinary integration of sensing methodologies are highlighted. We also critically assess the essential roles of acoustic physical modeling, standardized testing frameworks, and gold-standard databases in evaluating system performance. Finally, the review highlights cutting-edge applications of bioacoustics in human health monitoring, medical diagnostics, human-computer interaction, and precision agriculture. By synthesizing current technological convergences and outlining future trajectories, we provide a comprehensive perspective on the promising directions and pivotal challenges facing bioacoustic research.

Keywords: bioacoustics, signal processing, sensing, application

Highlights

- Establishes a unifying framework bridging medical, ecological, and agricultural bioacoustics through the complete continuum of biological acoustic information flow.
- Systematically contrasts diverse acoustic transduction mechanisms and signal acquisition and processing strategies across biological species, backed by comprehensive performance-comparison tables.
- Integrates advanced signal processing workflows—spanning traditional filtering, spatial acoustic imaging (acoustic cameras), and deep learning algorithms—with rigorous physical models.
- Outlines a holistic roadmap addressing critical technical bottlenecks, data scarcity, and generalization challenges to guide future bioacoustic research.

Abstract TOC



1. Introduction

Bioacoustics, as an interdisciplinary field, focuses on the mechanisms underlying the production, propagation, and reception of sounds by living organisms, as well as the functions and significance of these sounds in ecosystems. It integrates theoretical and methodological approaches from acoustics, biology, ecology, and signal processing, deciphering sound signals to reveal patterns of biological behaviors, population dynamics, and environmental changes^[1-3]. Its research subjects encompass taxa such as insects,

birds, mammals, and marine organisms, while also extending to acoustic signals from plants and humans^[4-6], providing a pivotal tool for investigating biological communication, habitat selection, reproductive behaviors, and organism–environment interactions. In recent years, advances in sensing technologies and computing capabilities have driven innovations in research methodologies^[7-9], evolving from traditional spectrogram analysis to the application of machine learning algorithms. This progression has enabled bioacoustics to emerge as a crucial pillar for ecological monitoring, biodiversity conservation, and medical diagnosis.

Physiological acoustics (or biomedical acoustics), a core interdisciplinary subfield bridging bioacoustics and medical health, aims to elucidate the principles governing the generation, propagation, and body surface detection of sound within the human body. Internal human physiological sounds originate from the physiological activities of multiple organs: respiratory sounds arise from pulmonary ventilation, heart sounds are generated by the opening/closing of cardiac valves and blood flow pulsation, speech is formed via vocal cord vibration and vocal tract modulation, and gastrointestinal sounds are induced by gastrointestinal peristalsis^[10-12]. When propagating toward the body surface, these signals must penetrate complex biological tissues, including muscle, bone, body fluids, and adipose tissue, encompassing reflection and refraction driven by differences in acoustic impedance, tissue elastic responses dictated by acoustic modulus, and frequency-dependent propagation characteristics. Specifically, high-frequency sound waves undergo rapid attenuation and shallow penetration due to tissue absorption and scattering, whereas low-frequency sound waves exhibit slow attenuation and tend to resonate within body cavities^[13-15]. A defining feature of human physiological signals is their intrinsic association with physiological and pathological states: abnormal organ function alters the generation and propagation pathways of these signals. Precise deciphering of such changes not only provides a noninvasive window for investigating human physiological mechanisms but also lays a theoretical foundation for the early diagnosis, health monitoring, and rehabilitation assessment of disorders (e.g., cardiovascular and cerebrovascular diseases, digestive system diseases). In this regard, physiological acoustics serves as a critical link in translating bioacoustic research into precision medicine.

Bioacoustic monitoring exhibits considerable potential and value in scientific research and practical applications. It provides a noninvasive, efficient method for assessing biodiversity and monitoring ecosystem health. By recording and analyzing environmental sound signals over time, researchers can track species distribution, population dynamics, and habitat quality, offering a robust basis for ecological conservation and environmental management^[16-19]. Moreover, bioacoustic technology is instrumental in wildlife conservation, enabling precise monitoring of endangered species' activity ranges and behavioral patterns through species-specific vocalizations, thus supporting evidence-based conservation policies^[20-22]. Beyond ecology, bioacoustics shows broad applicability in agriculture, fisheries, and medicine. For example, in agriculture, it facilitates targeted pest control by detecting pest sound signals; in medicine, it aids in studying vocal mechanisms across species, enhancing the diagnosis and treatment of speech disorders^[23-25]. These diverse applications not only advance fundamental science but also deliver innovative solutions to real-world challenges.

In recent years, bioacoustics has advanced significantly, propelled by rapid developments in detector technology, signal processing algorithms, and computational power^[26,27]. The proliferation of high-sensitivity microphone arrays and autonomous acoustic devices has enabled large-scale, long-term environmental acoustic monitoring^[28,29]. Meanwhile, the adoption of AI algorithms has greatly enhanced the precision of sound signal classification and recognition^[30,31]. Nevertheless, critical challenges remain. Background noise in complex environments often interferes with target signal isolation, posing a persistent difficulty^[32]. Overlapping acoustic traits across species further complicate accurate identification, requiring additional investigation. Moreover, the absence of standardized data-sharing frameworks restricts cross-regional and interdisciplinary collaboration. Finally, the limited alignment between physical models of bioacoustic signals and real biological processes hampers the accuracy of noninvasive monitoring predictions. These unresolved issues not only drive technological innovation but also open broad avenues for future research.

To effectively address the aforementioned challenges and achieve technological breakthroughs, it is particularly urgent to break down the long-standing disciplinary barriers. In most cases, bioacoustic research in the fields of medical health, wildlife ecology, and agriculture tends to operate in isolation due to significant differences in their signal characteristics and evaluation criteria. However, the in-depth integration of flexible electronics and artificial intelligence technologies in recent years has revealed a common underlying technological core across these fields. For instance, whether detecting weak fetal heartbeats or monitoring xylem cavitation in plants under drought stress, “acoustic impedance matching” at the tissue-sensor interface remains a universal physical challenge.

Based on this, this review breaks traditional boundaries and integrates these seemingly disparate fields into a unified framework, systematically summarizing the latest advances in bioacoustic sensing and signal processing technologies. Following the main thread of acoustic information flow, this paper elaborates on the mechanisms of bioacoustic signal generation and propagation, as well as the hardware system covering sensing materials and structural design; it then focuses on the innovative progress in signal preprocessing, feature extraction, and physical acoustic modeling, and highlights the potential of interdisciplinary cross-innovation (e.g., how robust denoising technologies developed for complex marine environments can inspire the design of wearable human diagnostic devices, or how high-sensitivity MEMS architectures in human-machine interaction can be transformed into tools for plant health monitoring); furthermore, it discusses the typical applications and existing bottlenecks in fields such as medicine, ecology, and agriculture; finally, it looks forward to frontier directions including the miniaturization of sensing platforms, multisource data fusion, and intelligent algorithms. By synthesizing the above content, this review aims to provide a comprehensive reference for researchers in related fields and effectively build a bridge between basic bioacoustic research and cross-domain practical applications.

2. Characterization and classification of bioacoustic signals

2.1 Classification by biological source and ecological function

The nature and origin of sound exhibit remarkable diversity across biological taxa but fundamentally rely on the physiological basis of mechanical vibration. Bioacoustic signals, characterized by their diverse physical properties and profound biological relevance, have emerged as a focal point of interdisciplinary research. Their systematic classification and characterization not only underpin our understanding of bioacoustic phenomena but also provide a theoretical foundation for the development of advanced sensing and signal processing technologies.

Acoustic signals can propagate through diverse media, including gas, liquid, and solid (**Figure 1(a)**), and physical parameters such as sound velocity, wavelength, reflection/refraction behaviors, and energy dissipation vary with differences in the mechanical properties (e.g., density, elastic modulus) of the medium during propagation. Based on frequency ranges, acoustic signals are categorized into three major classes: infrasound (<20 Hz), audible sound (20 Hz–20 kHz), and ultrasound (>20 kHz). Infrasound, benefiting from its long-wavelength attribute, is often employed by large-bodied animals (e.g., elephants, whales) for long-distance information transfer; audible sound represents the primary frequency band for intraspecific communication in humans and most terrestrial animals (e.g., birds); and ultrasound plays a pivotal role in processes such as echolocation in bats and navigation in dolphins. As illustrated in **Figure 1(b)**, bioacoustic signals are most effectively categorized by their biological source and ecological function, as these parameters define the underlying sensing requirements. These signals are generally divided into three major classes: human-derived (e.g., physiological sounds), animal-derived (e.g., vocalizations and echolocation), and plant-derived signals (e.g., stress emissions). This source-centric classification emphasizes the biological relevance of the acoustic data, ranging from clinical diagnostic biomarkers in humans to behavioral and survival

indicators in animals and plants.

2.2 Acoustic information flow across biological species

As previously discussed, from the perspective of biological sound sources and ecological functions, bioacoustic signals can be categorized into human-derived, animal-derived, and plant-derived signals (**Figure 1(b)**). The generation of human physiological signals is closely linked to the anatomical structure and physiological functions of sound-producing organs. For instance, speech arises from airflow-driven vibration of the vocal cords, modulated by resonance in the oral and nasal cavities; in contrast, heart sounds and bowel sounds reflect the physiological activities of the cardiovascular and digestive systems, respectively.

Acoustic signals within the human body and most animals primarily originate from the coupling of viscoelastic tissue motion and multiphase fluid dynamics. These signals propagate through multiple layers of internal media, including deep tissues, muscles, and adipose tissue, before reaching the body surface, where they are ultimately captured by detectors or sensors^[10,33]. During propagation, human acoustic signals are shaped by tissue density, acoustic impedance, viscoelasticity, and interstitial fluids, leading to reflection, refraction, attenuation, and frequency-dependent transmission^[13,34,35]. Respiratory sounds provide a representative example: they travel through the tracheobronchial tree and surrounding tissues before reaching the chest wall. Because the lung parenchyma strongly attenuates high-frequency components, body-surface respiratory sounds are mainly concentrated in the low-frequency range, making the thoracic cavity behave as a low-pass acoustic filter^[36-38]. Heart sounds, generated by valve motion and blood-flow-induced vibration, similarly propagate through multilayer viscoelastic thoracic tissues before being detected at the body surface^[11]. Therefore, accurate detection of internal human sounds requires sensor designs and signal-processing strategies that account for organ-specific sound generation, tissue-mediated propagation, and frequency-selective attenuation.

Beyond the human body, animal acoustic signals also exhibit high diversity and complexity. Their production and propagation mechanisms share certain similarities with those in humans while simultaneously displaying species-specific features. Animal acoustic signals encompass a wide range of modalities, from bird songs to echolocation in marine mammals, reflecting propagation and coupling mechanisms that integrate multiple physical processes^[39]. Overall, in both humans and animals, acoustic signals not only serve as tools for information exchange but are also intrinsically linked to survival strategies and social behaviors.

In contrast, the mechanisms underlying acoustic generation and propagation in plants exhibit striking differences from those in humans and animals. Plants lack specialized sound-producing organs; their “acoustic signals” are inherently passive vibrations induced by physiological activities, primarily originating from cavitation events during water transport in xylem vessels (i.e., impacts on vessel walls from bubble collapse or migration), cell division, turgor pressure fluctuations, and tissue vibrations under stress such as herbivory or mechanical stimulation (as shown in the right of **Figure 1(b)**)^[40]. Within plants, acoustic vibrations propagate mainly through the vascular system (xylem and phloem) and contiguous cell walls, forming the core pathway for internal vibration signal transmission. Compared to propagation through air or soil, vibrations travel faster within plant tissues (e.g., reaching 3 000–5 000 m·s⁻¹ in xylem) with slower energy attenuation, enabling long-distance signal transmission. Moreover, propagation efficiency is regulated by the plant’s own structure: lignified tissues (e.g., stems, roots) facilitate vibration conduction more effectively than succulent tissues (e.g., leaves, petals), while embolism or disruption of the vascular system due to adverse conditions such as drought significantly impedes vibration transmission. Through the propagation of internal acoustic signals, plants can rapidly transmit local stress signals (e.g., herbivory-induced vibrations perceived by leaves) to roots or other organs, thereby triggering systemic defense responses (such as the synthesis of anti-herbivore proteins and stomatal closure) to adapt to stressful environments.

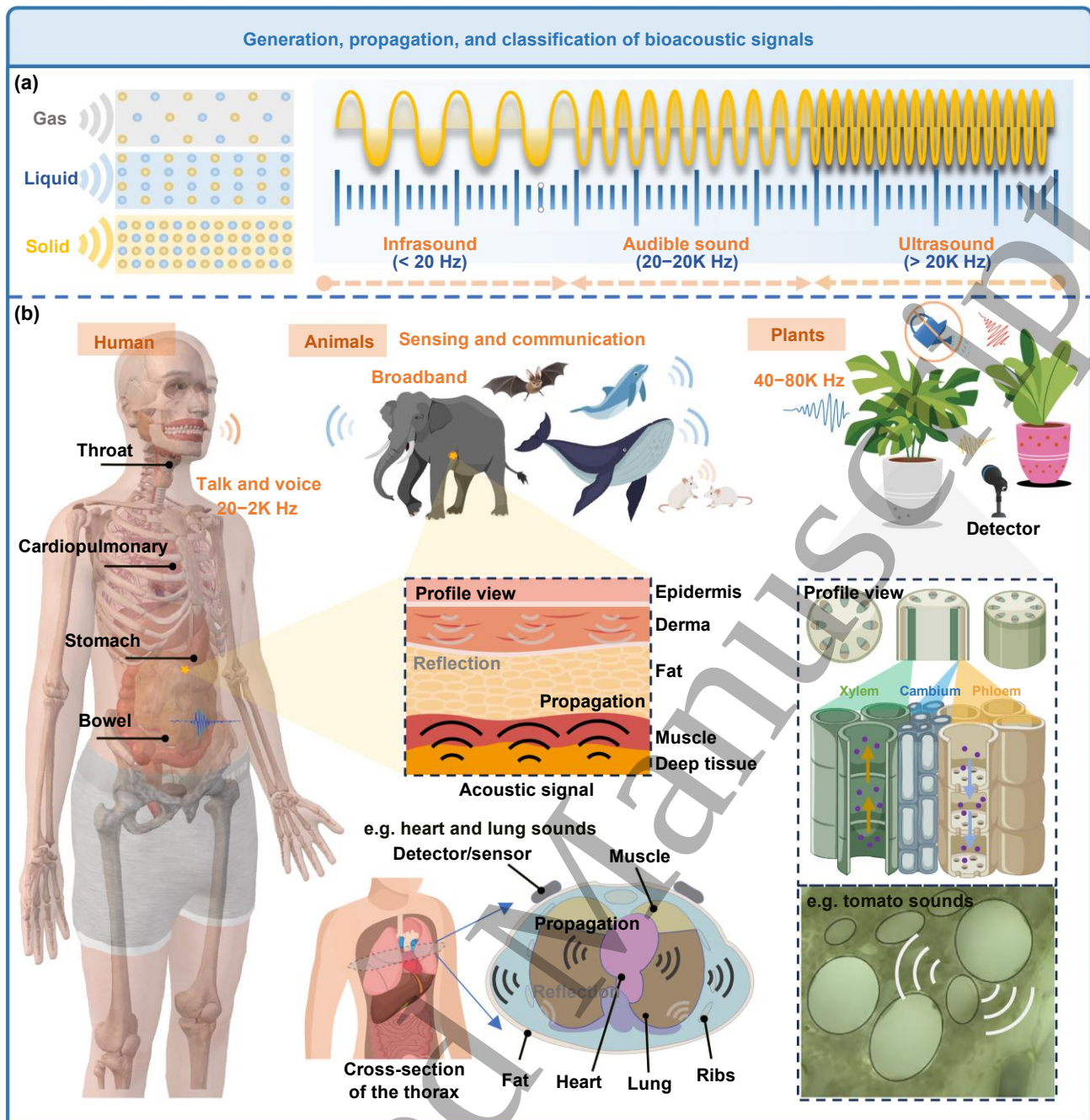


Figure 1. Overview of the origin, classification, and propagation of bioacoustic signals in biological species. (a) Origin and classification of sound. (b) Sound propagation in biological species. Depicts sound propagation characteristics and acoustic signal acquisition strategies across humans, animals, and plants. For humans, it includes vocal sounds from the throat (ranging from 20 to 2 kHz) and internal acoustic signals generated by organs such as the cardiopulmonary system, stomach, and bowel. For animals, it covers broadband acoustic signals utilized in sensing and communication. For plants (exemplified by tomatoes, etc.), it illustrates acoustic signal propagation within internal structures (e.g., xylem, cambium, phloem) and corresponding detection methods. Additionally, cross-sectional/profiled views (e.g., of the human thorax and plant tissues) visualize how acoustic signals propagate, reflect, and transmit in biological tissues.

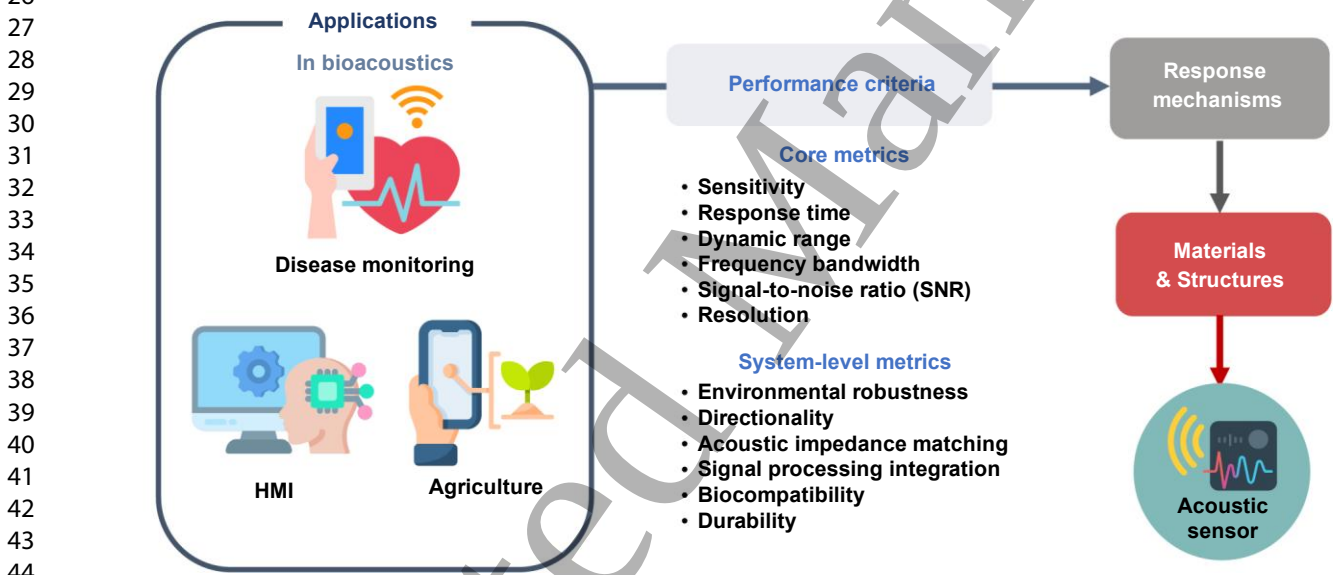
3. Bioacoustic sensors: from applications to sensor selection

3.1 Performance requirements

Over the past few decades, the field of bioacoustic sensing has undergone a significant paradigm shift from pursuing universal performance benchmarks to adopting application-specific design principles. This

1 evolution recognizes that optimal sensor performance must be defined relative to operational contexts rather
2 than abstract metrics. A representative example is cardiac sound monitoring for early diagnosis of heart failure
3 and valvular disorders, where bioacoustic sensors must capture subtle, low-amplitude components such as the
4 third (S3) and fourth (S4) heart sounds in addition to the primary S1 and S2 beats. These low-frequency signals
5 (20–150 Hz) often exhibit tiny amplitudes and are easily masked by respiratory or motion noise, requiring
6 sensors with high sensitivity, flat low-frequency response, superior SNR and good signal postprocessing
7 capability. Moreover, detecting abnormal murmurs or gallops demands a wide dynamic range (30–1 000 Hz)
8 to simultaneously resolve faint pathological events and stronger normal heart tones^[41]. Hence, optimal sensor
9 design must balance broad bandwidth with noise suppression while avoiding resonance within the diagnostic
10 band. Such distinct requirements necessitate a more nuanced approach to sensor development.

11 This section provides a concise overview of the core selection criteria in some specific application
12 scenarios (healthcare, human–machine interaction (HMI) and agriculture), including benchmarks such as
13 sensitivity, response times, frequency and dynamic range, signal-to-noise ratio (SNR), and resolution. In
14 parallel, system-level metrics such as environmental robustness, directionality, acoustic impedance matching,
15 signal processing integration, biocompatibility, and long-term stability are considered in diverse deployment
16 environments (**Figure 2**). Establishing these evaluation criteria facilitates more rational selection of sensor
17 mechanisms, materials, and structural designs, thereby promoting systematic and application-specific
18 development of future bioacoustic sensors.



46 **Figure 2.** Performance requirements and selection criteria in various bioacoustic scenarios.

49 Sensitivity, expressed in $\text{mV} \cdot \text{Pa}^{-1}$ or $\text{dBV} \cdot \text{Pa}^{-1}$, quantifies the electrical response per unit of sound
50 pressure. In bioacoustic sensing, it should be evaluated in conjunction with dynamic range and frequency
51 bandwidth, as accurate physiological sound detection depends on a balanced trade-off among these parameters.
52 Physiological monitoring presents a clear illustration of these considerations, where the low-frequency
53 components of heart sounds require high sensitivity to detect subtle signals, while the broader frequency
54 distribution is more essential for respiratory noises^[42]. These varying characteristics necessitate sensor designs
55 that not only maximize sensitivity but also ensure alignment with the target signals. As such, achieving both
56 high sensitivity and appropriate frequency response constitutes a dual design imperative in the development
57 of acoustic sensors for complex bioacoustic environments. Notably, to achieve a flat frequency response, the
58 resonance frequency of the designed acoustic sensor should be positioned outside the target frequency range.
59 Nevertheless, in specific applications involving a narrowband signal, a tailored resonance band may be
60

introduced to detect weak acoustic signals^[43].

Response time and resolution determine the ability of a sensor to capture transient and subtle acoustic events. A short response time minimizes waveform distortion during rapidly varying signals^[44]. High resolution enables the discrimination of small acoustic or mechanical changes associated with physiological or pathological states. However, improved resolution often increases susceptibility to noise, requiring coordinated optimization of sensing materials, device structure, and signal processing^[45]. The SNR reflects the distinguishability of target acoustic signals from environmental and circuit noise. A high SNR is essential for detecting weak physiological sounds and ensuring reliable feature extraction. It can be improved through sensor-level response enhancement, acoustic shielding, low-noise front-end circuits, appropriate amplification, and filtering strategies such as power-line interference suppression.

Beyond the core sensing metrics discussed above, several system-level parameters should be considered. Environmental robustness and directionality are critical for bioacoustic sensors used in complex deployment scenarios. Field monitoring requires resistance to dust, wind, water, corrosion, and temperature fluctuations, whereas medical ultrasound, auscultation localization, and voice recognition often benefit from directional sensitivity or array-based spatial discrimination.

Acoustic impedance matching governs the efficiency of acoustic energy transfer across interfaces. In bioacoustic applications, particularly in medical ultrasound and wearable acoustic sensors, the inherent impedance mismatch between conventional sensor materials and biological tissues necessitates strategies such as integrating specialized acoustic windows or employing multilayered gradient-impedance structures. Alternatively, the acoustic impedance of the sensor materials can be tailored by incorporating fiber-reinforced composites or microsphere-embedded architectures that increase porosity to reduce bulk density. Such material modifications effectively reduce intrinsic acoustic impedance, thereby enhancing acoustic coupling efficiency and minimizing reflective losses^[46].

Biocompatibility, wearability, mechanical compliance, and long-term stability are essential for wearable and implantable bioacoustic systems. For skin-mounted sensors, especially those used for long-term physiological monitoring, wearability involves stable skin conformity, stretch adaptability, breathability, and comfort during prolonged attachment. Stretch adaptability allows the device to accommodate dynamic skin deformation during breathing, speaking, or daily movement, while breathable substrates or porous encapsulation layers help reduce sweat accumulation and maintain stable acoustic coupling. Hypoallergenic materials are also required to minimize skin irritation, and antibacterial surface coatings may be needed for specific scenarios, such as infant bowel sound monitoring^[47]. For implantable or minimally invasive ultrasonic transducers, inert encapsulation and immunological inertness are essential to avoid foreign body responses. For continuous monitoring, electrical and mechanical stability must be maintained through fatigue-resistant structures, material optimization, and protective encapsulation.

Integrated signal-processing capability is increasingly important for intelligent bioacoustic systems. Hardware modules should provide mechanism-specific signal conditioning, low-noise amplification, multichannel synchronization, and wireless transmission. When additional sensors such as IMUs are used for motion-artifact suppression, synchronized multimodal acquisition becomes necessary^[48]. On the software side, time-domain, frequency-domain, and ML-assisted analysis can support robust feature extraction and classification, while edge processing with compact MCUs can reduce data transmission and power consumption.

Once the performance requirements under specific application scenarios are identified, the appropriate acoustic sensing mechanisms and active materials can be selected in a more targeted manner, and sensor structures can then be designed accordingly based on the intended parameter ranges (**Figure 2**). For instance, in applications requiring flexibility and a high-frequency response, piezoelectric polymers such as polyvinylidene difluoride (PVDF) are often prioritized due to their mechanical compliance and fast response. In contrast, bioinspired designs incorporating microstructure piezoresistive sensors can offer frequency

selectivity through structural tuning for artificial cochlear applications^[49]. In scenarios demanding spatial resolution, miniaturized sensor arrays are typically realized using microelectromechanical (MEMS) architecture.

3.2 Major acoustic transduction mechanisms

Recent advances in bioacoustics have been largely driven by innovations in acoustic transducers capable of capturing weak, low-frequency physiological sounds under dynamic and complex human conditions. Unlike conventional microphones optimized for air-conducted speech, sensors for biomedical monitoring must address distinct challenges, such as low sound pressure (<10 Pa), tissue damping, motion artifacts, and acoustic-impedance mismatch between the sensor and skin. To meet these requirements, research has increasingly focused on flexible and microelectromechanical (MEMS) acoustic sensors that combine high sensitivity, biocompatibility, and long-term stability.

Among diverse transduction strategies, piezoelectric and triboelectric mechanisms are particularly attractive for their potential self-powered operation and high SNR, whereas piezoresistive and capacitive sensors offer simple readout, high sensitivity to small mechanical deformations, and compatibility with flexible wearable architectures^[50,51]. Together, these mechanisms dominate current developments in wearable bioacoustic sensing by offering complementary advantages in sensitivity, power consumption, mechanical compliance, and system integration. Meanwhile, electromagnetic and optical sensors, although capable of exceptional sensitivity, remain hindered by bulkiness, alignment complexity, and poor skin conformity. Ultrasonic sensing, extending into the kilohertz–megahertz regime, further enables high-resolution imaging and internal organ motion tracking, bridging the gap between surface acoustics and deep-tissue diagnostics.

In recent years, several high-quality reviews^[9,26,27,52,53] have comprehensively summarized the fundamental principles, material systems, structural designs, and fabrication techniques of acoustic sensors, including their emerging applications in bioacoustics and human health monitoring. Therefore, this section does not revisit the transduction mechanisms in detail. Instead, we focus on the recent progress, representative examples, and comparative performance of various acoustic sensor types, emphasizing their practical advantages, limitations, and application prospects in physiological sound detection. By integrating materials innovation, structural engineering, and MEMS-level miniaturization, we aim to elucidate how different sensing strategies evolve toward high-fidelity, flexible, and intelligent platforms capable of long-term wearable bioacoustic monitoring.

3.2.1 Piezoelectric and triboelectric acoustic sensors

Piezoelectric transducers have become indispensable in bioacoustic sensing owing to their self-powered operation, rapid response, and efficient electromechanical coupling^[54,55]. Their performance is governed by fundamental piezoelectric equations:

$$S = s^E T + d E \tag{1}$$

$$D = d T + \epsilon^T E \tag{2}$$

where the piezoelectric constant (*d*) critically determines the energy conversion efficiency^[56]. Therefore, the selection of piezoelectric materials is of paramount importance for optimizing sensor performance.

Early rigid ceramics such as lead zirconate titanate (PZT) and lead magnesium niobate-lead titanate (PMN-PT) offered high electromechanical coupling but suffered from rigidity and limited skin conformity. Polymeric systems such as PVDF and PVDF-TrFE improved mechanical compliance and are therefore widely used in flexible and wearable acoustic sensors^[57,58]. More recently, molecular ferroelectrics have emerged as a promising class of piezoelectric materials for acoustic sensing because they combine high piezoelectric output, structural tunability, low density, and potentially lower acoustic impedance than conventional ceramics. Representative hybrid organic–inorganic molecular ferroelectrics, such as trimethylchloromethyl ammonium trichlorocadmium (TMCM-CdCl₃) and related TMCM-based halide systems, have shown large piezoelectric

coefficients and can be incorporated into polymer matrices to improve flexibility and stretchability. However, their intrinsic brittleness, environmental stability, and large area processability still require further optimization before broad wearable deployment^[59].

Device design strongly influences the piezoelectric response; traditional piezoelectric acoustic sensors commonly adopt a sandwich structure, with electrodes placed on both sides of the active layer. Recently, various structural strategies have been employed to amplify mechanical deformation and thus enhance signal output. In bioacoustic applications, thin membranes operate in mixed d_{31}/d_{33} working modes where in-plane shear stress dominates under acoustic excitation. A typical example is that thermally drawn PVDF fibers doped with barium titanate (BTO) ceramics show a high d_{31} value ($\approx 46 \text{ pC}\cdot\text{N}^{-1}$) with anisotropic structures (**Figure 3(a)**). When woven into acoustic textiles, it can detect nanoscale chest vibrations and sound pressure as low as 10^{-7} atm ^[60]. The bioinspired device also significantly boosted the performance. A printed microcone array based on a lead-free perovskite material mimicking cochlear hair cell geometry achieved $150 \text{ mV}\cdot\text{Pa}^{-1}$ sensitivity over 100–3 000 Hz and exhibited directional responsiveness^[61]. Beyond thin films, piezoelectric nanofibers with low acoustic impedance further improved conformity and breathability for long-term physiological sound monitoring^[62]. Porous piezoelectrets with engineered voids and tunable dipole alignment are gaining attention for their mechanical flexibility and adjustable piezoelectric performance. A porous fluorinated ethylene propylene (FEP) patch incorporating an electromagnetic-shielding layer reduced harmonic interference by $> 20 \text{ dB}$ and demonstrated reliable detection of heartbeats, respiration, coughing, and swallowing^[56]. These examples underscore how material selection and structure design at the micro level reconcile flexibility with high piezoelectric output.

Triboelectric acoustic sensors (TAS) complement piezoelectric systems by generating electrical signals through contact electrification and electrostatic induction. This mechanism involves (1) contact-induced charge transfer between dissimilar materials (characterized by electron affinity differences), (2) separation-induced potential generation and (3) alternating current production through cyclic operation. Their ability to operate without external power and their high voltage output make them promising for voice recognition^[63,64], human–machine interaction (HMI)^[65,66], and health monitoring^[67,68].

Early in 2018, a porous FEP/Au-Kapton multilayer TAS was designed to achieve a broadband response from 0.1 to 5 kHz for speech perception^[69]. By engineering internal boundaries and spacers, devices later shaped multiresonant behavior to enhance frequency selectivity, extending detection to the full human auditory range for sound localization and tracking^[70]. These demonstrations reveal how structure can be intentionally designed to emphasize clinically relevant bands (e.g., 100–1 500 Hz for lung and bowel sounds).

Beyond conventional thin-film architectures, microstructure engineering has become a key strategy for enhancing triboelectric acoustic sensors. Surface morphology modulation, such as cochlea-inspired conical micropillar arrays made of PVDF-HFP, effectively enlarges the contact area and amplifies acoustic-to-electric conversion, enabling the detection of subtle vibrations within a 50–80 dB range^[71]. At the nanoscale, hierarchical and gradient porous composites based on BTNP@SiO₂/PVDF-TrFE systems (**Figure 3(b)**) provide broadband (145–9 000 Hz) and frequency-selective responses for noise-robust voice recognition^[72]. Together, these hierarchical designs demonstrate how structural control from micro- to nanoscale enables efficient and selective acquisition of complex acoustic information, paving the way for bioinspired and intelligent sound-sensing platforms.

Overall, both piezoelectric and triboelectric acoustic sensors exhibit excellent adaptability in the bioacoustic frequency range and compatibility with flexible substrates. Piezoelectric devices feature superior linearity and long-term stability, whereas triboelectric devices deliver higher voltage and broader frequency response. The critical challenges ahead lie in suppressing dielectric and charge-leakage noise, improving encapsulation against humidity, and balancing softness with acoustic fidelity for continuous bioacoustic monitoring.

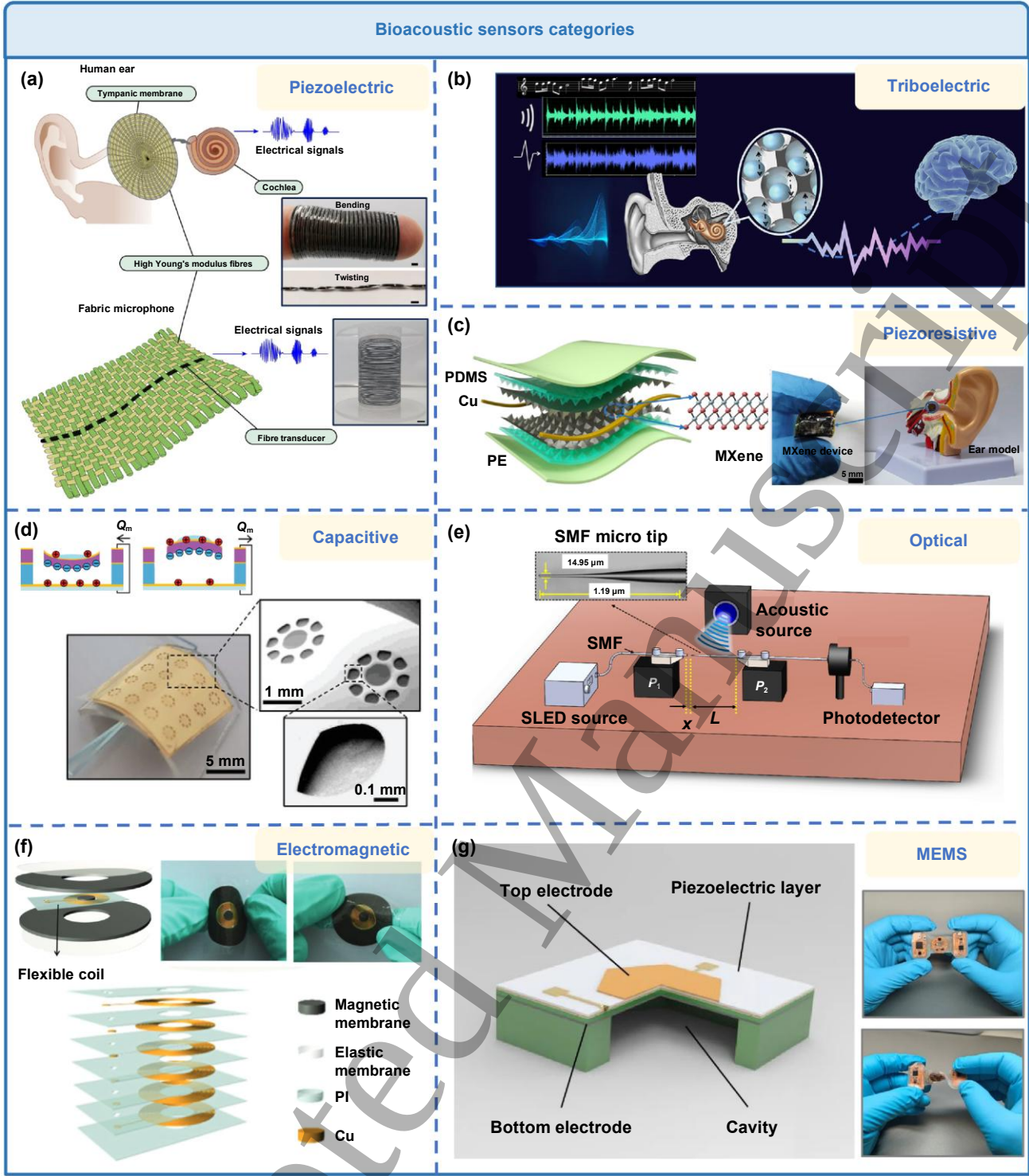


Figure 3. Overview of bioacoustic sensors based on different sensing mechanisms. (a) Piezoelectric nanofiber-based intelligent hearing system with a spiral trampoline-like structure. Reproduced from^[60], with permission from Springer Nature. (b) An acoustic core-shell resonance harvester from nanoscale material optimization. Reproduced with permission from^[72]. Copyright © 2022 The Authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original U.S. Government Works. Distributed under a Creative Commons Attribution NonCommercial License 4.0 (CC BY-NC). (c) Schematic illustration of the MXene eardrum device with a micropyramid structure. Reproduced with permission from^[73]. Copyright © 2022 The Authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original U.S. Government Works. Distributed under a Creative Commons Attribution NonCommercial License 4.0 (CC BY-NC). (d) Diagram of a skin-attachable electret-powered capacitive auditory sensor. ^[74]John Wiley & Sons. © 2022 Wiley-VCH GmbH. (e) Microcantilever acoustic sensor schematic: tapered optical fiber microtip (14.95 μm) and single-mode fiber, with adjustable <https://mc04.manuscriptcentral.com/ijem-caep>

cantilever for sound-to-light conversion. Reproduced from^[75]. CC BY 4.0. (f) MEMS micro dynamic microphone with a double-layer spiral coil, semicircular pad, and polyimide diaphragm, integrated via photolithography on a frame. ^[76]John Wiley & Sons. © 2020 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim. (g) Three-dimensional MEMS acoustic sensors based on SOI wafers are embedded in elastomeric substrates to achieve localized rigidity for sensing while maintaining global flexibility. Reproduced from^[77]. CC BY 4.0.

3.2.2 Piezoresistive and capacitive acoustic sensors

Piezoresistive acoustic sensors convert mechanical deformation into resistance changes through strain-induced modulation of carrier mobility, tunneling distance, or conductive-path geometry. The resistance variation follows^[78, 79]:

$$\frac{\Delta R}{R_0} = \varepsilon(1 + 2\nu) + \frac{\Delta\rho}{\rho} \quad (3)$$

where ε represents the mechanical strain ($\Delta L/L_0$) and ν is the material's Poisson's ratio. The equation's first term captures geometric deformation effects, while the second term reflects intrinsic resistivity changes from electronic structure modifications. Their simple readout circuitry and high signal amplitude make them attractive for wearable bioacoustic monitoring, particularly for throat, laryngeal, and speech-related vibrations^[78, 79]. Early designs employing doped silicon diaphragms or metallic strain gauges provided linear responses but limited stretchability and low gauge factors, which hindered the detection of faint, low-frequency biological sounds. Crack engineering in metallic films enhances piezoresistive sensitivity by forming reversible microcrack networks that dynamically modulate conductive pathways under acoustic stress, enabling amplified response to weak vibrations with straightforward fabrication and excellent reproducibility.

The emergence of nanocomposite elastomers incorporating graphene, MXene, or carbon nanotubes (CNTs) has markedly improved flexibility and sensitivity. These conductive networks respond to microstrains induced by acoustic pressure through tunneling and percolation effects, achieving detection thresholds^[80, 81]. One representative example is employing polydimethylsiloxane (PDMS) as a flexible elastomeric substrate, embedding regularly arranged micropyramidal structures formed by doping 2D MXene nanosheets into the matrix (**Figure 3(c)**)^[73]. Through synergistic geometric and electronic modulation, the MXene–elastomer tympanic membrane realizes hierarchical signal amplification, achieving a 0.1–3 kHz bandwidth and ~50 dB SNR within the key 200–900 Hz range while maintaining a linear acoustic response.

Capacitive acoustic sensors function by transducing mechanically induced vibrations from incident sound waves into periodic fluctuations in capacitance, thereby enabling acoustic perception and signal conversion ($C = \epsilon A/d$)^[82]. Their inherent linearity and scalability through MEMS processing have made them dominant in commercial microphones and increasingly relevant for biomedical applications^[83, 84]. These structures suffer from brittleness and fabrication complexity, rendering them less suitable for emerging applications in flexible wearable electronics. To address these limitations, several strategies, such as deformable diaphragms^[85, 86], compliant electrodes^[80, 87, 88], and soft dielectric layers^[89–91] have been developed. Polymeric diaphragms such as polydimethylsiloxane (PDMS), polyimide (PI), or thermoplastic polyurethane (TPU) have replaced rigid silicon to achieve lower acoustic impedance and better skin conformity. The incorporation of electret materials provides built-in polarization, eliminating the need for external bias and reducing power consumption. A representative electret-based membrane achieved sensitivities up to $2.2 \text{ V} \cdot \text{Pa}^{-1}$ with a flat 80–3 000 Hz response, enabling accurate speech and breath detection (**Figure 3(d)**)^[74].

Recent designs exploit microstructured diaphragms and compliant electrodes to enhance performance. Corrugated or spring-arm supports relieve residual stress and increase membrane compliance, while graphene or Ag-nanowire electrodes ensure conductivity without sacrificing flexibility^[90]. Ionic-gel dielectrics further expand versatility by forming electric double layers that strongly couple with acoustic pressure. Such ionic-

capacitive hybrids exhibit high sensitivity and acoustic-impedance matching with wet or soft tissues, demonstrating potential for underwater and mucosal sound sensing^[92].

Comparatively, capacitive sensors deliver a broader frequency range and superior linearity, whereas piezoresistive sensors excel in low-frequency, high-deformation scenarios such as voice, cough, or heart-sound monitoring. Both types benefit from MEMS-compatible fabrication, allowing miniaturization and array integration. However, challenges persist. Piezoresistive devices suffer from hysteresis, power drift, and temperature sensitivity; capacitive types require stable biasing and often exhibit reduced output under low-frequency, small-amplitude vibrations. From a biomedical perspective, the fundamental trade-off lies between mechanical compliance and electrical precision: increasing softness improves comfort but weakens electromechanical coupling.

3.2.3 Optical and electromagnetic acoustic sensors

Optical and electromagnetic acoustic sensors represent two high-precision yet nonideal approaches for wearable bioacoustic monitoring. Although both have achieved remarkable sensitivity and frequency response in laboratory or industrial environments, their fundamental physical configurations impose severe limitations on flexibility, miniaturization, and skin conformity, which are critical requirements for human physiological sound detection. Nevertheless, their exceptional stability and noncontact operation render them suitable for nonhuman bioacoustic applications, including the detection of plant acoustic emissions and other low-frequency biological vibrations.

Optical acoustic sensors transduce sound waves into optical signals by modulating light intensity or wavelength through reflective or interferometric mechanisms. Typical configurations, including intensity-modulated films, fiber Bragg gratings (FBGs), and Fabry Perot interferometers, offer ultrahigh sensitivity and broad bandwidth. Dass fabricated a diaphragm-free microcantilever fiber acoustic sensor (MCM), which realizes low-frequency sound wave detection based on the principle of tapered microtip single-mode fiber light field coupling (**Figure 3(e)**)^[75]. This microphone uses a tapered fiber microtip cantilever. Acoustic waves deflect the cantilever, altering the light coupling efficiency with a fixed fiber. It achieves high sensitivity ($10.63 \text{ mV} \cdot \text{Pa}^{-1}$) in the 0–400 Hz range, which is ideal for low-frequency bioacoustics such as visceral sounds. However, its macroscopic cantilever length limits miniaturization and practical integration for wearable applications. Mechanical fragility and temperature sensitivity are also concerns.

Electromagnetic acoustic sensors generate voltage through electromagnetic induction between a vibrating coil and a magnetic field. Traditional microphones using permanent magnets and wound coils are mechanically robust and capable of broadband operation up to tens of kilohertz, but the necessity of magnetic components inherently results in bulky, rigid structures. When applied to biological monitoring with subtle signals and the dominant frequency lying under 1 kHz, the induced voltage is often in the microvolt range, far below the threshold required for high signal-to-noise detection^[26, 93]. Recent works employing microscale coils and flexible NdFeB magnetic films have slightly improved miniaturization and conformability (**Figure 3(f)**), yet the output remains limited with measurements during chest heartbeat tests yielding approximately 12 microvolts^[76]. Furthermore, magnetic hysteresis, temperature drift, and potential interference from external electromagnetic fields complicate stable long-term measurements.

Overall, optical and electromagnetic sensors exemplify the trade-off between measurement precision and mechanical adaptability. Their outstanding sensitivity and bandwidth make them valuable reference technologies for benchmarking signal accuracy or exploring specialized noncontact diagnostics, but their incompatibility with soft and deformable limits their translation into wearable applications. Future designs are expected to shift support to long-term static monitoring, such as plant acoustics, in an energy-efficient, preamplifier-free manner^[94].

3.3 Integration platforms and sensing modalities

Beyond the fundamental transduction mechanisms discussed above, bioacoustic sensors also differ in their integration platforms and operating modalities. Microelectromechanical systems (MEMS) technology represents a miniaturized fabrication and integration platform that can incorporate piezoelectric, capacitive, piezoresistive, or electromagnetic transduction mechanisms. Ultrasonic sensing, in contrast, refers to an acoustic frequency regime and application modality, typically relying on piezoelectric or capacitive transducers for high-frequency wave emission and reception. Therefore, MEMS integration and ultrasonic sensing are discussed here as cross-cutting architectures rather than parallel sensing mechanisms.

3.3.1 Toward MEMS integration and hybridization

MEMS has revolutionized acoustic sensing through miniaturization, mass production, and integration of multiple transduction mechanisms within a unified platform. By leveraging mature silicon processing and thin-film deposition, MEMS microphones achieve excellent reproducibility and low power consumption. However, their conventional rigid architecture and high resonant frequency (typically >1 kHz) are poorly matched to the low-frequency regime of human body acoustics. To address this, recent works have introduced flexible MEMS architectures that integrate soft materials (PI, PDMS, TPU) with functional films (PZT, PVDF, graphene). For example, an origami-inspired flexible electromagnetic MEMS sensor surrounded by a flexible coil and a circular vibrator can achieve a broadband (1–10 kHz) response and conformal adhesion to the skin^[95]. Although its output voltage (~ 12 μV) remains low, the design demonstrates the feasibility of integrating magnetomechanical conversion in wearable form. Similarly, multilayer graphene diaphragms incorporated into MEMS microphones have achieved sensitivities up to $24 \text{ mV} \cdot \text{Pa}^{-1}$ with reduced stiffness and improved compliance^[96]. Piezoelectric MEMS sensors represent another major advance, combining self-powered operation with microfabrication precision. Flexible single-crystal PZT films and rib-structured diaphragms significantly enhance piezoelectric coupling and mechanical stability, achieving sensitivities of $36 \text{ mV} \cdot \text{Pa}^{-1}$ and SNRs exceeding 50 dB^[96]. Such devices are particularly promising for miniaturized stethoscopes, wearable patches, and implantable health monitors. Piezoresistive MEMS variants using graphene or porous semiconductors further extend capabilities for detecting subtle laryngeal or vocal cord vibrations^[97].

Despite these achievements, critical challenges persist. Rigid MEMS structures are inherently limited in accommodating the skin's curvature, leading to signal distortion and poor acoustic coupling. Moreover, the sensitivity and bandwidth trade-off remains a major bottleneck: thin diaphragms offer a wide frequency response but reduce the displacement amplitude, while thick films improve the sensitivity but narrow the bandwidth. Future research must therefore adopt hybrid transduction strategies such as integrating piezoelectric and piezoresistive effects within a single MEMS diaphragm to exploit complementary advantages in sensitivity and dynamic range. Another emerging direction is soft-hard hybrid integration, where MEMS chips are embedded in elastomeric substrates to achieve localized rigidity for sensing while maintaining global flexibility (**Figure 3(g)**). This strategy not only improves user comfort but also facilitates array-level integration for spatially resolved bioacoustic imaging^[77]. Additionally, the combination of MEMS with edge algorithms and wireless communication modules is redefining wearable diagnostics toward intelligent, self-adaptive sensing systems capable of real-time decision-making under complex physiological conditions.

3.3.2 Ultrasonic sensing transducers and high-frequency acoustic modalities

Ultrasonic sensors are devices that exploit the propagation characteristics of ultrasonic waves in a medium to achieve sensing functionality, with their operational principle primarily rooted in the processes of emission, propagation, and reception of ultrasonic waves. In most biomedical and wearable ultrasound systems, piezoelectric transducers remain dominant because of their high electromechanical coupling

efficiency and fast response. Thus, piezoelectric ultrasonic sensors are better understood as piezoelectric transducers operating in the ultrasonic frequency range and optimized for both wave emission and echo reception^[98]. Most piezoelectric ultrasonic transducers adopt a sandwich structure composed of a piezoelectric active layer, top and bottom electrodes, acoustic matching layers, and backing layers^[99]: PZT or piezoelectric composites are commonly used as active elements. Matching layers reduce acoustic impedance mismatches and improve coupling with the target medium, while backing layers absorb backward waves and enhance axial resolution. Therefore, the material composition and thickness of these layers are key parameters for balancing sensitivity, bandwidth, and imaging resolution^[100].

With the continuous advancement of intelligent sensing technologies, rigid ultrasonic transducers, distinguished by their high precision and stability, have assumed a pivotal role in the field of biomedical engineering. Yin et al. proposed a typical rigid ultrasonic transducer based on 1-3 piezoelectric composites, which is used for prosthetic control^[101]. This is a common sandwich structure with a customized thickness of the matching layer, backing layer and shell, which has already been used in their previous research^[102-104]. To address the limitations of rigid transducers, Sgambato et al. explored a flexible modular A-mode ultrasound armband with dry silicone coupling, enabling wrist-hand tracking and demonstrating the great potential and generalization capacity of ultrasound-based technology as a virtual reality interface (**Figure 4(a)**)^[105].

Despite the technical foundation laid by these works, the inherent limitations of rigid transducers remain prominent, including fixed geometric dimensions, inflexible structural forms, and poor conformal contact with human skin, which severely restrict their long-term wearable applicability in unconstrained daily scenarios. When transitioned to flexible or soft architectures, ultrasonic transducers retain muscle status monitoring capabilities while achieving better conformal contact with human skin; compared to rigid counterparts, flexible substrates exhibit enhanced wear resistance and long-term comfort. For instance, Gao et al. developed a fully integrated wearable ultrasonic myography system with a thickness reduced to approximately 2 mm, significantly improving comfort and usability (**Figure 4(b)**)^[106]. Another type of ultrasound transducer was proposed by Toymus et al. for bladder volume monitoring. It has a design of air-backing, which eliminates energy loss due to absorptive backing layers and allows ultrasonic sensing with lower voltages, realizing high accuracy detection in volunteers with various bladder shapes and sizes (**Figure 4(c)**)^[107]. Similarly, Zhang et al. proposed another kind of conformable ultrasound sensor based on multiple phased arrays for bladder volume monitoring (**Figure 4(d)**)^[1]. In this work, two matching layers are designed for better matching with human skin, and a thick nonconductive backing layer outside the electrodes is set for sound wave oscillation absorption, which is different from the structure developed by Gao et al.^[106], whose conductive backing layer is between the piezoelectric component and electrodes.

Moreover, for blood flow or blood pressure monitoring, ultrasonic sensors can also play an important role. As an example, a transducer array was designed by Wang et al., which is still a sandwich structure with a flexible substrate and electrode (**Figure 4(e)**)^[108]. To improve the breathability and wearability, Noda et al. proposed a kind of textile-based ultrasound imaging probe (**Figure 4(f)**). Compared to previous probes, this probe can show good performance in B-mode imaging because of the size and arrangement of piezoelectric components. On the other hand, it shows great potential to integrate ultrasound sensors into common cloth for daily monitoring^[109].

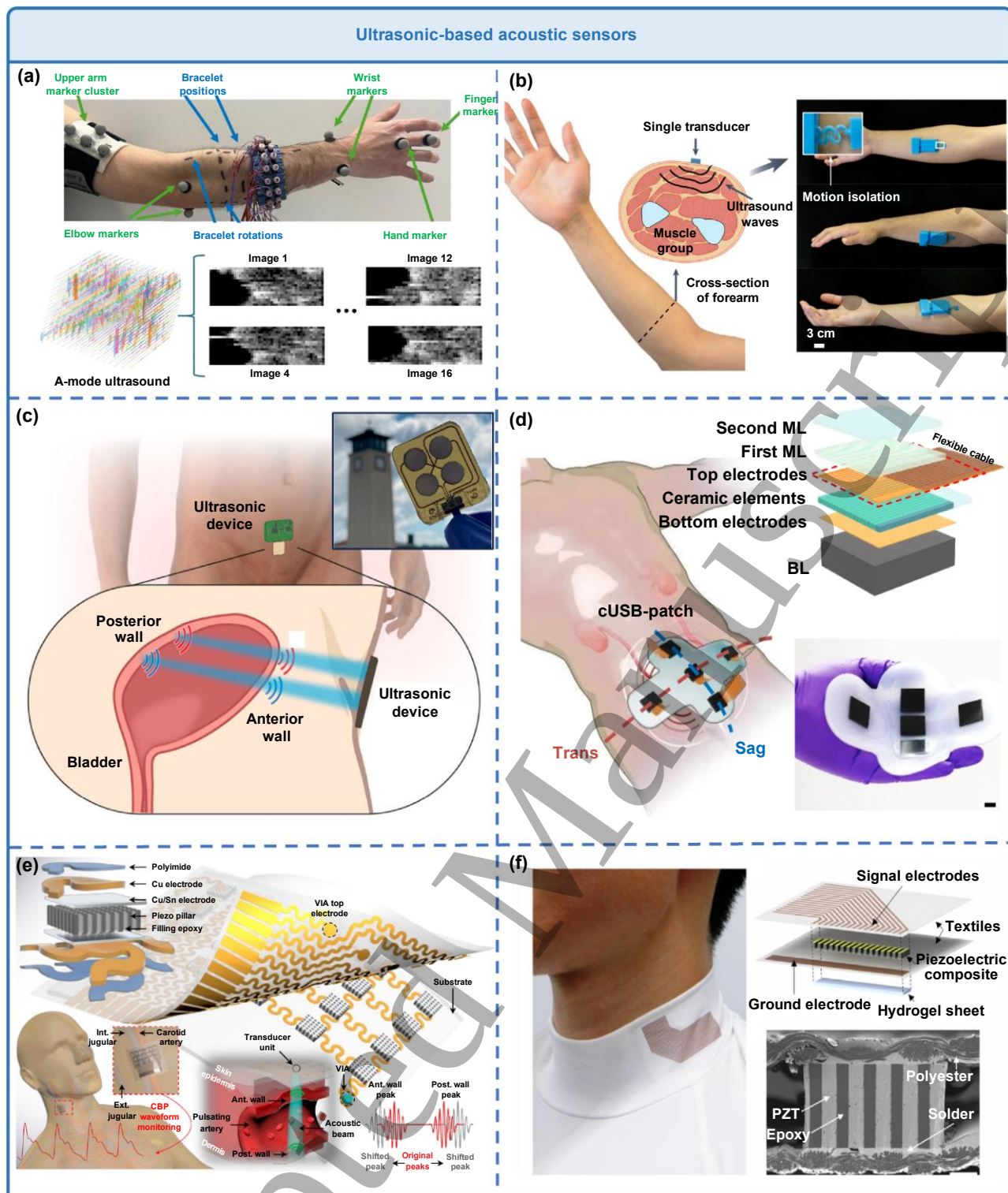


Figure 4. Ultrasonic-based acoustic sensors. (a) A-mode 32-channel dry-coupling ultrasound bracelet for wrist and hand tracking. Reproduced from^[105]. CC BY 4.0. (b) Flexible A-mode ultrasound sensor for wearable echomyography. Reproduced from^[106], with permission from Springer Nature. (c) Flexible ultrasound sensor array for bladder volume monitoring. Reproduced with permission from^[107]. CC BY-NC-ND 4.0. (d) Flexible ultrasound patch for bladder volume monitoring. Reproduced from^[1], with permission from Springer Nature. (e) Flexible ultrasound sensor array for central blood pressure waveform monitoring. Reproduced from^[108], with permission from Springer Nature. (f) Fabric-based ultrasound sensor array for vascular monitoring. Reproduced with permission from^[109]. CC BY-NC-ND 4.0.

Table 1. Application-oriented comparison of representative bioacoustic sensing technologies.

Quantitative values are extracted from cited representative studies and may vary with material composition, device structure, acoustic source, measurement protocol, and signal-processing method. MEMS is treated as an integration platform, whereas ultrasonic sensing is treated as a high-frequency acoustic modality rather than an independent transduction mechanism.

Sensing mechanism/modality	Representative examples	Suitable bioacoustics scenarios/frequency range	Representative quantitative performance	Merits and limitations	Design guideline
Piezoelectric sensing [55,59-61]	PVDF/PVDF-TrFE fibers; PZT/BTO composites; piezoelectret patches; TMCM-CdCl ₃ fibers	Heart/lung/bowel sounds, voice, wearable auscultation; low-frequency to several kHz	TMCM-CdCl ₃ ferroelectric fibers: bandwidth 30–5 000 Hz; SNR >20 dB; Sensitivity 30.16 mV·dB ⁻¹	High sensitivity, fast response, self-powered; limited by polarization stability, coupling, and encapsulation	Suitable for weak physiological sounds requiring high sensitivity and compliance
Triboelectric sensing [68-72, 110]	FEP/Au-Kapton TAS; triboelectric stethoscope; microstructured TENG	Voice, HMI, cardiac auscultation; 0.1 Hz–20 kHz	FEP/Au-Kapton auditory sensors: bandwidth 0.1-5 kHz; dynamic range: 50-80 dB; sensitivity 1 215 mV·Pa ⁻¹ ; SNR > 36 dB	High voltage and self-powered; limited by humidity, charge leakage, drift	Suitable for self-powered voice/HMI or weak-sound enhancement, long-term encapsulation and charge stability are critical
Piezoresistive sensing [73, 78-81]	CNT/graphene/MXene elastomers; microcrack sensors; MXene eardrum	Throat vibration, speech, cough; low-frequency to kHz	MXene-elastomer membrane: bandwidth 0.1–3 kHz; SNR approx. 50 dB at 200-900 Hz	Simple readout and high deformation sensitivity; limited by hysteresis, drift, temperature sensitivity	Best for contact deformation rather than high-fidelity airborne sound
Capacitive sensing [74, 82, 83, 85, 90, 91, 111]	Electret auditory sensors; soft capacitive microphones; iontronic hydrogel microphones	Speech, breath, underwater/mucosal sensing; structure-dependent frequency	Electret-based membrane: sensitivity: 2.2 V·Pa ⁻¹ with bandwidth 80–3 000 Hz;	Good linearity and array compatibility; limited by bias stability and parasitic capacitance	Suitable for miniaturized microphones and soft e-skin arrays
Optical acoustic sensing [75]	FBG; Fabry–Pérot; optical microcantilever; LDV	Plant emissions, animal biomechanics, noncontact detection; device-dependent frequency	Microcantilever sensor: sensitivity 10.63 mV·Pa ⁻¹ , bandwidth 0–400 Hz,	EMI immunity and high displacement resolution; limited by bulky, fragile, alignment-sensitive	Suitable when noncontact and high precision outweigh wearability
Electromagnetic /magnetoelastic sensing [76, 93, 112]	Flexible magnetic membranes; dynamic microphones; liquid magnet sensors	Voice, laryngeal vibration, motion-artifact-resistant sensing; 1 Hz–10 kHz	Flexible electromagnetic sensor: bandwidth 1 Hz to 10 kHz sensitivity 0.59 mV·μm ⁻¹ at 1.7 kHz.	Noncontact magnetic readout, motion-artifact resistance; limited by low output, hysteresis, electromagnetic interference	Suitable for motion-artifact suppression or noncontact readout scenarios, amplification and magnetic shielding are important
MEMS-	MEMS	Portable auscultation,	Graphene MEMS:	Miniaturized	Suitable for scalable

Sensing mechanism/modality	Representative examples	Suitable scenarios/frequency range	Representative quantitative performance	Merits and limitations	Design guideline
integrated acoustic sensing platform [77, 94-96]	microphones; piezoelectric MEMS sensors; capacitive MEMS microphones; flexible MEMS chips	arrays, wildlife recording; target 20 Hz–20 kHz	sensitivity 24 mV·Pa ⁻¹ ; SNR >50 dB;	scalable; limited by rigidity and skin-coupling mismatch	arrays and portable acquisition
High-frequency ultrasonic sensing modality [1, 98-100, 105, 106]	Flexible ultrasound patches; A/B-mode probes; textile ultrasound probes	Deep tissue, bladder, blood pressure, muscle motion; >20 kHz–MHz	Wearable systems: thickness down to ~2 mm; A-mode/B-mode support monitoring	Depth-resolved and high spatial resolution; limited by rather than coupling/matching/calibration complexity	Suitable for depth-resolved monitoring surface auscultation

Note: the listed values represent reported or representative ranges from the cited studies; direct comparison should consider differences in device configuration, testing setup, and signal source.

3.4 Critical analysis and trade-offs in sensor selection

Different transduction categories with various mechanisms exhibit distinct advantages for specific bioacoustic targets. Piezoelectric and triboelectric sensors, characterized by high sensitivity, fast response, and mechanical compliance, are most effective for human physiological monitoring, such as heart, lung, bowel, and laryngeal sound detection, where low-frequency (<2 kHz) vibrations dominate. Piezoresistive and capacitive sensors provide high resolution for microdeformation and are ideal for vocal or throat vibration monitoring and wearable e-skin systems requiring continuous contact and stretchability. MEMS microphones, with miniaturized and low-power operation, extend applications to both human and animal studies, supporting portable auscultation and wildlife sound recording.

Furthermore, bioacoustic sensors cannot be implemented as a one-size-fits-all universal solution. Their selection is inherently governed by a tripartite trade-off among mechanical compliance, electromechanical coupling efficiency, and frequency bandwidth. For instance, in human physiological monitoring, piezoelectric polymers such as PVDF or triboelectric nanogenerators are commonly adopted^[60,69], rather than rigid ceramic materials such as lead zirconate titanate (PZT). Such design trade-offs deliberately sacrifice the ultrahigh energy conversion efficiency of rigid materials to achieve superior mechanical conformability, which is critical for minimizing acoustic impedance mismatch at the skin-sensor interface. In contrast, acoustic signals in plant research are extremely weak and fall within the high-frequency range of 20–150 kHz, thereby requiring

priority to be placed on detection sensitivity and bandwidth. Accordingly, rigid ultrasonic transducers or noncontact optical and electromagnetic sensors are widely utilized, despite their inherent lack of flexibility and stringent requirements for installation and alignment. Additionally, microelectromechanical system (MEMS) technology exhibits unparalleled miniaturization advantages for large-scale array deployment. Nevertheless, a critical design compromise remains: a thinner MEMS diaphragm is essential to broaden the frequency response range, which inevitably reduces vibration displacement and degrades the detection sensitivity toward typical ultralow-frequency vibrations (<100 Hz), such as heart sounds and bowel sounds. Consequently, MEMS sensors are rationally applicable only when the spatial resolution performance of sensor arrays outweighs the demand for ultralow-frequency peak sensitivity.

Collectively, the choice of transduction mechanism should align with the biological target, dominant frequency range, and required mechanical compliance, ensuring reliable signal acquisition across diverse bioacoustic systems. For reference, Table 1 summarizes representative sensor types, their primary targets, detection frequency ranges, key performance metrics, and potential applications. While advanced acoustic sensors provide the essential hardware foundation for capturing high-fidelity bioacoustic vibrations, the raw hardware output is frequently contaminated by environmental noise, motion artifacts, and overlapping cross-species signals. Therefore, the leap from capturing raw physical vibrations to extracting meaningful biological insights cannot rely on hardware alone; it necessitates robust signal acquisition strategies and sophisticated downstream processing algorithms. To address this, Section 4 systematically explores the methodologies for bioacoustic signal acquisition and the artificial intelligence-driven analytical frameworks used to decode these complex acoustic signatures.

3.5 Technical bottlenecks in sensing performance

While the aforementioned transduction mechanisms provide the physical basis for bioacoustic sensing, translating these principles into high-performance wearable and field-deployable devices faces several fundamental technical bottlenecks. These challenges are primarily manifested in the inherent trade-offs among three key performance metrics: SNR, sensitivity, and frequency bandwidth.

(1) Signal quality bottlenecks: the primary bottleneck for SNR in bioacoustics is the presence of high-intensity ambient noise and internal electronic noise floors. For wearable biomedical sensors, $1/f$ noise (flicker noise) at low frequencies severely interferes with the capture of heart and bowel sounds (typically <100 Hz). Moreover, as sensors are miniaturized to improve comfort, the reduction in active sensing area inherently leads to a higher thermal noise floor (Brownian noise), which limits the ability to resolve ultraweak signals, such as those from fetal heartbeats or deep-tissue acoustics. Current research aims to address this by developing differential noise-canceling architectures and specialized low-noise front-end amplifiers, yet achieving a high SNR in unstructured, chaotic environments remains a persistent challenge.

(2) Sensitivity vs. dynamic range trade-offs: high sensitivity is essential for detecting subtle bioacoustic events such as plant xylem cavitation or early-stage pathological murmurs. However, increasing sensitivity often comes at the expense of a dynamic range. For instance, sensors optimized for high-sensitivity detection of weak signals may suffer from saturation or nonlinear distortion when exposed to high-amplitude sounds (e.g., loud vocalizations or industrial agricultural noise). The challenge lies in engineering materials with high electromechanical coupling coefficients (e.g., optimized PZT or high-performance ferroelectric polymers) that can maintain linear sensitivity across several orders of magnitude in pressure amplitude.

(3) Frequency bandwidth constraints: Bioacoustic signals span a vast frequency range, from infrasonic animal communications (<20 Hz) to ultrasonic plant emissions (>20 kHz). Designing a single sensor that provides a flat, high-fidelity frequency response across this entire spectrum is a major engineering hurdle. High-sensitivity sensors often rely on mechanical resonance to amplify signals, which inherently limits their effective bandwidth. Conversely, broadening the bandwidth through mechanical damping inevitably degrades

the sensitivity and SNR. Furthermore, for MEMS-based acoustic sensors, the trade-off between the diaphragm's stiffness (required for high-frequency response) and its compliance (required for low-frequency sensitivity) represents a critical structural bottleneck that currently necessitates the use of multisensor arrays or heterogeneous integration strategies to achieve broad-spectrum coverage.

4. Bioacoustic signal acquisition and analysis

4.1 Bioacoustic signal acquisition

Bioacoustic signals, serving as acoustic carriers of dynamic interactions within living systems, facilitate cross-disciplinary breakthroughs spanning life sciences to engineering applications. This section focuses on the acquisition of acoustic signals from humans, animals, and plants, systematically discussing the characteristic differences across biological systems, the innovative principles underlying acquisition technologies, and their representative application scenarios. Emphasis is placed on human physiological signal acquisition, high-sensitivity MEMS and flexible electronics integration paradigms within wearable devices, and the technological innovations of heterogeneous microphone arrays and fiber-optic sensing in ecological monitoring of fauna and flora. In summary, this delineates the continuum of bioacoustic technology from physiological signal capture to ecological behavior decoding. While the operational environments differ from the controlled clinical settings for human auscultation to the chaotic underwater acoustic channels for marine tracking, the fundamental goal of signal acquisition remains the same: maximizing the SNR and capturing high-fidelity acoustic signals.

4.1.1 Acquisition systems for human physiological sounds

As a dynamic mapping of physiological states, the precise acquisition of human physiological signals represents a core challenge in bioacoustics. Unlike the isolated sensor transduction mechanisms detailed in Section 3, modern acquisition systems emphasize the synergy between sensing elements, front-end signal conditioning, and wireless architectures.

To capture inherently weak physiological signals with high fidelity, system-level integration of functional materials and front-end circuitry is paramount. For instance, wearable acoustic systems based on micromechanical arrays^[77] (**Figure 5(a)**) or self-powered auditory components^[69] (**Figure 5(b)**) increasingly rely on integrated transimpedance amplification and deep negative-feedback circuits. These front-end architectures are essential to optimize signal conversion, broaden bandwidth, and minimize raw distortion^[113-115]. Furthermore, ensuring a high-quality acoustic interface requires conformal system designs. Epidermal sensing systems^[116] (**Figure 5(c)**) and ultrathin vibration-sensitive electronic skins^[117, 118] (**Figure 5(d)**) achieve excellent acoustic impedance matching with the human body, facilitating multimodal acquisition (e.g., concurrent speech and neck motion) while physically suppressing environmental noise loss at the skin-sensor interface.

Beyond the analog front-end, the integration of digital modules has enabled autonomous, wireless diagnostic platforms. Soft, wearable stethoscopes now routinely integrate signal processing electronics to support real-time disease diagnosis directly on the edge^[111] (**Figure 5(e)**). Similarly, Wang introduced a wireless acoustic patch featuring a microbattery, Bluetooth module, and a bidirectional microphone configuration^[119] (**Figure 5(f)**). This architectural design employs system-level signal separation algorithms to isolate internal bowel acoustics from ambient interference, pushing the boundary of unobtrusive monitoring.

At a higher level of complexity, multidevice networks and active noise-suppression architectures are redefining physiological acquisition. Prof. Rogers' group developed an epidermal acoustic interface system^[48] enabled by a multidevice time-synchronization network (**Figure 5(g)**). It utilizes dual-microphone adaptive filtering to achieve real-time, synchronized spatiotemporal mapping of cardiopulmonary dynamics across multiple anatomical sites. Additionally, addressing motion artifacts at the system level is crucial; Prof. Chen's

group designed a self-filtering acoustic system utilizing a permanent fluidic magnet network ^[112, 120] (**Figure 5(h)**). By leveraging acoustic-magnetic coupling, the system inherently suppresses low-frequency motion artifacts below 30 Hz, transforming acoustic acquisition into a robust platform capable of functioning in chaotic real-world environments.

Despite recent advancements, human physiological signal acquisition still faces challenges. Persistent issues such as environmental noise interference, interindividual physiological variability, limitations in device portability and user comfort, and concerns over data privacy continue to hinder widespread clinical and translational adoption. Specifically, divergent acquisition strategies across humans, animals, and plants are driven by the specific operational constraints of each domain. In human wearable acoustics, the primary design decision is to combat motion artifacts. Researchers often choose dual-microphone architectures (inward-facing and outward-facing); the trade-off is that while this significantly improves the SNR through active noise cancellation, it doubles the data throughput and power consumption, challenging the battery life of miniaturized patches. In ecological monitoring, the paradigm shifts from single-node fidelity to spatial coverage. Heterogeneous microphone arrays are predominantly chosen to track wildlife. However, the critical trade-off in array design is spatial resolution versus data transmission bottleneck. A dense 64-micron array provides exceptional localization but generates massive, continuous audio files that cannot be transmitted over limited wireless networks in remote habitats, necessitating high-power local storage solutions. For plant ultrasonic acquisition, the necessity to sample at >300 kHz (to capture up to 150 kHz signals) strictly dictates the use of wired, lab-based high-speed data acquisition systems, explaining why autonomous, field-deployable plant acoustic sensors remain remarkably scarce.

Addressing these hurdles requires a holistic approach that integrates innovations in functional material design, system-level integration, and intelligent algorithmic frameworks. Future efforts should aim to enable real-time sensing with high sensitivity and low noise under complex physiological conditions while accommodating personalized needs. Such advances are expected to catalyze transformative progress in medical diagnostics and human-machine interaction.

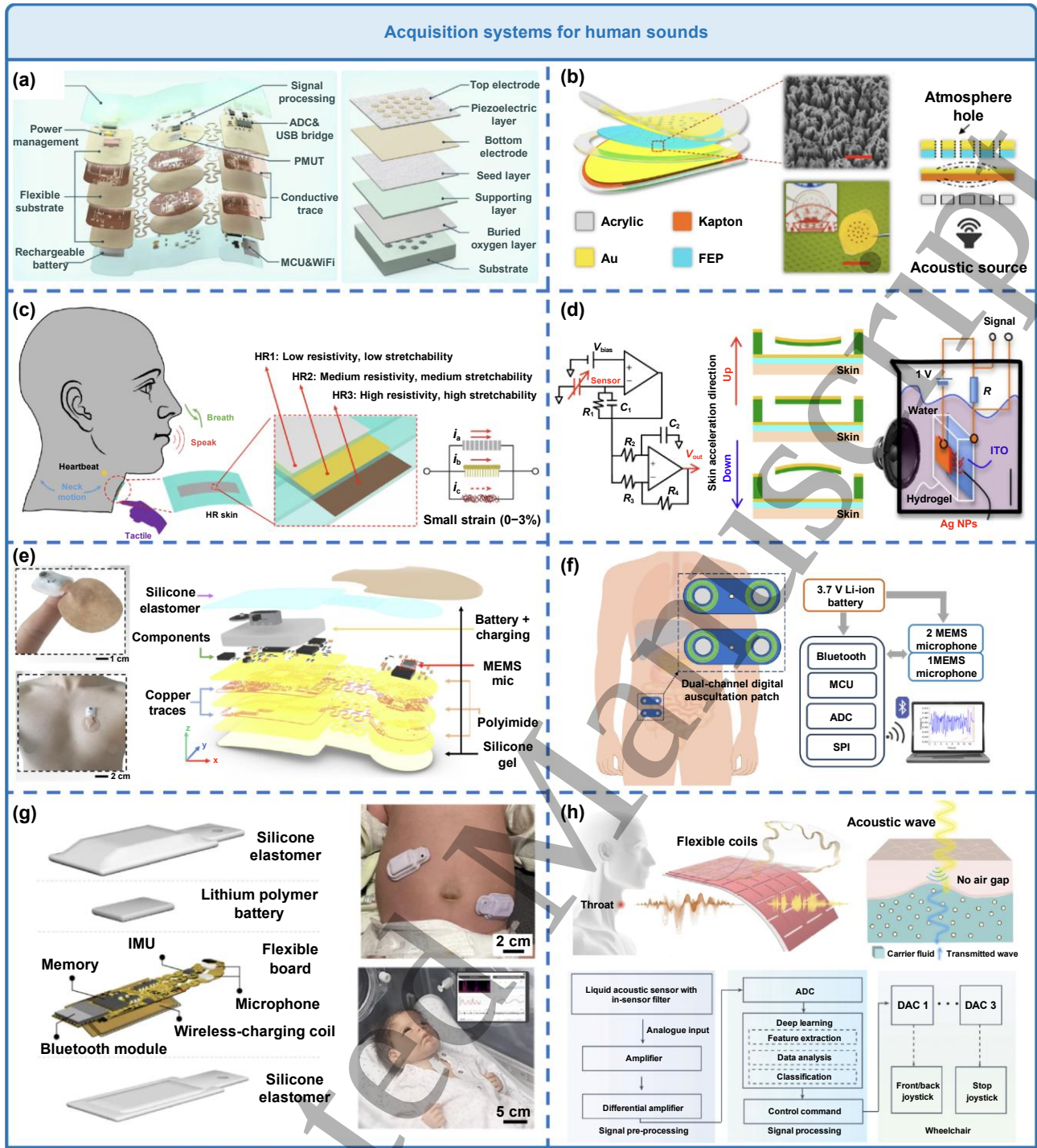


Figure 5. Acquisition systems for human sounds. (a) A wearable, skin-conformal acoustic system based on piezoelectric ultrasonic transducers. Reproduced from^[77], CC BY 4.0. (b) A triboelectric auditory system for human social robotics and hearing aids. From^[69]. Reprinted with permission from AAAS. (c) A hierarchical resistive epidermal sensing system for human physiological and bodily activities. Reproduced from^[116], with permission from Springer Nature. (d) Capacitive acoustic acquisition system for vibration sensing and speech detection. Reproduced from^[118], CC BY 4.0. Reproduced from^[117], CC BY 4.0. (e) A soft, wearable stethoscope for health monitoring and clinical assessment. Reproduced with permission from^[111]. Copyright © 2022 The Authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original U.S. Government Works. Distributed under a Creative Commons Attribution NonCommercial License 4.0 (CC BY-NC). (f) A wireless acoustic patch for auscultation and evaluation of bowel sounds. (g) An epidermal acoustic interface system for monitoring human physiological sounds.

Reproduced from^[48], with permission from Springer Nature. (h) Acoustic detection system based on acoustic-magnetic coupling for speech recognition. Reproduced from^[112], with permission from Springer Nature. Reproduced from^[120]. CC BY 4.0.

4.1.2 Acquisition systems for animal sound signals

The acquisition of animal acoustic signals has evolved from opportunistic single-species recordings to structured, large-scale ecological surveillance systems. The core of these macroscopic acquisition systems lies in scalable, high-spatial-resolution architectures rather than individual sensor performance. Heterogeneous microphone arrays stand out as a cornerstone for high-spatial-resolution and noninvasive terrestrial monitoring. Verreycken developed a scalable, dense 64-microphone array to localize vocalizing animals in natural habitats (**Figure 6(a)**)^[121]. This system resolved fine details of echolocation beams in foraging pale bats and simultaneously tracked multiple songbird species within a 75-meter radius, offering a robust tool for long-term, large-scale biodiversity conservation. Complementarily, Mennill designed a portable wireless array—comprising four GPS-synchronized stereo recorders with omnidirectional microphones—for field-based multichannel data capture^[122]. Deployed in open and forested environments, it achieved a sound source localization accuracy of (1.87 ± 0.13) m across 25 animal vocalizations, clarifying how microphone spacing, habitat type, and source position influence precision and providing a low-cost solution for spatial behavior studies.

Cross-species auditory research leverages specialized tools to unravel biomechanical mechanisms. Ding used a laser Doppler vibrometer (LDV) to quantify stapes vibration changes in live guinea pigs with endolymphatic hydrops (**Figure 6(b)**)^[123], revealing the biomechanical basis of Meniere's disease-related hearing loss and establishing a reference for cross-species auditory studies. For insect monitoring, Turov employed distributed acoustic sensors (DAS) with fiber-optic arrays to capture signals from Madagascar hissing cockroaches, achieving centimeter-scale localization (**Figure 6(c)**)^[124] and opening new avenues for insect behavioral research. Frommolt further integrated long-term monitoring with automation: a 5-year wetland study used four quad-channel recorders with cardioid microphones to track Eurasian bittern nocturnal calls^[125]. A 140 Hz sine wave template enabled automatic call detection, while time-difference-of-arrival (TDOA) algorithms and microphone directionality facilitated individual spatial tracking, forming an integrated framework for cryptic bird population monitoring.

Animals perceive a broader acoustic spectrum (infrasound <20 Hz to ultrasound >20 kHz) than humans, but capturing these signals has long been constrained by issues such as cost, power consumption, and portability. Okamoto proposed a compact, energy-efficient 4-channel microphone array (ChirpArray) designed specifically for long-term outdoor recording, which offers significant advantages in ecoacoustic monitoring (**Figure 6(d)**)^[126]. Its ultralow power consumption enables efficient and flexible deployment, making it an ideal solution for expanding field monitoring and large-scale landscape recording. Stoeger combined acoustic cameras with linear predictive coding (LPC) and machine learning (ML) to capture African elephant vocalizations during separation and reunion (**Figure 6(e)**)^[127]. Frame-by-frame analysis distinguished “trunk calls” from “oral calls,” with >97% classification accuracy, providing a multiscale paradigm for studying low-frequency acoustic biomechanics.

Marine bioacoustic detection is more complex due to water's variable sound-propagation properties, driving the integration of sonar and flexible electronics. Kinck used a hydrophone-equipped autonomous underwater glider (Seaglider) for near-real-time monitoring of deep-sea species such as beaked whales (**Figure 6(f)**)^[128]. Month-long deployments integrated salinity/temperature data to map cetacean distributions,

1
2 although bandwidth limits and ocean reverberation remain unresolved.

3 Recent advances have expanded marine capabilities: Song combined CT scanning and ultrasonic
4 measurements to reconstruct sound velocity in Indo-Pacific humpback dolphin forehead tissues (**Figure 6(g),**
5 **upper**)^[129], suggesting that dolphins optimize sound focusing via thermoregulation (37 °C body temperature),
6 informing biomimetic temperature-compensated acoustic devices. Duarte reviewed anthropogenic impacts on
7 marine soundscapes (**Figure 6(g), lower**)^[3], highlighting vessel/industrial noise disruption of cetacean/fish
8 communication and positioning acoustic monitoring as key for ecological health tracking. Barroso applied
9 ML to classify fish vocalizations^[130], enabling large-scale marine surveys. Tian developed a self-powered, all-
10 optical underwater ultrasonic sensor (SAWS) via triboelectric-induced electroluminescence^[131], pioneering
11 microscale acoustic visualization. For deep-sea benthic monitoring, Masmitja deployed a mobile robotic
12 platform with acoustic receivers (**Figure 6(h)**)^[132], using long-baseline (LBL) positioning and TDOA
13 algorithms to track tagged Norway lobsters (400 m depth, 3 months). Clock drift calibration via static
14 receivers/dynamic robots achieved meter-scale precision, offering a scalable solution for deep-sea acoustic
15 acquisition.

16 When evaluating these ecological sensing strategies, distinct trade-offs emerge between static passive
17 acoustic monitoring (PAM) and mobile autonomous platforms. While PAM nodes (e.g., ChirpArray) offer
18 low-power, long-term temporal continuity, they are limited by a fixed “listening radius” and struggle with
19 spatial blind spots in complex terrains^[126]. In contrast, mobile platforms such as Seaglidors provide spatial
20 flexibility but introduce significant self-noise from propulsion systems and flow, necessitating advanced blind-
21 source separation algorithms. Furthermore, the validation of these systems remains a major bottleneck. Unlike
22 clinical diagnostics that use ECG as a gold standard, ecological validation relies on expert manual annotation
23 of spectrograms or visual confirmation via camera traps, both of which are labor intensive and prone to
24 observer bias. Future systems must move toward “multimodal ground-truth” fusion, integrating acoustic data
25 with environmental DNA (eDNA) or GPS-tagged biomechanical sensors to cross-validate species presence
26 and behavioral interpretation.

27 In conclusion, animal acoustic acquisition technologies now span terrestrial and marine domains:
28 heterogeneous arrays enable precise terrestrial localization, LDV/DAS unravel auditory
29 mechanisms/microbehaviors, and low-cost infrasound/acoustic cameras overcome frequency detection
30 barriers. Marine innovations (gliders, robots, all-optical sensors) extend deep-sea monitoring. Future progress
31 requires addressing environmental interference and cross-scale coordination; low-cost, intelligent solutions
32 will drive integrated “terrestrial-marine” networks, supporting global ecological conservation.

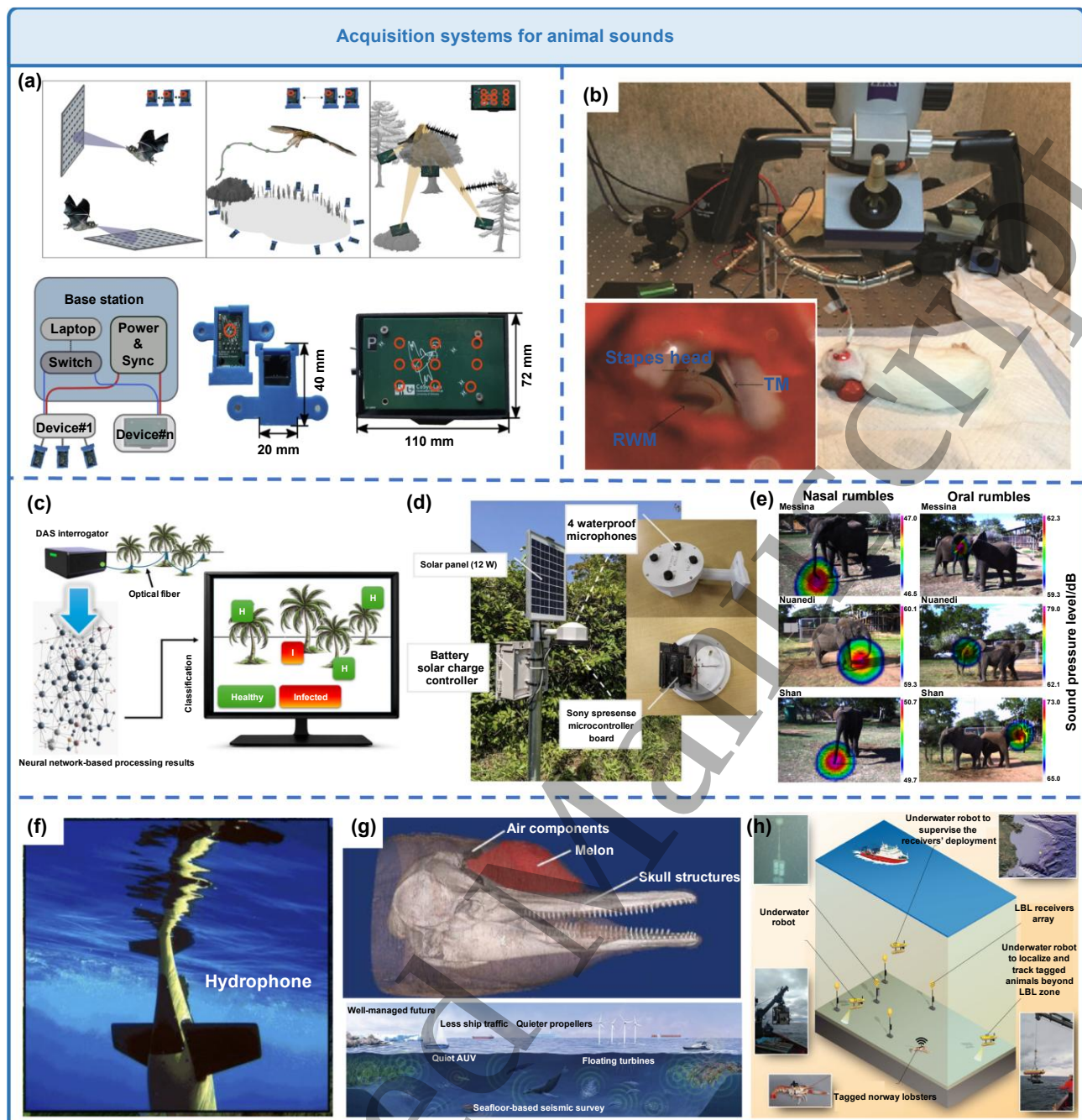


Figure 6. Acquisition systems for animal sounds. (a) A heterogeneous, scalable, dense microphone array to localize and track vocalizing animals in their natural habitats. Reproduced from^[121], CC BY 4.0. (b) Measurement of the impact of endolymphatic hydrops on sound transmission in live guinea pigs using a laser Doppler vibrometer. Reproduced from^[123], CC BY 4.0. (c) Recording sounds emitted by cockroaches via a distributed acoustic sensor. Reprinted from^[124], CC BY 4.0. (d) An affordable, low-cost microphone array (ChirpArray) for long-term ecoacoustic monitoring. Reproduced with permission from^[126], CC BY-NC 4.0. (e) Recording African elephant sounds with acoustic cameras. Reproduced from^[127], CC BY 4.0. (f) Near-real-time acoustic monitoring of deep-sea species with Seaglider™. Reproduced from^[128], CC BY 4.0. (g) Computed tomography scanning with ultrasonic measurements to reconstruct the sound velocity distribution in Indo-Pacific humpback dolphins (upper image). Reprinted from^[129], with the permission of AIP Publishing. Anthropogenic impacts on the marine soundscape (below image). From^[3]. Reprinted with permission from AAAS. (h) An acoustic-based mobile robotic platform for deep ocean benthic fishery track. From^[132]. Reprinted with permission from AAAS.

4.1.3 Plant acoustics and health monitoring systems

Plant acoustics, a pivotal frontier in bioacoustics, is unraveling hidden acoustic communication networks. Technologically, modern plant bioacoustic acquisition is increasingly characterized by the development of AIoT (artificial intelligence of things) platforms, which require high-speed data acquisition architectures capable of capturing ultrasonic emissions (>20 kHz). Understanding plant acoustic behavior begins with perception mechanisms. Demey elucidated that plants sense environmental acoustic signals (e.g., insect wingbeat frequencies) via mechanically sensitive membrane ion channels (**Figure 7(a)**)^[133]. This perception modulates physiological processes (e.g., stomatal movement) and plant-insect ecological networks, laying a molecular foundation for functional studies of plant acoustic signals. Toyota revealed that glutamate-triggered, calcium-dependent long-distance defense pathways are enhanced by stimulus-induced ion channels, amplifying defense gene expression (**Figure 7(b)**)^[134]. This advances defense research from signal description to mechanistic analysis, hinting at uncharacterized links between acoustic vibrations and plant defense. Mishra further synthesized plant acoustic responses, showing that acoustic vibrations alter membrane protein structures, microfilament arrangement, and calcium signaling, initiating molecular cascades that regulate gene expression and growth^[135]. Their proposed signal transduction model (**Figure 7(c)**) provides a theoretical framework for acoustic signal acquisition and plant health monitoring.

Stress-induced plant acoustics is a core focus. Khait used a microphone array to capture 20–150 kHz ultrasonic signals from drought-stressed tomato plants, with spectral traits correlating with water deficit (**Figure 7(d)**)^[136], establishing a noncontact monitoring paradigm. Waqas, Khait, and colleagues further found that plants emit species-, stress type-, and severity-specific ultrasonic signals under drought, wounding, or viral infection (**Figure 7(e)**)^[40]; these signals, potentially perceivable by animals, hold promise for precision agriculture^[137]. Technologically, Bonisoli developed a low-cost ultrasonic microphone system (20–100 kHz, covering key plant emission bands)^[138], leveraging open-source software and a single microphone to boost practical applicability. Nardini reviewed over a century of plant acoustics research, noting that sounds primarily originate from physical processes such as xylem cavitation (**Figure 7(f)**)^[139], but direct evidence for plant-animal/plant–plant acoustic communication remains speculative. Research has extended to marine ecosystems: Ostrovsky used high-frequency echosounding to monitor cyanobacterial blooms, analyzing echo differences between intact and collapsed *Microcystis* gas vesicles (**Figure 7(g)**)^[140]. Gas vesicles enhance acoustic scattering, enabling real-time spatiotemporal tracking of *Microcystis*—supporting cyanobacterial bloom prediction and aquatic ecosystem protection.

Current plant acoustics have a framework spanning molecular perception to ecological applications but face hurdles: signal complexity (environmental noise, multisource coupling) and unclear cross-species communication mechanisms. Future work should integrate genomics, acoustic engineering, and AI, build stress-specific standardized acoustic databases, and translate laboratory findings to precision agriculture/ecological conservation, deepening plant biology understanding and offering transformative solutions for global agricultural and ecological challenges.

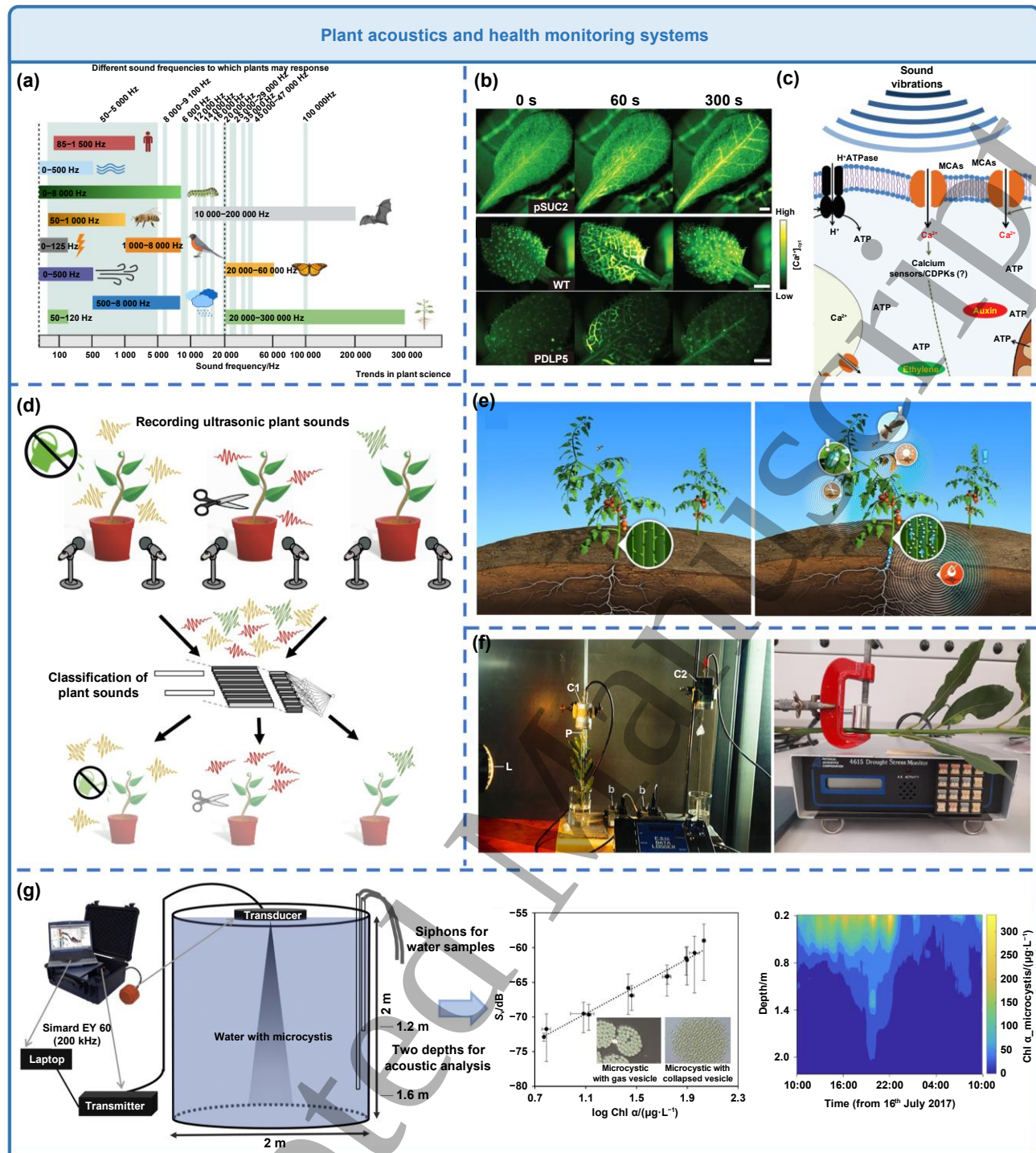


Figure 7. Plant acoustics and health monitoring systems. (a) Overview of plant sensing of sound frequencies. Reproduced with permission from^[133]. CC BY-NC-ND 4.0. (b) Glutamate initiates calcium dependent, long-range signaling for plant defense. From^[134]. Reprinted with permission from AAAS. (c) Mechanisms of acoustic signaling in plants. (d) Plants emit acoustic signals under stress and (e) may be detected by other organisms. Reprinted from^[135], Copyright (2016), with permission from Oxford University Press. Reproduced from^[136], CC BY 4.0. Reprinted from^[40], Copyright (2023), with permission from Elsevier. (f) The original setup for measuring plant acoustic emissions. Reproduced from^[139], CC BY 4.0. (g) Monitoring of the cyanobacterium Microcystis by a high-frequency echosounder. Reprinted from^[140], Copyright (2020), with permission from Elsevier.

The selection of an appropriate acoustic sensor is paramount in bioacoustic research, dictated by the

unique interplay between the target organism, the physical characteristics of the sound, and the specific application environment. For human health monitoring, the field is dominated by a trend toward multimodal, flexible, and wearable electronics. **As illustrated in Table 1**, sensors based on piezoelectric, triboelectric, and capacitive principles are at the forefront, engineered for high sensitivity, low noise, and conformal skin contact to capture faint physiological acoustics such as heart and lung sounds. These devices are increasingly integrated with MEMS technology for miniaturization and system-on-chip capabilities, with emerging paradigms such as magnetoelastic sensing offering noncontact alternatives that mitigate motion artifacts.

In contrast, animal bioacoustics demands a vastly different technological approach, emphasizing scalability, environmental resilience, and often spatial resolution. The arsenal here ranges from sophisticated heterogeneous microphone arrays that enable precise localization and tracking of vocalizing species such as bats and birds in their natural habitats to specialized tools such as laser Doppler vibrometers for detailed biomechanical studies. For aquatic environments, hydrophones deployed on autonomous gliders or static moorings are indispensable. Recent innovations include distributed acoustic sensing (DAS), repurposing fiber-optic cables for large-scale seismic and acoustic monitoring, which shows promise for detecting insect and other animal activities over long distances. Furthermore, the development of low-cost infrasound detector networks demonstrates a push toward affordable, large-scale monitoring solutions for species such as elephants.

Plant acoustic monitoring, a nascent but rapidly advancing field, primarily relies on ultrasonic microphones capable of detecting stress-induced emissions in the 20-150 kHz range. This noncontact approach offers a potential paradigm shift for precision agriculture by enabling early diagnosis of drought or pathogen attack. For aquatic flora, high-frequency echosounders provide an active acoustic method to map and monitor populations such as cyanobacteria blooms.

Beyond hardware innovations, a critical analysis of plant sensing strategies reveals that signal acquisition is fundamentally governed by the trade-off between contact-based and noncontact methods. Contact-based piezoelectric sensors^[141, 142] offer superior sensitivity to internal xylem cavitation events but suffer from significant signal attenuation at the bark-sensor interface, often requiring invasive debarking that alters the plant's physiological state. Noncontact ultrasonic microphones avoid tissue damage and enable high-throughput monitoring but are highly susceptible to ambient wind and machinery noise, demanding complex beamforming for signal isolation. Crucially, the validation of plant acoustic markers requires rigorous correlation with physiological ground truths, such as leaf water potential, stomatal conductance, and xylem tension measured via pressure chambers. Currently, the lack of standardized, synchronized acquisition protocols that link high-frequency ultrasonic emissions with these slow-varying physiological parameters prevents the establishment of a robust "acoustic-physiological" diagnostic framework for precision agriculture.

In conclusion, while the core physical principles of acoustics unite these subfields, sensor technologies have divergently evolved to meet specific biological and environmental constraints. The future trajectory points toward the fusion of these technologies into intelligent, networked systems. The integration of advanced materials for improved sensitivity, machine learning for automated signal analysis, and the development of low-power, ubiquitous sensing nodes will be crucial in building a comprehensive understanding of the biosphere through its sounds, from the human body to the depths of the ocean and the vastness of forests.

4.2 Processing and analysis of acoustics signal

Efficient processing and analysis of acoustic signals serve as a critical enabler for the application of bioacoustic technologies. During the preprocessing and feature recognition stages, the time-frequency

characteristics of acoustic signals are extracted through filtering, denoising, and time-frequency transformations, which are subsequently coupled with artificial intelligence algorithms to achieve precise identification of pathological and behavioral patterns. Multimodal acoustic processing integrates cross-physical-field data such as electromyographic and mechanical signals, enabling the interpretation of multidimensional coupling mechanisms in biological systems via joint modeling. Furthermore, acoustic camera technology, leveraging microphone arrays and beamforming, facilitates sound field visualization and enables sound source localization and dynamic monitoring, demonstrating unique advantages in behavioral studies and noise source identification.

4.2.1 Acoustic signal preprocessing, feature extraction and recognition

Preprocessing, feature extraction, and classification of bioacoustic signals are critical steps in deciphering such complex information. Early studies widely employed mel-frequency cepstral coefficients (MFCC) and its variants; for instance, Abdusalomov enhanced the efficiency of MFCC extraction through parallel computing, achieving 98.4% accuracy in speech recognition, although their adaptability to high-frequency nonstationary signals remains limited (**Figure 8(a)**^[143]). Li further introduced first- and second-order delta coefficients combined with a ResNet model, strengthening the representation capacity for both static and dynamic features (**Figure 8(b)**)^[144]. In the domain of speech recognition, Yang utilized a wearable artificial throat device and performed classification of spectrograms via transfer learning with AlexNet^[145], attaining 91% recognition accuracy in laryngectomy patients (**Figure 8(c)**). This hybrid-modal architecture demonstrated strong robustness under noisy conditions and outperformed conventional microphone-based approaches.

Convolutional neural networks (CNNs) have become a pivotal tool for the image-based processing of acoustic signals, owing to their exceptional capability in extracting localized features. In the context of underwater target recognition, Ke proposed a one-dimensional convolutional autoencoder-decoder architecture (**Figure 8(d)**)^[146]. This model employs depth wise separable convolutions and a hierarchical pretraining strategy to extract features of high resonance components in an unsupervised manner. Subsequently, when integrated with a supervised feature separation algorithm, it significantly enhances recognition performance.

The field of engineering and intelligent sensing has witnessed the considerable cross-disciplinary potential of multimodal acoustic signal processing. For instance, Ma Meng introduced coupled and deep-coupled autoencoders to effectively fuse vibrational and acoustic signals (**Figure 8(e)**)^[147]. Through frequency-domain transformation and high-level feature learning, their approach achieved a classification accuracy of 94.3% in gear fault diagnosis, demonstrating the advantage of multimodal coupled modeling for improving diagnostic precision.

Another notable contribution comes from the team of KunHyuck Lee, who developed an IMU-based mechano-acoustic wireless sensing device (**Figure 8(f)**)^[148]. Deployed on the sternoclavicular joint, this device synchronously captures diverse physiological signals, including respiration, heartbeats, and swallowing. By combining frequency-domain analysis, waveform transformation, and a hidden Markov model, the system enables the extraction of multiple physiological parameters and the classification of sleep stages. This research highlights the promising application of lightweight devices for unobtrusive and continuous monitoring.

Beyond traditional CNNs and RNNs, the landscape of bioacoustic signal processing is currently being reshaped by the advent of transformers, self-supervised learning (SSL), and contrastive learning^[149, 150].

Traditional supervised deep learning models are fundamentally constrained by the scarcity of high-quality, annotated bioacoustic datasets. To circumvent this bottleneck, SSL paradigms, such as Wav2Vec 2.0 and HuBERT, have been recently adapted to bioacoustics. By pretraining on massive amounts of unlabeled physiological or ecological audio, these models autonomously learn universal, high-dimensional acoustic representations. These robust base representations can subsequently be fine-tuned for specific downstream tasks (e.g., detecting subtle pathological heart murmurs or identifying rare bird calls) using only minimal labeled data, drastically reducing annotation costs.

Furthermore, Transformer architectures, particularly the audio spectrogram transformer (AST), have demonstrated superior analytical performance. Unlike CNNs, which rely on inductive biases and localized receptive fields, transformers leverage self-attention mechanisms to capture long-range temporal and spectral dependencies globally across the entire acoustic sequence. This global context is crucial for analyzing prolonged physiological events, such as continuous bowel peristalsis or extended whale songs. Additionally, in highly complex environments characterized by heavy background noise and overlapping cross-species vocalizations, contrastive learning approaches offer powerful solutions. By pulling representations of similar acoustic events closer together while pushing dissimilar ones apart in a joint latent space, contrastive learning significantly enhances the discriminability of weak biological signals against chaotic noise. This strategic shift from localized supervised training to global, foundation-model-based paradigms represents the most critical trajectory for overcoming the intrinsic data scarcity and noise interference bottlenecks in modern bioacoustics.

4.2.2 Critical evaluation of artificial intelligence bottlenecks

While the above summarizes various algorithms, a critical evaluation reveals several persistent bottlenecks that hinder the translation of deep learning from laboratory benchmarks to real-world bioacoustic deployments.

(1) Data dependency and the labeling crisis: despite the promise of self-supervised learning (SSL), most state-of-the-art bioacoustic models remain heavily dependent on high-quality, annotated datasets. Unlike general audio, bioacoustic labeling (e.g., identifying specific heart murmurs or rare avian vocalizations) requires highly specialized expertise from clinicians or ecologists, making large-scale supervision prohibitively expensive and time-consuming. This data dependency creates a “scarcity bottleneck” that often leads to overfitting on small, biased cohorts.

(2) Robustness to nonstationary noise: although contrastive learning improves feature discriminability, the robustness of current models to nonstationary, unpredictable noise (e.g., overlapping conversations in a NICU or machinery noise in precision agriculture) remains insufficient. Most CNN-based models suffer from significant performance degradation when the SNR drops below specific thresholds, as they struggle to distinguish between stochastic environmental interference and weak biological targets.

(3) Generalization and domain shift: a fundamental limitation is the lack of generalization across different hardware and environments. A model trained on high-fidelity electronic stethoscopes often fails when deployed on low-cost flexible sensors due to a “domain shift” in acoustic impedance and frequency response. Achieving seamless cross-dataset generalization requires advanced domain adaptation techniques that are still in their nascent stages in bioacoustics.

(4) The interpretability gap: in clinical and HMI applications, the “black box” nature of deep neural networks (DNNs) poses a major barrier to trust. While a transformer may achieve high accuracy, its inability to provide a physiologically grounded explanation for its decision (e.g., why it flagged a specific S3 gallop) limits its clinical utility. Future research must prioritize explainable AI (XAI) to bridge the gap between high-dimensional latent representations and human-understandable biological mechanisms.

As summarized in Table 2, the selection between conventional signal processing techniques (e.g., MFCC and support vector machine (SVM)) and deep learning methods depends primarily on these practical deployment conditions and trade-offs.

In summary, the field of acoustic signal processing is progressively shifting from traditional filtering and manual feature design toward end-to-end deep learning and self-supervised representation. As summarized in Table 2, a variety of deep learning algorithms are comprehensively compared. The selection between conventional signal processing techniques (e.g., MFCC-SVM) and deep learning methods (e.g., CNNs and transformer models) depends primarily on practical deployment conditions rather than the unilateral pursuit of classification accuracy. Deep learning models consistently achieve superior classification accuracies because they can autonomously extract high-dimensional features from raw spectrograms, overcoming the manual bias of traditional feature engineering. However, the design trade-off involves computational latency, power consumption, and interpretability. For wearable medical devices functioning on edge-computing MCUs, lightweight traditional algorithms (or shallow neural networks) are often chosen over deep transformers to conserve battery life and ensure real-time alerts. Conversely, in marine or agricultural bioacoustics where data are postprocessed offline on cloud servers, the massive computational cost of deep learning is a justifiable trade-off for its robustness in handling extreme environmental noise and overlapping cross-species signals.

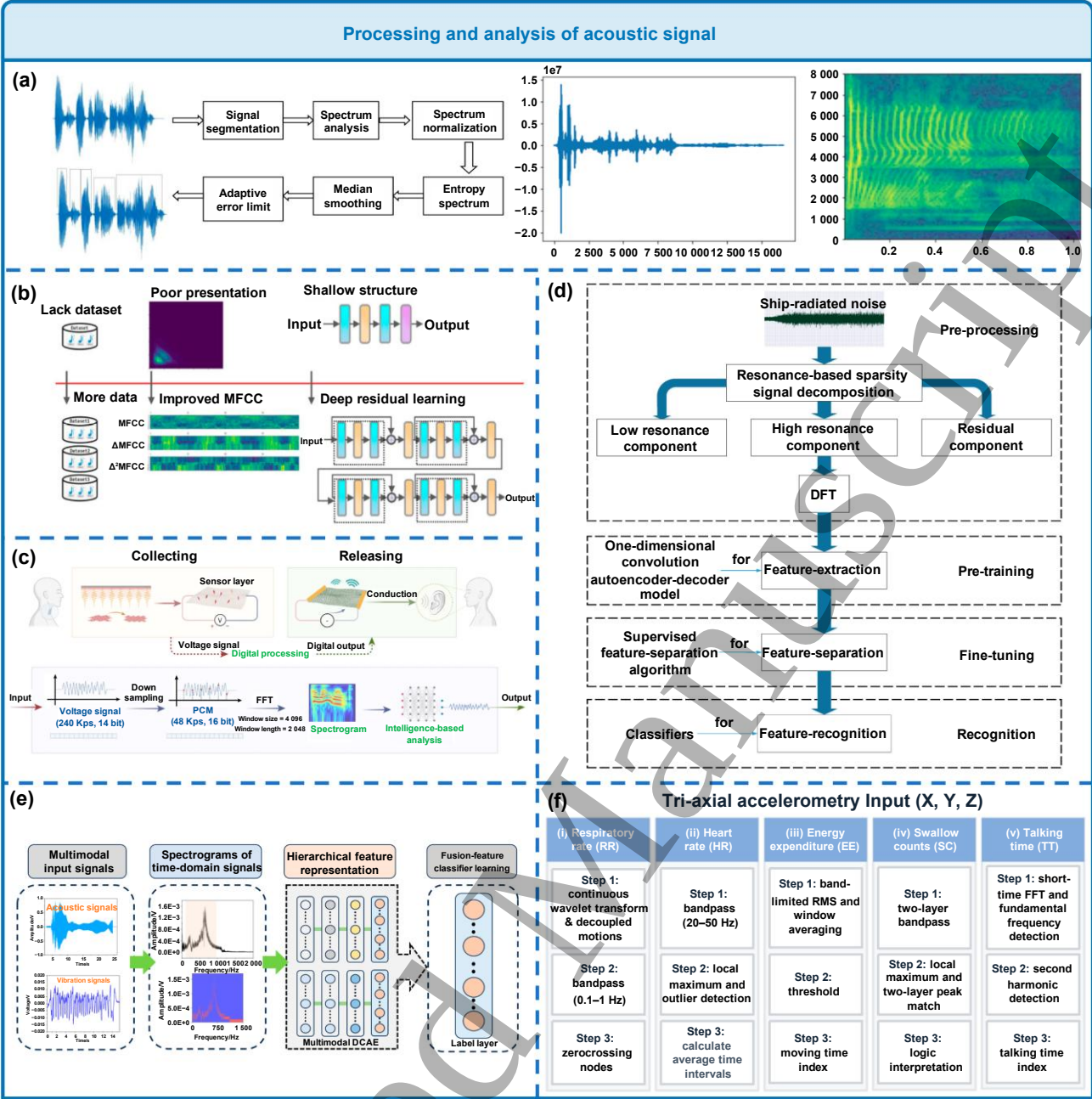


Figure 8. Processing and analysis of acoustic signal. (a) An ML-based rapid extraction method for MFCC speech feature parameters. Reproduced from^[143], CC BY 4.0. (b) Heart sound classification using enhanced MFCC and deep ResNet. Reproduced from^[144], CC BY 4.0. (c) Wearable artificial throats for mixed-modal speech recognition and interaction. Reproduced from^[145], with permission from Springer Nature. (d) Underwater acoustic target recognition using a supervised feature discrimination approach. Reproduced from^[146], CC BY 4.0. (e) Deeply integrated autoencoder for fault detection via multimodal sensor information. (f) Multimodal signal (breathing, heartbeat, swallowing, etc.) analysis based on frequency domain analysis and the ML model.

Table 2. Overview of typical methods for acoustic signal processing, feature extraction and classification.

Domain	Method category	Representative techniques	Key advantages	Major limitations
Acoustic signal processing	Traditional physical methods [151, 152]	Wiener filtering, wavelet transform, spectral subtraction, nonnegative matrix factorization (NMF), etc.	Low computational complexity, high interpretability	Performance degrades in nonstationary noise
	Deep learning-based methods [146, 153]	DNN-based mask learning, CNN-RNN hybrids, GAN transformer, etc.	Robustness to nonstationary noise, end-to-end optimization	High annotation dependency, limited generalization
Acoustic feature extraction	Traditional hand-crafted features [143, 154]	MFCC, filter bank, LPC, teager energy operator, etc.	Computational efficiency, clear physical interpretation	Manual design, low discriminability in complex scenes
	Deep neural representations [144-146]	Latent embeddings (via autoencoders), self-supervised features (Wav2Vec), etc.	High-dimensional discriminative features, multiscene adaptability	Weak interpretability, high training costs
Acoustic classification	Statistical models [154]	Gaussian mixture model-hidden markov model (GMM-HMM), SVM, random forests, etc.	Effective for small datasets, lightweight	Feature engineering dependency, poor nonlinear modeling
	Deep learning-based classifiers [145, 155]	CNN-RNN, transformer-encoder, self-attention, etc.	End-to-end classification, long-sequence modeling	Overfitting risk, requires regularization

4.2.3 Spatial visualization via acoustic cameras

While the aforementioned preprocessing and feature extraction algorithms (Section 4.2.1) focus on the temporal and spectral dimensions of single-channel signals, the acoustic camera represents the system-level implementation of these algorithms for spatial visualization. As a multidisciplinary innovation integrating acoustic sensing, signal processing, and image analysis, the acoustic camera enables the transformation of sound signals from the time domain to a visual spatial domain through microphone array acquisition, beamforming sound source localization, and color-encoded acoustic image overlay. This technology overcomes the limitations of traditional acoustic measurements by mapping sound energy distribution into

1 image data, remarkably enhancing the spatial resolution and analysis efficiency of sound source localization.

2 In terrestrial animal behavior research, acoustic cameras demonstrate unique value. The ultrasonic
3 vocalization camera system (USVCAM) system developed by Matsumoto constructs an ultrasonic
4 vocalization localization model based on the phase-lag principle^[156], combined with an improved overlapping
5 vocalization segmentation algorithm, thereby reducing the localization error of mouse ultrasonic vocalizations
6 (**Figure 9(a)**). This system successfully resolves the dynamic social vocalizations of ICR mice, providing a
7 powerful tool for researching the mechanisms of social behavior.

8 In marine exploration, where traditional optical imaging is often constrained by turbid water and light
9 attenuation, acoustic cameras serve as an ideal alternative by leveraging the favorable propagation
10 characteristics of sound in water. Zhou proposed a convolutional neural network (CNN)-based superresolution
11 reconstruction algorithm for acoustic images, substantially improving underwater target recognition
12 capabilities and laying a technical foundation for applications such as marine debris identification and fish
13 monitoring^[157] (**Figure 9(b)**). Furthermore, researchers, including Garcia and Connolly, have utilized acoustic
14 cameras coupled with CNN models to achieve efficient and automated fish detection and underwater
15 biological monitoring^[158, 159] (**Figure 9(c) and (d)**).

16 Furthermore, the applications of acoustic cameras have extended to the research of marine mammals,
17 aquatic plants, and other aquatic animals. From behavioral surveys of harbor seals and biomass counting in
18 deep scattering layers to tracking the distribution of marine mammals using passive acoustic monitoring
19 techniques and to Rogissart's real-time observations of winter aggregation behaviors in *Hemimysis anomala*
20 (**Figure 9(e)**)^[160], acoustic cameras have provided new perspectives for acquiring high temporal-resolution data
21 and deepening the understanding of biological behaviors.

22 In specialized and extreme environment monitoring, acoustic cameras effectively circumvent optical
23 interference through their sound-to-image conversion mechanism, demonstrating irreplaceable technical
24 advantages. With the deepening integration of artificial intelligence algorithms, acoustic cameras are evolving
25 from mere localization tools into intelligent monitoring platforms, poised to continuously drive innovative
26 applications in bioacoustics, industrial diagnostics, and ecological conservation.

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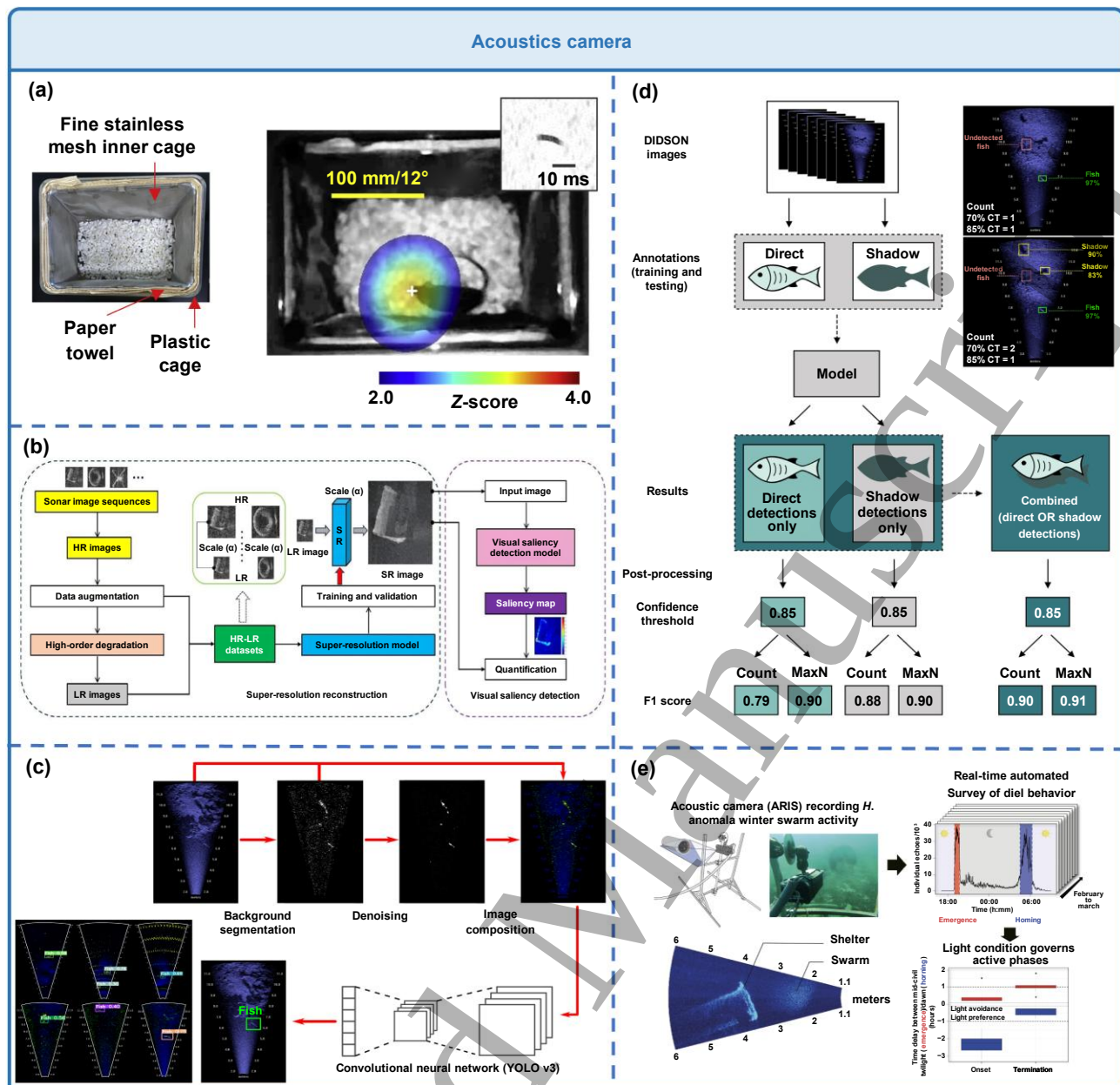


Figure 9. Applications of acoustic cameras in underwater target monitoring and detection. (a) USVCAM monitoring custom inner cage. Reproduced from^[156], CC BY 4.0. (b) The framework of sonar image SR reconstruction for marine debris detection. Reproduced from^[157], CC BY 4.0. (c) Detection with the CNN model. Reproduced from^[158], with permission from Springer Nature. (d) Deep learning models trained on acoustic imagery to detect fish and fish shadows. Reproduced from^[159], CC BY 4.0. (e) Recording an anomala winter swarm with an acoustic camera. Reprinted from^[160], Copyright (2024), with permission from Elsevier.

4.3 Active ultrasonic probing and image analysis

Although ultrasonic signals fall within the bioacoustic frequency spectrum by physical definition, their acquisition and analytical paradigms differ fundamentally from the passive listening techniques discussed in the preceding sections. Bioacoustic monitoring of animal vocalizations, cardiopulmonary sounds, and other physiological events primarily relies on passive listening to signals inherently generated by biological processes. In contrast, ultrasonic sensing typically employs active pulse-echo probing, in which the transducer actively emits high-frequency acoustic waves and detects the reflected echoes, combined with B-mode spatial image reconstruction. This transition from one-dimensional time-series data to multidimensional visual

information necessitates not only distinct hardware transceiver architectures but also analytical models grounded in computer vision. The convergence of ultrasound technology with portable and wearable devices is reshaping medical diagnostics and industrial monitoring. This shift demands a fundamental redesign of ultrasonic systems, moving beyond the capabilities of conventional, bulky platforms such as the Vantage NXT or S-Sharp^[161, 162]. The central challenge lies in developing highly integrated, low-power hardware and intelligent signal processing algorithms to enable high-fidelity imaging and monitoring in resource-constrained environments.

Modern system design prioritizes miniaturization and efficiency across three core modules: power, processing, and transceivers. Power management increasingly relies on DC-DC converters to provide the high-voltage excitation required for ultrasonic transducers. This is coupled with integrated transmitter chips, which often feature built-in multiplexers to drive multielement arrays from a compact form factor, as exemplified by Lin's wearable ultrasonic system based on a 32-channel linear array probe (**Figure 10(a)**)^[163].

Echo reception strategies offer a trade-off between performance and cost, ranging from fully integrated receiver chips to discrete amplifier-ADC combinations. The computational core presents a critical choice between processor architectures: power-efficient Microcontrollers (MCUs) for control tasks, specialized digital signal processors (DSPs) for complex analysis, and highly parallel field-programmable gate arrays (FPGAs) for real-time, customizable processing. As exemplified by the TinyProbe wearable ultrasound system, such heterogeneous architectures integrate compact 32-channel TX/RX front-ends, low-power operation (100-1240 mW), and multimodal imaging capabilities (A-mode, B-mode, Doppler) to enable continuous physiological monitoring and high-frame-rate applications (**Figure 10(b)**)^[101, 104, 164, 165]. This diversity allows for application-optimized design, from continuous physiological monitoring to high-frame-rate imaging.

Processing methodologies are dictated by data modality. For time-series signals (A-mode), the standard pipeline involves time-frequency transformation, feature extraction (e.g., envelope detection, muscle thickness), and subsequent analysis. This can be achieved through lightweight model-based approaches, such as quadratic fitting for joint torque estimation^[166] (**Figure 10(c)**), or more complex deep learning (DL) models. In contrast, image signals (B-mode) are analyzed through the lens of computer vision. Techniques such as image classification and object recognition are directly applied, often yielding powerful results that merge physical imaging principles with data-driven intelligence. For example, transformer models have been successfully deployed for the simultaneous recognition of hand motion and force from sonomyographic images (**Figure 10(d)**)^[167].

Despite rapid progress, significant bottlenecks remain. Key frontiers include data bottlenecks: escalating sensor frequencies and sampling rates create a data deluge, demanding innovative real-time compression and on-edge processing techniques that are currently underdeveloped. Algorithmic robustness: A-mode ultrasound analysis for musculoskeletal applications suffers from motion artifacts, poor cross-subject generalization, and performance degradation under muscle fatigue, necessitating more adaptive and robust models. Calibration of flexible systems: a critical hurdle for conformable ultrasound arrays is calibration on curved surfaces. Current approaches are split between hardware-assisted methods using auxiliary sensors^[58, 168, 169] and purely algorithmic solutions^[170-172]. A promising path forward is the fusion of simple, cost-effective sensors with efficient, generalizable algorithms to create practical calibration frameworks. In conclusion, the future of ultrasound hinges on overcoming these interdisciplinary challenges. Success will unlock the full potential of intelligent, miniaturized systems for decentralized healthcare and advanced industrial inspection.

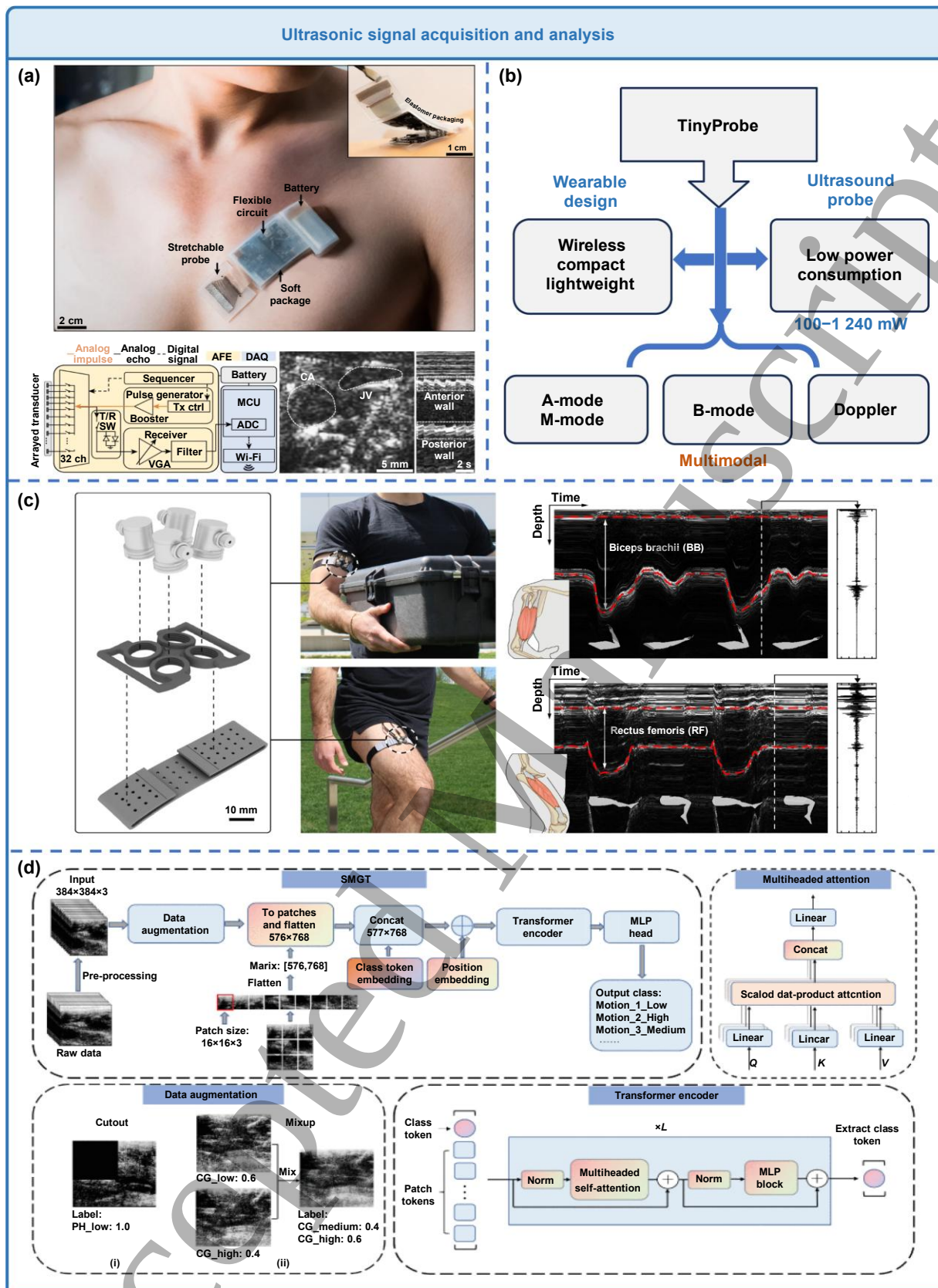


Figure 10. Ultrasonic signal acquisition and analysis. (a) A fully integrated ultrasound system monitoring deep tissues. Reproduced from^[163], with permission from Springer Nature. (b) A compact, lightweight, and low-power wearable multimode ultrasound system (TinyProbe). (c) Estimation of joint torque with the A-mode ultrasound system. Reproduced from^[166], CC BY 4.0. (d) Transformer-based ultrasound signal processing method. Reproduced from^[167], CC BY 4.0.

5. Bioacoustic physical modeling

Effective signal acquisition and analysis depend on a clear understanding of the generation, propagation, and measurement of bioacoustic signals. Bioacoustic signals are generated by tissue vibration and fluid dynamics within living systems. However, the physical inaccessibility of internal organs has consistently posed challenges for conducting *in vivo* investigations of sound generation and propagation mechanisms. Based on anatomical configurations, bioacoustic physical models provide a critical framework for understanding the physics of sound generation and propagation in living organisms. In physiological applications, physical modeling can reveal tissue-dependent attenuation transmission pathways and spectral variations in acoustic signals. In plants, it can also help explain how water transport or structural changes within plants influence the generation and propagation of acoustic signals. More broadly, physical modeling provides a mechanistic basis for sensor design and signal interpretation. Such understanding supports sensor placement, frequency-band selection, and the identification of the physiological sources of abnormal acoustic features.

Despite the remarkable success of data-driven signal processing and deep learning in classifying bioacoustic features, these analytical models often operate as 'black boxes', lacking mechanistic interpretability. Relying solely on algorithmic extraction is insufficient for understanding the physiological or anatomical origins of sounds. To fundamentally comprehend how these complex acoustic features are generated within biological tissues and to theoretically guide the structural optimization of the aforementioned sensors, it is imperative to bridge data-driven analysis with physics-based understanding. Consequently, the subsequent section delves into bioacoustic physical modeling, elucidating the fluid-structure interactions and wave propagation mechanisms intrinsic to living systems.

5.1 Physiological acoustics models

Human voice production is governed by the vibration of vocal folds (VFs) and involves a complex interaction between fluid dynamics, tissue motion and acoustics. VF adduction closes the glottis, thereby allowing airflow expelled from the lungs to create a critical pressure differential across the glottal region. This pressure gradient triggers self-sustained oscillations of the VFs, generating a pulsatile glottal jet. The frequencies of oscillation in the modal (or chest) register in normal voiced speech concentrate on 80-220 Hz, with the acoustic energy dominated by the fundamental phonation frequency (f_0) and its higher harmonics^[173]. There are three primary mechanisms of voice generation: (1) a dipole sound source is produced by net forces that the surfaces of the glottal walls exert on the airflow, which contributes most to the sound radiation. (2) A monopole sound source originates from the VF motion. (3) A quadrupole source is produced by turbulent vortices generated by unsteady glottal jet flow downstream of the VFs, playing a comparatively minor role in sound radiation^[173, 174]. Traditional theory of phonation, known as Fant's linear source-filter approach, assumes that VF vibrations are not influenced by vocal tract acoustics. However, this assumption neglects the nonlinear interaction between the source and the filter^[175]. Building upon this foundation, Ishizaka and Flanagan^[176] modeled VFs as a two coupled spring-mass-damper system. This model provided a simple and effective manner for computing VF dynamics. Titze^[177] studied the nonlinear interaction between the sound source and vocal tract, which explained the effects of the cross-sectional area of the laryngeal vestibule. More recently, reduced-order models (lumped-element models) were used to demonstrate that the evolutionary loss of human VFs contributes to enhanced phonation stability^[178] (**Figure 11(a)**). Computational fluid dynamics (CFD) models are also used to analyze phonatory dynamics. Yang^[179] proposed a full coupling flow-structure interaction (FSI) model for studying the dynamic interaction between compressible fluids and VFs. Seo^[180]

introduced an immersed boundary method (IBM) for solving linearized perturbed compressible equations (LPCE) in complex geometries. The method accurately captures flow-induced noise (**Figure 11(b)**). Link^[181] developed a flow-structure-acoustic interaction (FSAI) model, which provided new results of the impact of the Coanda effect (**Figure 11(c)**). Notably, mammals beyond humans also rely on analogous fluid-tissue interactions for phonation (**Figure 11(d)**)^[182]. Nevertheless, high-precision models face two challenges: excessive computational costs and the need for precise characterization of tissue mechanical properties.

Heart sounds are generated by the closure of heart valves and their flow-induced vibrations throughout the cardiac cycle. These acoustic signals propagate through multilayered viscoelastic tissues and reach the chest surface. Pathological murmurs often arise from turbulent flow, which is caused by valve stenosis, vascular obstructions, or structural defects. Turbulent pressure fluctuations resulting from blood flow through the constriction produce sound and vibration^[183]. Modeling of cardiac hemodynamics faces challenges due to large-scale chamber motion, complex valve dynamics induced by flow, and complex turbulence^[184]. The conventional approach to CFD simulation of vascular flow employs the finite volume method (FVM) and finite element method (FEM)^[185]. The use of lumped parameter network (LPN) models of the heart and the circulatory system as boundary conditions for CFD simulations improves the reliability of hemodynamic simulation in major vessels (e.g., aorta, pulmonary artery)^[186]. More innovatively, a novel hybrid approach combines incompressible Navier–Stokes equations with linearized perturbed compressible equations (LPCE) to simulate flow-induced sound generation^[187, 188]. Modeling heart sound propagation in the human body remains difficult. This is because thoracic tissues are viscoelastic media in which wave propagation involves both compression and shear waves, accompanied by dispersion, refraction, and energy dissipation. Existing models often neglect critical anisotropic heterogeneities that govern elastic wave scattering and mode conversion. Seo^[189] proposed a computational method to simulate the generation and propagation of blood flow-induced sounds. The blood flow is simulated by solving the incompressible Navier–Stokes equations based on IBM. Concurrently, heart murmur propagation is solved through a high-order finite-difference method in the time domain with vascular wall pressure incorporated as the acoustic source term (**Figure 11(e)**). Machine learning shows significant potential in murmur classification^[190, 191]. However, current studies predominantly rely on single auscultation point recordings, limiting their ability to localize pathological acoustic sources. Future progress requires the development of high-precision acoustic models and computationally efficient algorithms.

Lung sounds are produced by air flow through the pulmonary system during respiration. Normal respiratory sounds are classified into tracheobronchial and vesicular sounds. However, under pathological conditions, air and fluid abnormally occupy the lungs, which produces adventitious sounds, including crackles, wheezes, and rhonchi. The frequency range of adventitious sounds can reach 4 000 Hz^[111]. The propagation path of lung sounds involves multiple tissue types, including hard tissue, soft tissue, and the heterogeneous lung parenchyma, which is a mixture of soft tissue and air. Specifically, high-frequency wave components propagate preferentially through the airway, while low-frequency wave components couple more effectively with airway walls^[192]. Nevertheless, the chest acts as a low-pass filter due to the lung parenchyma's strong attenuation of high frequencies^[193]. To better understand lung sound generation and propagation mechanisms, several theoretical models have been proposed. Wodicka^[38] developed an equivalent acoustic circuit model in the frequency range from 100 Hz to 600 Hz, representing the vocal tract, trachea, and first five bronchial generations as electrical components. By simulating the lung parenchyma as a fluid medium with uniform air bubbles, they predicted the magnitude of wall motion. Royston^[194] developed an axisymmetric idealized model by modeling the airway as a single central cylindrical chamber and ignoring airway structural complexity. Ozer^[195] employed the boundary element method (BEM) to simulate wave propagation in porous

media, enabling sound source localization via multisensor arrays in phantom experiments. Dai^[196] integrated CT-based 3D reconstruction with Biot theory to establish a finite element model based on porcine lung geometric structures. Most recently, Royston’s team^[197, 198] proposed a one-dimensional (1D) branching acoustic waveguide approach to simulate sound propagation and airway wall vibrations in the subglottal bronchial tree (**Figure 11(f)**). More in-depth simulation studies of specific pathological conditions are needed to improve the accuracy of lung disease diagnosis.

Compared to cardiovascular sound signals, research on gastrointestinal sounds remains limited, primarily due to the complexity of intestinal contents and digestion dynamics. Bowel sounds are known to originate from intestinal peristalsis^[199]. The generation mechanism of bowel sounds shares similarities with VF vibration, as intestinal contractions compress luminal fluids and gases to generate pressure gradients that induce wall vibrations. Furthermore, distinct gastrointestinal segments and luminal contents have been shown to significantly influence the quantity, energy, and frequency characteristics of bowel sounds^[200], highlighting the need for segment-specific acoustic modeling. Building on this physiological foundation, Du^[201] developed a bowel sound generation model based on a spring-mass-damper system. Specifically, this model quantifies bowel activity through four biomechanical parameters: the individual wave component (IWC), pressure index (PI), component quantity (CQ), and component interval time (CIT) (**Figure 11(g)**). Sinnott^[202] utilized CFD models to quantitatively characterize intestinal fluid-structure interactions, laying a foundation for elucidating bowel sound emission mechanisms. Dimoulas^[203] proposed a multichannel abdominal acoustic localization technique, enabling topographic analysis and mapping of gastrointestinal motility through multipoint auscultation. They employed a physical phantom comprising artificial acoustic sources, six sensors, and layered biomimetic materials to validate their localization results. These computational advancements are complemented by progress in machine learning-driven signal processing. By applying time-frequency domain analyses, researchers have achieved automated classification of bowel sounds^[199, 203]. A comparative summary of the major acoustic characteristics and frequency ranges of these physiological signals is provided in **Table 3**. Such information is valuable for sensor design and for the interpretation of physiological sounds. Nevertheless, in practical physiological monitoring, acoustic information alone is often insufficient to fully characterize complex biological activities. This motivates the development of multimodal signal coupling models.

Multimodal signal coupling models, which integrate acoustic, mechanical, electrophysiological, and other signals, demonstrate unique advantages in complex physiological monitoring. To suppress external interference caused by body movement, Silva^[204] developed a coupled microphone-accelerometer sensor pair. Specifically, this design employs the mechanical impedance mismatch between accelerometer and skin vibration signals to filter out noise. Liu^[205] proposed a mechanoacoustic electrophysiological sensing platform capable of concurrent acquisition of cardiovascular mechanical vibrations, heart sounds, and electrocardiography (ECG). This system improves diagnostic accuracy for cardiovascular pathologies. Lee^[148] employed triaxial accelerometers to track mechano-acoustic signatures and kinematics of body motion, obtaining real-time conventional biomarkers through frequency-domain analysis and machine learning. Furthermore, multimodal monitoring systems integrating physiological acoustics with visual signals, thoracic radiography, and electrophysiological data provide complementary pathological insights^[206, 207]. Recent studies have shown that this framework can overcome the limitations of single-modality signals and provide a more comprehensive understanding of the dynamic interactions within physiological systems.

Table 3. Comparative analysis of human physiological acoustic signals.

Generation	Propagation	Frequency	Primary feature
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range				
Voice	Vocal folds vibration coupled with fluid dynamics	Vocal tract filter	100-4 000 Hz ^[208]	Acoustic energy is dominated by the fundamental phonation frequency (f_0) and its higher harmonics
Lung sounds	Turbulent flow or vortices in the airway	Tracheobronchial tree- chest	100-4 000 Hz ^[111]	Lung sounds are correlated with respiratory cycle
Heart sounds	valve closure and turbulent flow	Chest (soft tissue, hard tissue)	20-650 Hz ^[67]	The first heart sound (S1) and the second heart sound (S2) alternate rhythmically
Bowel sounds	Peristaltic transport of gas–liquid mixtures	Propagation through abdominal wall fat and skin layers	100-1 000 Hz ^[201]	Frequency varies with digestive cycle

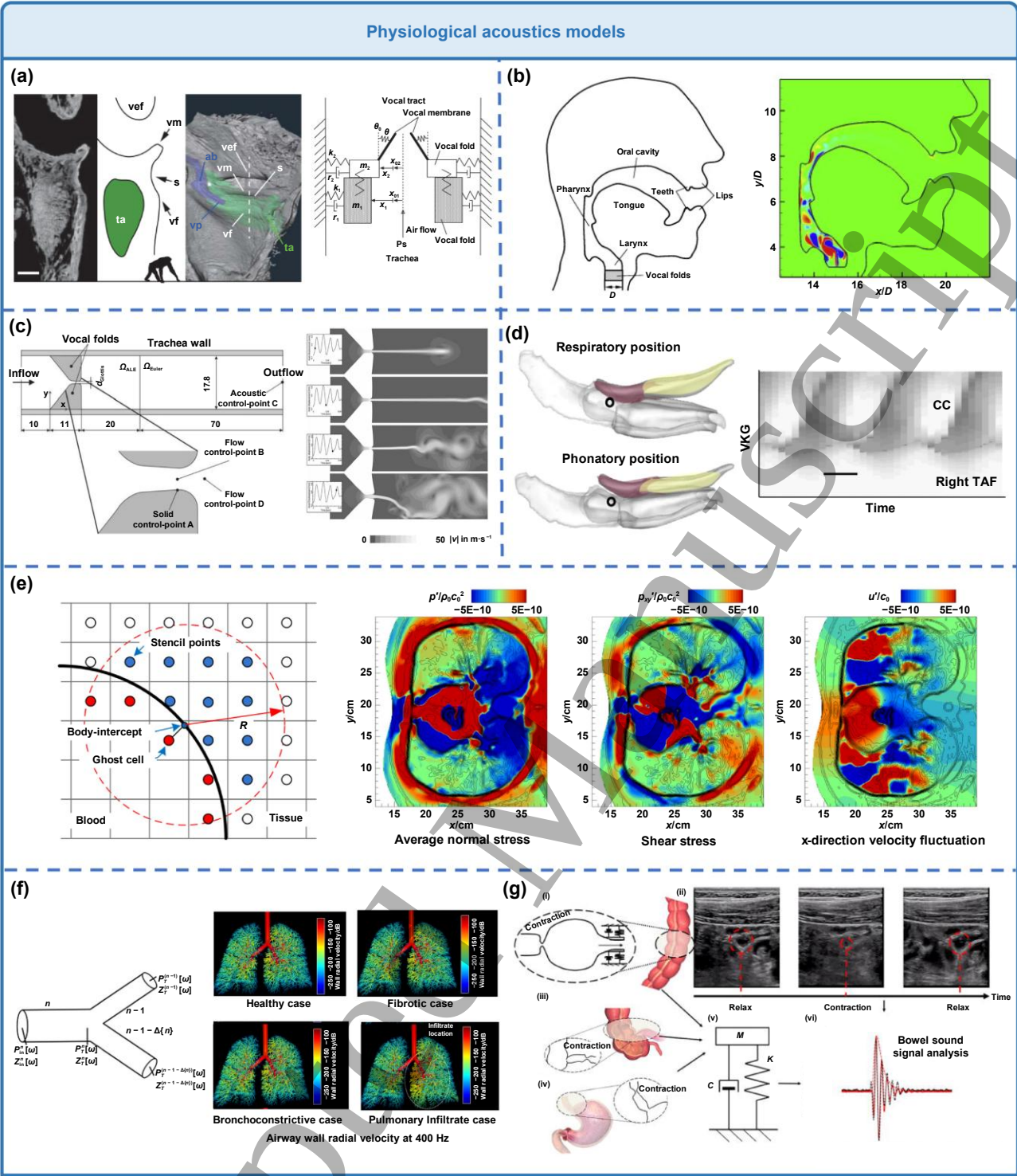


Figure 11. Physiological acoustics models. (a) 3D vocal tract reconstruction from MRI of chimpanzee (left) and schematic illustration of the nonhuman model with a vocal membrane (right). From^[178]. Reprinted with permission from AAAS. (b) Sagittal vocal tract morphology of humans and instantaneous vorticity field of pulsatile jet flow. Reprinted from^[180], Copyright (2011), with permission from Elsevier. (c) Fluid-structure interaction model of laryngeal dynamics (left) and development of the Coanda effect within three cycles (right). Reprinted from^[181], Copyright (2009), with permission from Elsevier. (d) Airflow-induced self-sustained mucosal oscillation in the baleen whale larynx. Reproduced from^[182], with permission from Springer Nature. (e) Direct simulation of flow-induced sound generation and propagation based on the immersed boundary method. Reprinted from^[189], Copyright (2017), with permission from Elsevier. (f) Multiscale analytical model

of bronchial airway acoustics. Reprinted from^[198], with the permission of AIP Publishing. (g) Mathematical model of bowel sound wave generation. Reproduced with permission from^[201]. © 2018 Acoustical Society of America.

5.2 Plant acoustics models

Significant progress has been made in the generation mechanism and detection technology of plant acoustic signals. Plants can emit specific acoustic signals closely associated with their physiological states under biotic or abiotic stress. As early as the 1960s, Milburn and Johnson^{[209][44]} detected audible acoustic emissions (AAEs) in *Ricinus communis* petioles and leaves using contact sensors, providing the first experimental evidence of plant sound production. Subsequently, Tyree and Dixon^[210] employed piezoelectric sensors to monitor xylem ultrasonic acoustic emissions (UAEs), revealing that these acoustic emissions correlate with xylem cavitation. More recently, Khait^[136] achieved remote detection of stress-induced UAEs in herbaceous plants such as tomatoes and tobacco. Their work demonstrated that drought and mechanical damage trigger ultrasonic signals (20–100 kHz) whose spectral characteristics are strongly correlated with xylem anatomical structures and stress types (**Figure 12(a)**)^[211].

Plant acoustic signals originate from multiple sources. One potential mechanism that may produce sounds is cavitation in the stem. Under drought and cutting stress, air seeding can lead to the entry of air via pit membranes, triggering water column rupture and tension energy release, which induce cell wall vibrations and generate acoustic waves^[139]. Similarly, freezing stress induces ice crystal formation that forces dissolved gases, with bubble nucleation releasing tension and producing acoustic emissions^[212]. Laschimke^[213] hypothesized that the tension in the xylem is caused by coherent bubbles adhering to vessel walls and that acoustic emissions are generated via bubble regrouping (**Figure 12(b)**). Beyond these, Fukuda^[214] observed acoustic emissions across latent, early, and advanced stages of pine wilt disease, linked to progressive hydraulic system disruption. Notably, plant acoustic signals do not originate only from the water conducting system^[139, 215, 216]. Lamacque^[217] recorded two acoustic emission phases in severely dehydrated lavender plants: the first phase correlated with rapid stem diameter loss and a significant increase in loss of xylem conductance, while the second phase aligned with cellular damage, indicating direct mechano-acoustic transduction from cell injury (**Figure 12(c)**). Cellular damage coupled with wall cracks, cell wall shrinkage, membrane rupture, and intracellular cavitation may generate acoustic signals. Khait^[136] hypothesized that ultrasonic emissions originate from cavitation in the stem. Hussain^[211] noted only partial frequency overlap between cavitation-associated emissions and recorded sounds, suggesting that other mechanisms contribute to acoustic signal generation.

The diversity of biomaterials in plant tissues complicates acoustic signal propagation pathways. The xylem, as a three-phase composite medium (water, gas, and cell walls), exhibits sound wave propagation velocities and attenuation characteristics heavily influenced by its microstructural organization^[218]. The damping of bark further attenuates signal intensity, which accounts for why Khait^[136] failed to capture reliable signals in two woody species^[139]. Even contact-based detection requires removal of a small portion of bark to obtain viable signals^[216]. Notably, some hypothesize that these signals may be perceived by adjacent plants or insects to elicit adaptive responses; however, such claims await empirical validation through experiments and related methodologies^[219, 220].

The development of bioacoustic physical models reveals a constant tension between computational fidelity and clinical/practical utility. For instance, fully coupled 3D flow-structure interaction (FSI) models of the vocal folds or cardiovascular system provide highly accurate visualizations of turbulent vortices and tissue dynamics. However, their excessive computational cost (often requiring days of supercomputing) renders

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2 them impractical for real-time patient diagnosis. Consequently, researchers often choose simplified 1D
3 lumped-parameter networks (LPNs) or equivalent acoustic circuits. The explicit trade-off here is sacrificing
4 the resolution of 3D spatial wave scattering (such as refraction through anisotropic tissues) to gain rapid,
5 predictive outputs that can guide the structural design of acoustic sensors or provide immediate diagnostic
6 parameters. Similarly, in plant acoustics, models must choose between simulating localized microcavitation
7 events versus macroscopic stem propagation, as bridging these scales simultaneously remains computationally
8 prohibitive.
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11 Therefore, the ultimate value of physical modeling lies in its deep synergy with the aforementioned
12 hardware and algorithmic innovations. By embedding these physical priors (e.g., tissue attenuation profiles or
13 wave dispersion rules) into deep learning networks (as discussed in Section 4.2), researchers can develop
14 physics-informed neural networks (PINNs) that overcome the “black-box” limitations of pure data-driven
15 methods. Furthermore, these theoretical models serve as essential blueprints for sensor hardware, directly
16 dictating the optimal frequency bandwidth, acoustic impedance matching strategies, and spatial deployment
17 of the sensor arrays discussed in Section 3.
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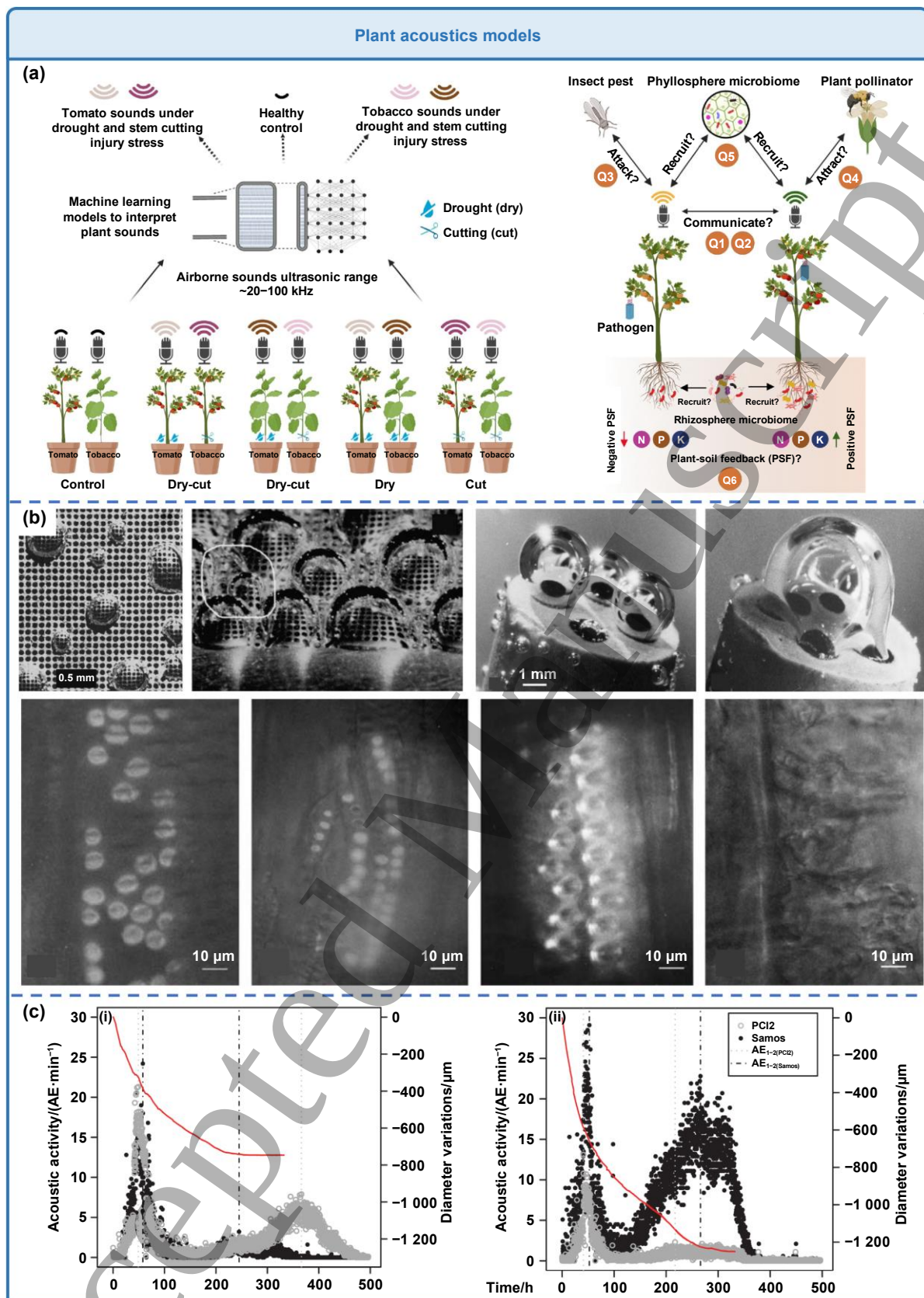


Figure 12. Plant acoustics models. (a) Stress-induced ultrasounds enable remote detection. Reprinted from^[211], Copyright (2023), with permission from Elsevier. (b) Water immersion micrographs: gas accumulations on xylem vessel walls during different expansion stages of *Ulmus glabra*. Reprinted from^[213], Copyright (2006), with permission from Elsevier. (c) Acoustic signals and branch diameter variations (red lines) in two uprooted lavender plants during dehydration stress. The PCI2 (gray points) and Samos (black points) systems are used

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2 to record acoustic signals, with peak activity phases (AE_1/AE_2) marked by dotted lines. Reproduced with
3 permission from ^[217]. © The Author(s) 2022. Published by Oxford University Press on behalf of the Society
4 for Experimental Biology. All rights reserved.
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9 **6. System performance evaluation**

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11 **6.1 Mannequin simulation sounds and validation**
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13 Physiological signals, characterized by their transient, sporadic, and nonrepeatable nature, pose
14 significant challenges for evaluating acoustic sensor performance. These signals vary across individuals and
15 within the same subject under different physiological or emotional states, necessitating reliable, repeatable,
16 and physiologically relevant inputs for sensor design and optimization. System-level evaluations should mimic
17 in vivo conditions, ensuring acoustic propagation through media with properties akin to human tissues to
18 minimize signal distortion from scattering or attenuation at the sensor-skin interface.
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21 To meet these requirements, high-fidelity auscultation mannequins offer an ideal solution for mimicking
22 the generation and transmission of physiological acoustic signals within the human body, which is essential
23 for benchmarking sensor performance under controlled and repeatable conditions. These manikins, commonly
24 used in medical education and clinical competency training, act as programmable “standardized patients”.
25 They are equipped with prerecorded audio libraries of heart, breath, lung crackles, and bowel sounds, which
26 can be selectively played at anatomically accurate auscultation points. They provide standardized and
27 repeatable sound sources by enabling programmable playback, which is critical for comparative sensor testing.
28 In addition, the outer materials of these mannequins, such as medical-grade silicone, are often selected to
29 closely match the acoustic impedance of human skin, thereby simplifying the design of the acoustic coupling
30 interface for wearable sensors.
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35 The earliest high-fidelity mannequin dates back to 1968, developed for cardio-vascular training with
36 capabilities such as carotid pulse simulation and controllable heart and respiratory sounds^[221]. Modern systems,
37 enabled by advances in digital media and simulation technology, now support programmable multisystem
38 physiological signal outputs, thereby enabling comprehensive evaluation of a sensor’s performance. For
39 instance, the lung sound auscultation trainer “LUNG” ver. 2 (Kyoto Kagaku Co. Ltd., Kyoto, Japan) integrates
40 five independently programmable lung sound modules distributed across different thoracic zones (**Figure**
41 **13(a)**). The system contains a variety of lung sound scenarios and supports synchronized heart sounds as
42 background, creating a clinically realistic, composite acoustic environment. Further developments in high-
43 fidelity auscultation manikins have led to the emergence of advanced, multimodal platforms such as SimMan
44 3G Plus (Laerdal Medical), which now offer a mature integration of multiorgan systems, including cardiac,
45 respiratory, and gastrointestinal modules. These platforms enable customizable and programmable playback
46 of physiological sounds, allowing researchers to emulate a broad spectrum of clinical scenarios with high
47 temporal and spectral fidelity. Importantly, their modular configuration supports the synchronized output of
48 multiple signal types across anatomically distributed locations, enabling realistic signal superposition and
49 interference patterns. This level of programmability facilitates comprehensive benchmarking of acoustic
50 sensor performance, including sensitivity, signal discrimination, crosstalk resistance, and multisource
51 resolution under repeatable and physiologically relevant conditions.
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55 High-fidelity mannequins provide controllable, repeatable, and standardized physiological sound sources
56 for acoustic sensor evaluation. By providing ground-truth audio inputs with known spectral and temporal
57 characteristics, they facilitate rigorous, quantitative validation of acoustic sensors during early-stage
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development and calibration. However, human subject testing remains a critical step toward clinical translation, as it enables validation of sensor performance under real-world conditions. By benchmarking against clinical gold standards, such testing reveals practical challenges such as skin–sensor coupling stability, motion artifacts, interindividual variability and long-term wearability. Beyond confirming functional effectiveness, human testing also provides valuable insights from both a technical and user experience perspective, guiding iterative optimization of sensor design for real-life applications.

During human testing of acoustic sensors, it is essential to evaluate their ability to monitor physiological sounds across a range of anatomical sites and subject conditions. By placing the sensor on specific body locations, bioacoustic signals such as heart sounds, breath sounds, swallowing, pulse, and bowel activity can be captured. To assess interindividual adaptability, subject cohorts should span a broad demographic spectrum including variations in age, sex, body habitus, and health status. For instance, Rogers and colleagues demonstrated a wireless, broadband acoustic-mechanical sensing network that enabled pulmonary function assessment in individuals with chronic lung disease, as well as continuous monitoring of respiratory and gastrointestinal activity in neonates admitted to intensive care units^[48]. In the chronic lung study, 13 sensing modules were strategically placed on predefined anatomical landmarks to capture lung acoustics during respiration in both healthy volunteers (n=20) and patients (n=35). Notably, representative recordings from neonatal ICU patients were benchmarked against FDA-approved clinical monitors, confirming the accuracy of physiological parameter extraction under clinical conditions (**Figure 13(b)**).

Beyond static recordings, human trials should also capture physiological signals during both resting and dynamic states to reflect real-world variability and assess sensor robustness under motion because variations in body movement, posture, and activity can influence signal fidelity. An illustrative example is shown in **Figure 13(c)**, where heart sounds were recorded across multiple activity states (quiet sitting, talking, walking, and jogging)^[111]. The device maintained clean acoustic signals with negligible motion artifacts, enabled by its soft form factor and skin-conformal design. Such soft materials help accommodate skin-device contact variations during motion, minimizing acoustic impedance mismatches between the epidermis and the microphone diaphragm and suppressing high-frequency noise due to locomotion or vocalization.

In summary, human subject validation should evaluate not only technical parameters such as SNR, amplitude response, and dynamic robustness but also resistance to motion artifacts and ambient noise. Validation against clinical-grade equipment ensures output accuracy, while subject diversity across age, sex, body type, and health status supports generalizability. Incorporating both healthy and pathological cohorts further enhances the sensor's translational value for personalized and diagnostic healthcare applications. When combined with programmable auscultation mannequins that offer controllable and repeatable acoustic benchmarks, in vivo human testing provides a comprehensive framework for assessing sensor performance across both standardized and real-world physiological scenarios.

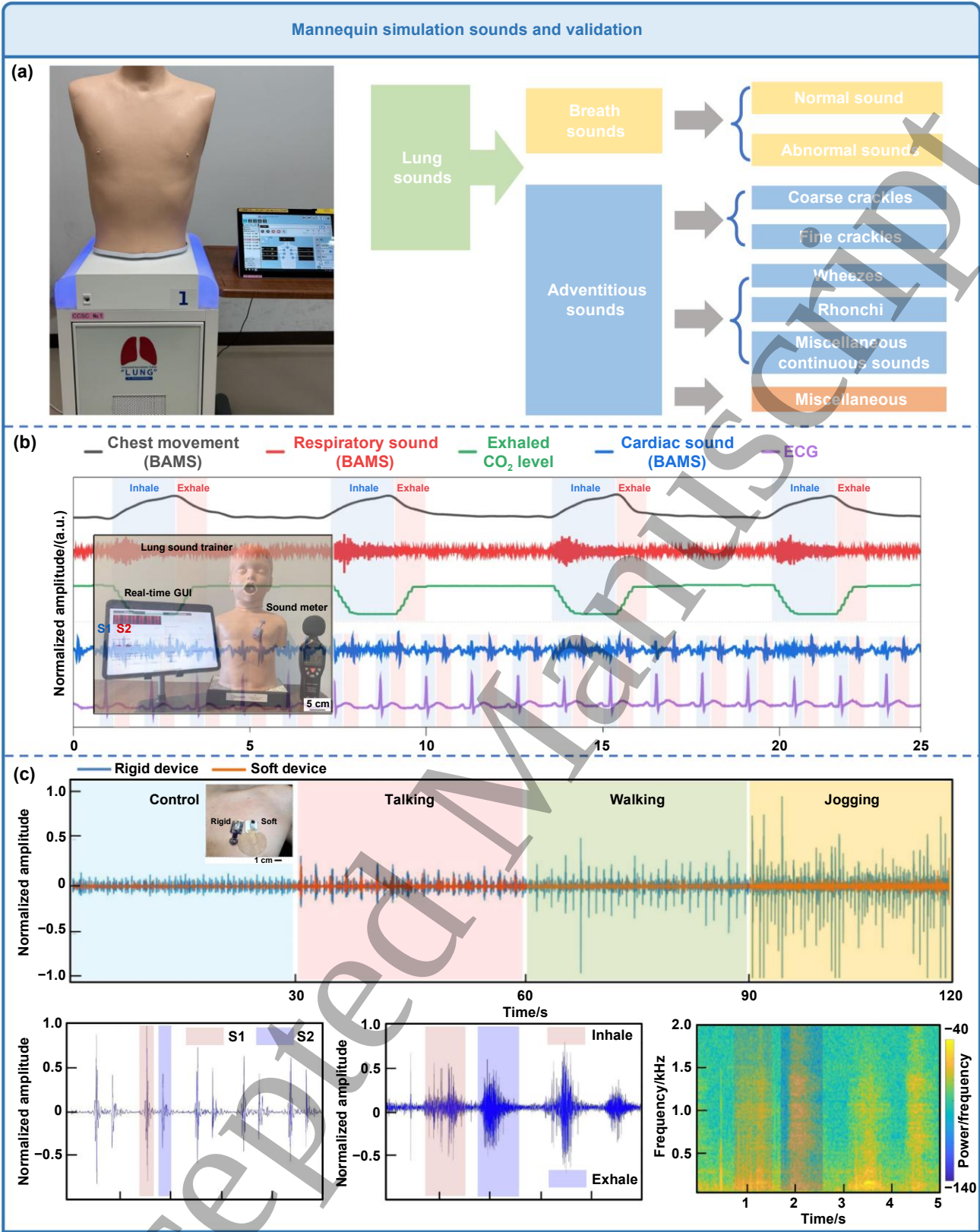


Figure 13. Mannequin simulation sounds and validation. (a) Lung sound auscultation trainer “LUNG” ver.2, a mannequin simulator for lung sound auscultation. Reproduced with permission from [221]. Copyright © 2011, Mary Ann Liebert, A Division of Sage Publications. (b) Wireless acoustic monitoring systems for neonatal respiration and chronic pulmonary disease patients. Right images indicate the monitoring of respiratory sound intensity in ten neonates in the neonatal intensive care unit (NICU) and comparative respiratory rate monitoring between healthy subjects and chronic pulmonary disease patients. Reproduced from [48], with

permission from Springer Nature. (c) Comparison of signal amplitudes between soft acoustic sensors and rigid devices under static and dynamic conditions, demonstrating noise immunity and motion artifact suppression capabilities. Reproduced with permission from^[111]. Copyright © 2022 The Authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original U.S. Government Works. Distributed under a Creative Commons Attribution NonCommercial License 4.0 (CC BY-NC).

6.2 Databases

Bioacoustic research elucidates behavioral, ecological, and evolutionary mechanisms through the analysis of animal acoustic signals. With technological advancements, bioacoustic databases have emerged as critical tools for integrating, storing, and analyzing complex acoustic data, providing substantial support for interdisciplinary research. This part primarily centers on the monitoring of bioacoustic signals associated with human physiological sounds and animal vocalizations.

Databases dedicated to human physiological sound monitoring predominantly encompass cardiac, pulmonary, and respiratory acoustic recordings. In addition, the surveillance of animal vocalizations within natural habitats is chiefly conducted through acoustic camera technologies or passive acoustic monitoring systems, with corresponding databases mainly comprising audio recordings of fish, avian species, and various other wildlife. For physiological acoustics, particularly heart and lung sounds, PhysioNet serves as a widely utilized and extensive database. It encompasses diverse collections, including both open-access and restricted components. Representative subdatabases frequently employed in research encompass the EPHNOGRAM database (featuring simultaneous electrocardiogram and phonocardiogram recordings), the SensiSmartTech cardiovascular signals database, and the MGH/MF waveform database. Other notable examples include HF_Lung_V1, a comprehensive lung sound database designed for respiratory sound analysis; the manikin-recorded cardiopulmonary sounds dataset, which facilitates research on heart and lung acoustic signals; and ZCHSound, which specifically targets pediatric heart sounds associated with congenital heart disease. Additionally, the CTU-UHB database provides valuable intrapartum cardiotocography recordings, while the heart sounds shenzhen corpus offers extensive heart sound data for cardiovascular research. The chest wall lung sound dataset captures lung sounds from various anatomical vantage points on the chest wall, and the classifying heart sounds challenge database serves as a benchmark for heart sound classification algorithms. Beyond heart and lung sounds, additional datasets cover respiratory and bowel sounds. The ICBHI 2017 challenge respiratory sound database, for example, focuses on diverse types of breath sounds, while VoB serves as a comprehensive platform that consolidates various human body sounds from respiratory to bowel sounds. Collectively, these resources provide a rich foundation for advancing algorithmic development and clinical applications in cardiopulmonary acoustic monitoring.

For acoustic monitoring of other wild animals in natural environments, there are also various types of databases available. SILIC is a typical cross-database framework for biodiversity information analysis that integrates online animal sound databases, PAM databases and object detection models for extracting sound information from multiple species. Then, there is also a database WABAD for PAM, which is a world annotated bird acoustic dataset. Similarly, databases such as Macaulay Library (sounds of birds), Xeno-canto (sounds of birds), and iNaturalist (sounds of cross-taxon) are also included for wildlife sound analysis.

We conducted a systematic statistical analysis of the aforementioned bioacoustic database, with the results compiled in **Table 4** for reference by relevant researchers. It should be noted that, due to the continuous growth in database size and the expanding scope of coverage, some data may not be fully incorporated. These databases serve as a cornerstone for bioacoustic research, providing robust data support for feature analysis,

algorithm development, and clinical applications of signals such as human heart and respiratory sounds. Additionally, they offer critical references for fundamental research and practical applications in fields such as animal vocal behavior studies and the exploration of plant acoustic response mechanisms.

Table 4. Bioacoustic database and the reference website.

Database	Data content	Scale (approx.)	Applicable scenarios	Website
PhysioNet	Multiparameter physiological signals (PCG, ECG, respiration)	Massive (>50 TB, thousands of datasets)	Cardiovascular & respiratory algorithm validation, physiological modeling	https://www.physionet.org/about/database/
HF_Lung_V1 [222]	Lung sounds (annotated crackles, wheezes, normal)	9 765 audio samples (261 subjects)	Respiratory disease screening, lung sound classification benchmarking	https://arxiv.org/abs/202.03049v3
Cardiopulmonary sounds dataset[223]	Heart and lung sounds recorded via electronic stethoscopes	~210 subjects under diverse conditions	Cardiopulmonary acoustic separation and signal denoising analysis	https://doi.org/10.17632/8972jxbpmp.1
ZCHSound[24]	Pediatric heart sounds (normal and congenital heart disease)	Thousands of subjects heart sound recordings	Congenital heart disease screening, pediatric cardiology	http://zchsound.ncrcc.h.org.cn/
Heart Sounds Shenzhen Corpus	Heart sounds (normal, murmurs, extra systole)	>800 recordings (170 subjects)	Heart murmur detection, automated cardiovascular disease diagnosis	https://www.researchgate.net/publication/342707934
Chest wall lung sound dataset[225]	Spatial lung sound recordings from various anatomical sites	Multichannel recordings from ~215 subjects	Spatial mapping of lung sounds, multisensor array localization validation	https://data.mendeley.com/datasets/jwyy9np4gv/2
Classifying heart sounds challenge	PCG recordings (normal, murmur, extra sound, artifact)	832 distinct recordings	Machine learning algorithm benchmarking for heart sound classification	https://peterjbentley.com/heartchallenge/
ICBHI 2017 challenge database	Respiratory sounds with annotated respiratory cycles	920 recordings (126 subjects), ~5.5 hours	Respiratory anomaly detection, deep learning model evaluation	https://bhichallenge.med.auth.gr/ICBHI_2017_Challenge
Voice of body	Comprehensive physiological sounds (respiratory, cardiac, bowel)	Multimodal body acoustic data	Wearable device validation, continuous gastrointestinal/cardiac monitoring	http://www.vob-bit.org/

Macaulay library	Animal vocalizations (birds, mammals, amphibians, insects)	>30 million media assets	Biodiversity monitoring, ecological behavior analysis, species identification	https://www.macaulaylibrary.org/
Xeno-canto	Global avian sounds and other wildlife acoustic signals	>900 000 recordings (>10 000 species)	Avian bioacoustics, ecological surveillance, training ML for bird calls	https://www.xeno-canto.org/
iNaturalist	Cross-taxon acoustic and visual observations	>150 million observations globally	Citizen science, global biodiversity tracking, conservation	https://www.inaturalist.org/

6.3 Gold standards and evaluation frameworks

6.3.1 Clinical medical diagnostics

In clinical medicine, the “gold standard” is the most authoritative diagnostic method, accurately identifying diseases and validating new technologies. It ensures diagnostic reliability, guides treatment, and serves as the reference for evaluating other methods’ sensitivity and specificity^[226]. Diseases affecting different systems require specific diagnostic criteria as the “gold standard” due to variations in pathology, anatomy, and clinical presentation^[227]. This chapter explores the application of acoustic technology as a potential diagnostic criterion for common diseases. Heart sounds, produced by cardiac mechanical activities such as valve movement and blood flow, can be heard or electronically recorded^[228]. They include the first (S1) and second (S2) heart sounds, with S3, S4, and murmurs indicating potential abnormalities^[229]. These sounds are central to cardiac auscultation, providing information on heart and valve function. **Table 5** summarizes the acoustic characteristics and clinical significance of various body sounds. Auscultation involves listening to heart sounds with a stethoscope at specific chest locations. Since its invention in 1816, the stethoscope amplifies heart vibrations, aiding in detecting abnormalities^[230]. As a low-cost, readily available “gold standard” for initial cardiovascular screening, it is vital in emergencies and remote areas. Physicians adjust the stethoscope position and pressure based on patient breathing and posture to capture transient signs. While dependent on experience, its noninvasive and immediate nature makes it key for gathering initial cardiac information^[231]. Electronic stethoscopes and phonocardiograms (PCGs) complement traditional auscultation. Electronic stethoscopes use sensors to convert body sounds into electrical signals, improving sound quality and enabling recording and analysis^[232]. PCGs create waveforms from these sounds, allowing AI algorithms to identify and classify murmurs and alerting providers to high-risk conditions. This objective analysis addresses limitations of traditional auscultation, providing quantitative parameters for screening^[233]. Integrated into wearables, these technologies enable monitoring of chronic or postsurgical patients. They can detect faint murmurs missed in standard exams. Combining PCGs with ECG or ultrasound data allows comprehensive cardiovascular evaluation^[234].

However, electronic stethoscopes and PCGs are not yet independent diagnostic standards. Variations in device algorithms lead to a lack of unified standards, and PCGs alone cannot diagnose disease causes, as different conditions may produce similar sounds^[235]. Sensor position, patient posture, and respiration also affect signal quality. Currently, they serve as auxiliary tools and must be combined with other technologies, such as ECG or ultrasound, for effective diagnosis. For respiratory diseases, high-resolution CT, CT

angiography, etiology tests, and pulmonary function tests are the gold standards for structural, vascular, infectious, and functional issues, respectively. Lung auscultation remains a key initial tool for rapid screening based on breath sound changes. Breath sounds originate from airflow in the airways. Normal sounds include bronchial (loud, high-pitched, 200–1 200 Hz), heard in the suprasternal fossa; alveolar (soft, low-pitched, 350–640 Hz)^[236, 237], heard in lung fields; and bronchoalveolar, a mixture heard near the sternum. Traditional lung auscultation is noninvasive and low-cost but subjective and experience-dependent and lacks objective metrics, thus requiring confirmation by imaging or function tests^[232]. Electronic stethoscopes and intelligent analysis digitize sounds, improving objectivity. Wearable sensors enable home monitoring. However, due to the nonspecific nature of lung sounds, auscultation cannot replace definitive diagnostics such as HRCT.

Bowel sound auscultation assesses abdominal conditions by listening to frequency, pitch, and intensity in the abdominal quadrants. Normal sounds occur 5–35 times/minute (220–530 Hz), increasing after meals^[238]. It allows real-time monitoring of motility, screening for acute conditions, and evaluating postoperative recovery. Abnormal sounds include hyperactive (>35 times/minute, 400–1 000 Hz), suggesting gastroenteritis or bleeding^[238, 239]; weakened/absent (<5 times/minute, <100 Hz), indicating paralysis; and special sounds such as metallic tones, suggesting mechanical obstruction. Limitations include subjectivity and low specificity, requiring confirmation via palpation or imaging. Future directions include intelligent auscultation using biosensors for quantitative analysis and wearable devices for continuous monitoring, reducing reliance on repeated imaging^[199].

6.3.2 Ecological monitoring

Ecological and veterinary monitoring environments are complex and lack a single, universal “gold standard.” In the field of ecoacoustics, the core of evaluating passive acoustic monitoring systems lies in benchmarking their outputs against “ground truth” derived from expert manual annotations or external surveys (e.g., observer-based monitoring, eDNA monitoring)^[240]. This evaluation model emphasizes the performance of automated identification tools in terms of precision, recall, and F1 score, rather than a single diagnostic indicator, to address variations in species and behavioral coverage among different monitoring methods. In veterinary monitoring, bioacoustics serve as a noninvasive, continuous tool for auxiliary screening or early warning. Acoustic markers, such as coughing and stress vocalizations, can effectively predict disease trends such as *Mycoplasma hyopneumoniae*^[241]. However, final diagnosis still relies on “gold standards” such as clinical examinations, pathological scoring, or qPCR assays^[242]. Currently, the core positioning of acoustic monitoring is for early warning and auxiliary screening, acting as a supplement to traditional diagnostics to ensure diagnostic accuracy across diverse ecological and farming environments^[241,242].

6.3.3 Agricultural applications

In the fields of plant pathology and precision agriculture, the diagnostic gold standard has evolved into a multiverification system comprising symptom observation, microscopic analysis, immunological assays (e.g., ELISA), and molecular diagnostics (e.g., qPCR, NGS). Especially for high-risk diseases, confirmation typically requires positive results from at least two methods based on different biological principles^[243,244]. Acoustic sensing, as a noninvasive supplementary tool, is transitioning from traditional contact-based ultrasonic detection to airborne acoustic analysis. Cutting-edge research confirms that stressed plants emit specific ultrasonic signals between 20–100 kHz, providing a highly valuable screening perspective for the early identification of drought or physical damage^[136]. The diagnostic efficacy of these acoustic markers primarily centers on sound emission rates and sound pressure levels, which are anchored to physiological “ground truths” such as leaf water potential or stomatal conductance^[245]. Plant acoustic technology, integrated

with AI-driven automated identification tools, can capture real-time critical trends as plants transition from healthy to pathological states. This serves to complement traditional molecular and pathological gold standards, establishing a comprehensive monitoring loop from physiological early warning to biological confirmation in agricultural applications^[136,245].

Table 5. Acoustic characteristics and clinical implications of internal body sounds.

Sound Category	Specific Sound	Frequency/ Hz	Range	Rate (events/min)	Pathological significance
Cardiac sounds	S1	20–180 ^[205,233]		Heart-rate dependent ^[205,233]	-
	S2	50–250 ^[205]		Heart-rate dependent ^[205,233]	-
	S3	25–70 ^[233]		Heart-rate dependent ^[233]	Heart failure, cardiomyopathy ^[246, 247]
	S4	20–30 ^[233,248]		Heart-rate dependent ^[233]	Coronary artery disease, hypertrophic cardiomyopathy ^[205,247]
	Murmur	100–600 ^[249]		Continuous/transient ^[249]	Aortic stenosis, mitral regurgitation ^[249]
Respiratory sounds	Normal breath	100–1 200 ^[236, 237]		12–20 ^[236]	-
	Wheezes	400–1 000 ^[237]		Respiratory-cycle dependent ^[237]	Asthma, COPD, foreign body obstruction ^[237]
	Crackles	<600 ^[237]		Respiratory-cycle dependent ^[237]	Pulmonary edema, pneumonia, bronchiectasis ^[237]
	Rhonchi	< 200 ^[237]		Respiratory-cycle dependent ^[237]	Acute exacerbation of COPD ^[237]
Bowel Sounds	Normal peristalsis	220–530 ^[238]		5–35 ^[239]	-
	Hyperactive	500–1 000 ^[238]		>35 ^[239]	Mechanical bowel obstruction, acute

(Metallic tinkling)				gastroenteritis ^[238, 239]
Hypoactive e/Absent	<100 ^[238]	<5 ^[239]	Paralytic peritonitis ^[238, 239]	ileus,

7. Applications

The synergistic integration of conformal acoustic sensors, intelligent signal processing algorithms, mechanistic physical models, and standardized evaluation protocols establishes a robust technological pipeline for modern bioacoustics. Ultimately, the true value of this comprehensive theoretical and methodological foundation is realized through its practical deployment in real-world scenarios. Building upon the core technologies discussed in the preceding sections, this section explores how bioacoustics is revolutionizing diverse fields. Specifically, we highlight its transformative applications in clinical diagnostics, where acoustic signatures enable unprecedented noninvasive disease detection; in human-machine interaction, where bioacoustic interfaces bridge human and artificial intelligence; and in precision agriculture, where sound-based monitoring optimizes crop and livestock health to promote sustainability.

7.1 Acoustic stethoscopy for diagnosis

The intellectualization and portability of acoustic stethoscopes represent a core direction in modern medical diagnostics, focusing on the precise capture of physiological signals and early disease warning. For instance, Lee developed a soft wearable stethoscope integrating nanomaterial printing and flexible circuits for continuous cardiorespiratory monitoring (Figure 14(a)). This device, featuring a MEMS-based microphone and silicone gel backing, achieves a 14.8 dB SNR on the chest wall, surpassing traditional digital stethoscopes. An embedded CNN enables 95% accuracy in diagnosing four lung diseases, including crackles and wheezes, and detects obstructive sleep apnea via respiratory anomaly analysis during sleep. This wearable, intelligent diagnostic platform overcomes the spatiotemporal constraints of manual stethoscope interpretation, enhancing telemedicine applications. In parallel, Jin introduced a photoacoustic blood stethoscope, extending auscultation to photoacoustic-hemodynamic monitoring through a microlens array and piezoelectric sensor (Figure 14(b)). Its 3D imaging algorithm, with 200 μm resolution, distinguishes arteriovenous oxygenation differences. In mouse models, it tracks rhodamine B metabolic decay, aligning with invasive measurements. Using a 532 nm laser to excite blood chromophores for acoustic signal generation, this device integrates 3D imaging to noninvasively quantify hypoxic states, vascular compliance, and pharmacokinetics, establishing a novel paradigm for deep vascular physiological assessment.

Along the trajectory of material innovation, Hui proposed a triboelectric stethoscope that harnesses the constitutive property of rapid charge saturation in fluorinated ethylene propylene (FEP)/polyimide (PI) membranes during friction (Figure 14(c)), boosting the sensing sensitivity to 1 215 mV·Pa⁻¹. Their power-function cross-sectional acoustic cavity further elevates the SNR to 36 dB via acoustic energy focusing, enabling a CNN model to achieve 97% classification accuracy for five cardiac states. Concurrently, Yan fabricated piezoelectric fibers via thermal drawing and woven them into acoustic textiles (Figure 14(d)). Exploiting the waveguide effect of aramid-cotton blend structures, their shirt-type sensor captures nanoscale chest wall vibrations, enabling sound source localization (angular error <1.73°) and identification of heart

1
2 sound splitting. These two technologies break through the hardware sensitivity bottleneck of traditional
3 stethoscopes from two dimensions: charge enhancement at nanomaterial interfaces and acoustic energy
4 conduction via textile structures.
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6 At the system integration level, Wang developed a flexible skin device that employs 3D-printed elastic
7 resonant cavities (wall thickness 2 mm) to enable dynamic monitoring of borborygmi for up to 5 hours (**Figure**
8 **14(e)**)^[251]. The study revealed periodic fluctuations in postprandial borborygmi peak counts and short-term
9 energy, demonstrating superior performance in capturing features of mechanical intestinal obstruction (MIO)
10 versus paralytic ileus. In a more interesting approach, Liu proposed an epidermal mechanoacoustic sensor
11 with a Silbione/Ecoflex core-shell structure (effective modulus 31.8 kPa)^[205] capable of synchronously
12 recording heart sounds, ECG, and carotid artery pulsations (**Figure 14(f)**). In the context of ventricular assist
13 device (VAD) monitoring, this sensor detects acoustic changes indicative of intrapump thrombosis.
14 Experimental results show that the main peak of the vibration spectrum broadens during thrombus formation,
15 providing a potential acoustic marker for fault warning in implanted devices.
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18 These abovementioned advancements in epidermal mounting, multimodal fusion, and expanded
19 physiological signal monitoring signify the transformation of acoustic stethoscopes from single diagnostic
20 tools to holistic physiological monitoring platforms.
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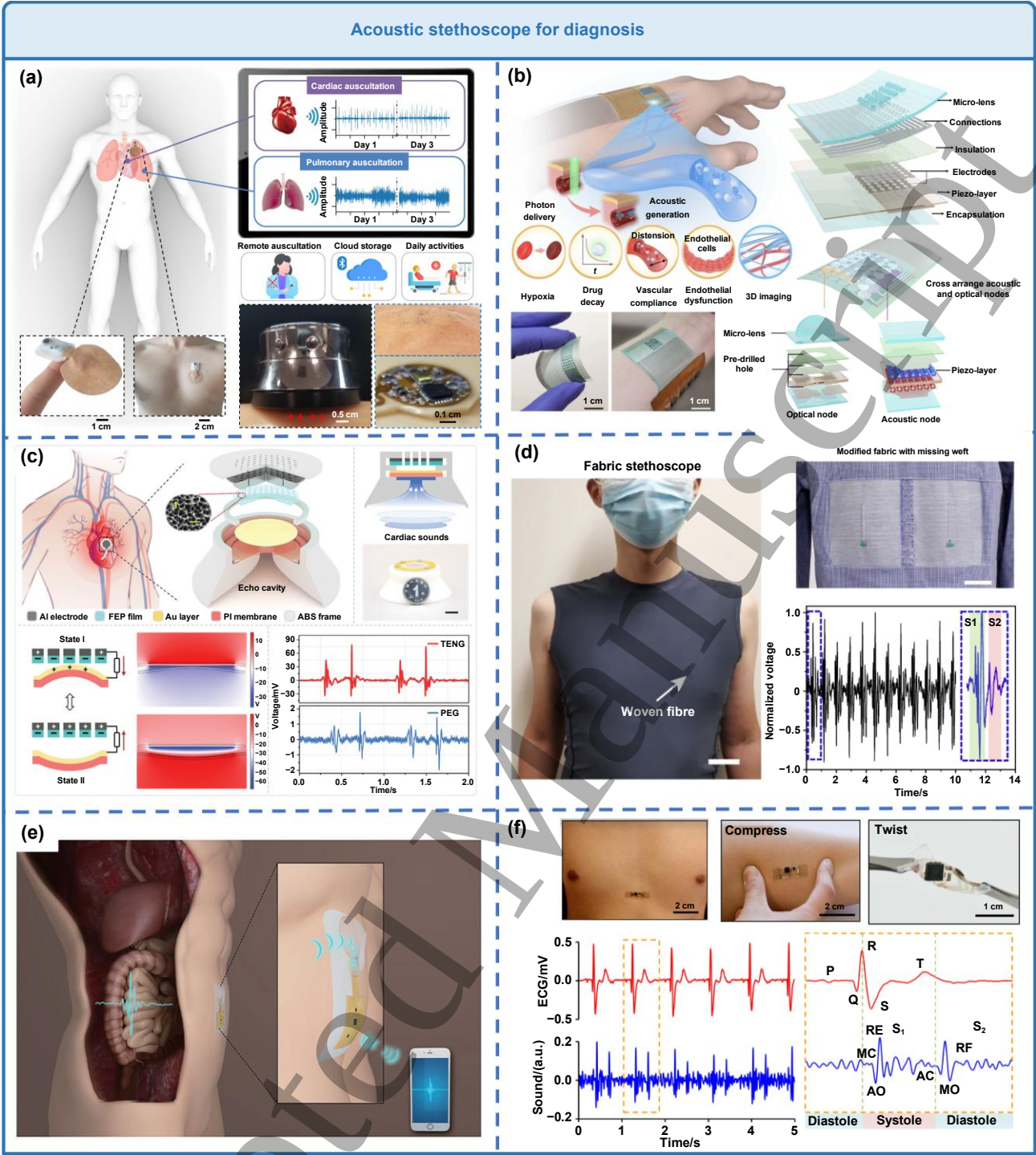


Figure 14. Acoustic stethoscope for diagnosis. (a) A soft portable stethoscope for automated disease diagnosis. Reproduced with permission from^[111]. Copyright © 2022 The Authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original U.S. Government Works. Distributed under a Creative Commons Attribution NonCommercial License 4.0 (CC BY-NC). (b) A flexible optoacoustic blood stethoscope for cardiovascular monitoring. Reproduced from^[250], CC BY 4.0. (c) Triboelectric-based stethoscope for physiological signal monitoring. ^[110]John Wiley & Sons. © 2024 Wiley-VCH GmbH. (d) Acoustic fabrics for source localization and cardiac sound detection. Reproduced from^[60], with permission from Springer Nature. (e) A flexible acoustic device for bowel sound monitoring. Reproduced from^[251], with permission from Springer Nature. (f) A mechano-acoustic device for cardiovascular diagnostics. Reproduced with permission from^[205]. CC BY-NC 4.0.

7.2 Human-computer interaction

In the field of human-computer interaction, acoustic devices have demonstrated enormous potential, with related research continuing to make breakthroughs. With some advanced examples, Wang developed Sono Textiles, an intelligent acoustic fabric that innovatively integrates glass microfibers with piezoelectric sensors (**Figure 15(a)**)^[252]. By monitoring acoustic energy attenuation, this fabric enables tactile perception, gesture recognition, and physiological monitoring, showing broad application prospects in distributed tactile sensing, medical health monitoring, etc. Zhao reported a liquid acoustic sensor for speech recognition, which relies on a 3D-oriented and branched magnetic network structure formed by suspended neodymium-iron-boron magnetic nanoparticles (**Figure 15(b)**)^[112]. Capable of distinguishing small pressures, the sensor features a self-filtering capability to eliminate low-frequency biomechanical motion artifacts. When combined with machine learning algorithms, it achieves 99% recognition accuracy in noisy environments. Moreover, Yang developed a graphene-based intelligent wearable artificial throat that exhibits high sensitivity to human speech and associated vocal movements (**Figure 15(c)**)^[145]. This device captures mixed-modal acoustic signals and mechanical motions, enabling the detection of low-frequency fundamental signals with robust noise resistance. It achieves an average accuracy of 99.05% in detecting fundamental speech elements and facilitates speech recognition and reproduction for laryngectomy patients. Furthermore, Park designed a biomimetic elastic robotic skin that integrates impedance and acoustic tomography for tactile sensing, achieving three-dimensional tactile localization and material identification^[253]. This bioinspired hierarchical sensing network provides a novel strategy for bionic tactile feedback in dexterous robotic manipulation (**Figure 15(d)**). Additionally, Wang engineered a biomimetic flexible piezoelectric mobile acoustic sensor featuring a multiresonant ultrathin structure (**Figure 15(e)**)^[254]. This sensor, tailored for machine learning-based biometric identification, holds significant potential for applications in biometric recognition and related fields. In addition to piezoacoustic applications, Qiao developed a highly sensitive omnidirectional acoustic sensor based on a triboelectric method capable of simultaneously recognizing multiple speakers in assistive conference systems and driving commands amidst background music in autonomous vehicles^[255], thereby enhancing human-machine interaction (**Figure 15(f)**). In addition, Yao introduced a noise-robust triboelectric acoustic sensor enhanced by DL^[256], tailored for human-machine collaboration in noisy environments.

These studies collectively highlight three major trends in the development of acoustic devices for human-machine interfaces. At the material level, a shift from rigid inorganic crystals to flexible composite materials, such as liquid magnets and porous polymers, is evident. At the functional level, there is a progression from single-modal sensing to multiphysics integration, exemplified by the synergistic perception of mechanical vibrations and acoustic signals. At the algorithmic level, the deep integration of ML with sensor design is driving advancements in noise-robust recognition and personalized interaction. Future challenges center on balancing sensor miniaturization with functional integration, optimizing robustness in complex environments, and enhancing the generalizability of cross-modal signal processing algorithms.



Figure 15. Acoustic sensor for healthcare and human-computer interaction. (a) A smart acoustic textile for human-machine interactions. Reproduced from^[252], CC BY 4.0. (b) A liquid-based acoustic sensor for human-machine communication. Reproduced from^[112], with permission from Springer Nature. (c) A wearable artificial throat for speech recognition and interaction. Reproduced from^[145], with permission from Springer Nature. (d) Biomimetic robot skin for human-robot interaction haptic perception. From^[253]. Reprinted with permission from AAAS. (e) Biomimetic and flexible piezoelectric acoustic sensors for biometrics and interaction. Reproduced with permission from^[254]. Copyright © 2021 The Authors, some rights reserved; exclusive licensee American Association for the Advancement of Science. No claim to original U.S. Government Works. Distributed under a Creative Commons Attribution NonCommercial License 4.0 (CC BY-NC). (f) An omnidirectional acoustic sensor for human-machine interaction. ^[255]John Wiley & Sons. © 2024 Wiley-VCH GmbH.

7.3 Agriculture

The application of acoustic technology in agriculture is rapidly evolving from traditional monitoring to intelligent and precision-oriented approaches. By integrating bioacoustics, sensor technology, and AI, it has provided innovative solutions for the sustainable development of modern agriculture. The primary prerequisite

for achieving agricultural sustainability lies in the precise assessment of ecological impacts. For instance, the study from Doohan explored the potential of acoustic monitoring in evaluating agricultural biodiversity^[257], emphasizing its advantages of low cost and operational simplicity (**Figure 16(a)**). The study indicated that species-specific acoustic metrics (such as the vocalization characteristics of bats and birds) based on automatic recognizers can effectively correlate farm management with biodiversity changes. Analysis using five-dimensional evaluation criteria showed that species-specific indices demonstrate higher diagnostic capability in assessing the impact of farm management on biodiversity, while general metrics such as acoustic indices are more suitable for overall biodiversity assessment at the regional scale. This research quantitatively links acoustic technology with agricultural sustainability goals, facilitating the agricultural sector in tracking its sustainability performance. In livestock farming, acoustic technology has facilitated behavioral analysis and health monitoring. As an example, Martinez-Rau proposed the noise-robust foraging activity recognizer (NRFAR), which discriminates between grazing and ruminant behaviors in cattle with high accuracy under low SNR ratios by analyzing the time-frequency features of bovine chewing vocalizations (**Figure 16(b)**)^[258]. This approach significantly reduces misjudgment rates. For avian species and poultry, vocal signals serve as biomarkers for health assessment. Mahdavian demonstrated that respiratory diseases in chickens alter vocal characteristics: e.g., changes in wavelet entropy values of vocalizations on the third day of bronchitis infection reflect disease progression^[259], with a misdiagnosis rate of less than 14% (**Figure 16(c)**). These findings provide acoustic feature references for developing intelligent poultry health monitoring systems.

Precise monitoring of crop water status represents a core challenge in water-saving agriculture. Álvarez-Arenas developed a noncontact resonant ultrasound spectroscopy (NC-RUS) technique that leverages variations in leaf thickness resonance frequencies within the 0.1–1.2 MHz band to enable real-time nondestructive monitoring of water status in crops such as grape (*Vitis vinifera*) and coffee (*Coffea arabica*)^[260]. Concurrently, Alzuhair constructed an AIoT platform integrating acoustic and environmental sensing, achieving low-power data transmission via LoRa networks^[261]. By employing Mel filterbank features and CNNs for classifying environmental sounds (including rain, fire, and livestock vocalizations), this system demonstrates 87.5% recognition accuracy in noisy environments, with single-node power consumption optimized to 3.8 mW findings that lay the hardware foundation for large-scale field deployment.

Moreover, acoustics can regulate plant growth and stress responses. Király reviewed the extensive applications of ultrasound in horticulture and agriculture (**Figure 16(d)**)^[262], demonstrating its role in promoting plant growth and development, enhancing yield, and improving crop quality. Meanwhile, integrating ML models and AI for data processing and interpretation enables targeted agricultural management interventions based on analytical results. Furthermore, the study by Choi revealed the multifaceted value of acoustic applications in agriculture (**Figure 16(e)**)^[263]. On the one hand, specific frequency acoustic vibrations act as mechanical stimuli to regulate plant development. For example, promoting seed germination and enhancing yield and quality. On the other hand, acoustics can activate plant defense systems, as research has shown that treating *Arabidopsis thaliana* with 1 000 Hz acoustic vibrations enhances its resistance to *B. cinerea* by modulating salicylic acid signaling pathways. These findings provide novel insights for green pest control and hold promise for broader agricultural applications in the future.

In agricultural engineering applications, Furriel achieved directional deposition by regulating the trajectory of suspended pesticide droplets through acoustic pressure fields (**Figure 16(f)**)^[264]. Experiments show that this method enhances leaf coverage by 44.97% and reduces drift by 58.82%, significantly outperforming traditional spraying technologies. Moreover, the study used specific frequency (42–557 Hz) sound waves to induce resonance in coffee fruits, enabling selective abscission of mature fruits. Unlike traditional mechanical vibration rods, this technology avoids physical damage to plants, providing a new

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paradigm for precision agricultural equipment development.

Turning to agricultural ecological restoration, Müller focused on tropical forests in Ecuador, integrating bioacoustic and metabarcoding techniques with acoustic index models and CNNs (**Figure 16(g)**)^[265]. They found that the composition of vocal vertebrate communities and automated acoustic metrics were significantly correlated with forest restoration gradients and could reflect the diversity of nonvocal insects. This offers a low-cost, repeatable new approach for biodiversity monitoring in abandoned agricultural land restoration.

In the realm of soil management, Adamo proposed an acoustic soil moisture measurement method^[266]. Based on the Brutsaert model, their theoretical analysis and prototype development yielded a low-cost, high-precision soil moisture sensor capable of measuring sound velocity in soil (and other materials such as fluids), providing a novel method for soil moisture monitoring in precision agriculture.

The application of acoustic technologies in agriculture has evolved from single-function monitoring to intelligent sensing and precision regulation. However, several challenges remain: (1) noise robustness in complex farmland environments requires further improvement, which can be addressed by integrating deep learning to optimize feature extraction algorithms. (2) The molecular mechanisms of plant vocalization remain unclear, necessitating integration with transcriptomics to decode signaling pathways underlying ultrasonic responses. (3) Inadequate device integration calls for the deep fusion of AIoT platforms with agricultural machinery. Looking ahead, with advancements in MEMS sensors and edge computing, acoustic technologies are poised to play a pivotal role in smart agriculture, providing technical support for carbon-neutral agricultural practices.

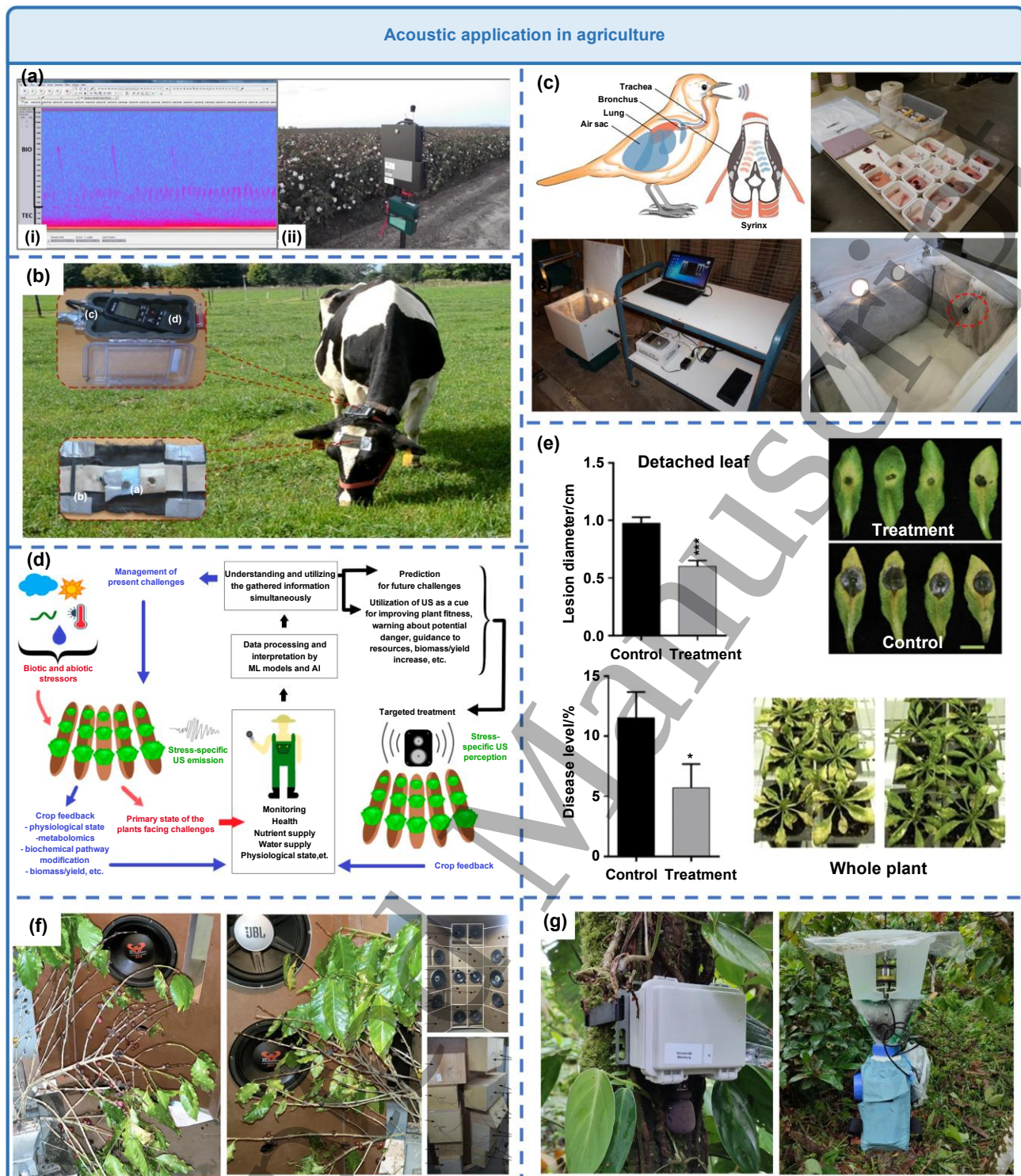


Figure 16. Acoustic application in agriculture. (a) Acoustic monitoring in agricultural industries. Reprinted from^[257], Copyright (2019), with permission from Elsevier. (b) Acoustic signal recording system based on microphones for the identification of grazing cattle activities. Reproduced from^[258], CC BY 4.0. (c) Acoustic recording system for poultry health monitoring. Reprinted from^[259], Copyright (2021), with permission from Elsevier. (d) Plant ultrasound in horticulture and agriculture. Reproduced from^[262], CC BY 4.0. (e) Positive regulation of plant antimicrobial activity by sound vibration treatment. Reproduced from^[263], CC BY 4.0. (f) Acoustic technology for coffee harvesting and handling. Reprinted from^[264], Copyright (2022), with permission from Elsevier. (g) Acoustic monitoring combined with DL facilitates the assessment of biodiversity restoration in tropical forests. Reproduced from^[265], CC BY 4.0.

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7.4 Comparative analysis of cross-domain technical challenges

While the abovementioned sections highlight the transformative applications of bioacoustics across clinical diagnostics, human–machine interfaces (HMI), and precision agriculture, a horizontal comparison reveals several shared, fundamental technical bottlenecks that currently limit broad commercial translation. Rather than isolated issues, these challenges manifest distinctively across different operational domains:

(1) **Motion artifact suppression and environmental noise robustness:** in real-world deployments, target bioacoustic signals are invariably corrupted by noise. For clinical stethoscopes (Section 7.1) and HMI devices (Section 7.2), the primary interference stems from low-frequency motion artifacts (e.g., body movements, skin friction) and ambient conversation. Current solutions rely heavily on acoustic impedance matching materials or differential microphone arrays to isolate signals. Conversely, in agriculture and ecology (Section 7.3), the noise profile is highly unstructured and dominated by wind, rain, and agricultural machinery. Suppressing these broad-spectrum interferences requires transitioning from physical shielding to adaptive, AI-driven noise cancellation algorithms capable of functioning in chaotic open-field conditions.

(2) **Trade-offs between power consumption and system integration:** the shift from wired data acquisition to wireless, continuous monitoring creates a severe power-integration dilemma. Wearable medical patches and HMI voice-recognition interfaces demand ultrasmall form factors, strictly limiting battery capacity. This necessitates the development of ultralow-power microcontrollers or self-powered triboelectric/piezoelectric acoustic sensors. For agricultural AIoT networks, while size constraints are relatively relaxed, the sensors must operate autonomously for months in remote fields. Therefore, integrating computationally intensive edge-AI modules for real-time acoustic classification (to avoid transmitting massive raw audio files via LoRa) without depleting the energy budget remains a critical architectural trade-off.

(3) **Cross-subject and cross-environment generalization:** a ubiquitous challenge across all bioacoustic applications is the degradation of machine learning model performance when transitioning from training datasets to unseen subjects or environments. In clinical and HMI applications, algorithms trained on specific cohorts often fail when applied to individuals with varying body mass indices (BMIs), anatomical variations, or distinct vocal timbres. Similarly, in agricultural applications, ultrasonic stress emission models trained on specific plant species in controlled chambers struggle to generalize to diverse crops in open fields with varying soil and atmospheric conditions. Overcoming this generalization gap urgently requires large-scale, standardized, and multicenter acoustic databases coupled with advanced domain adaptation and transfer learning techniques.

By recognizing these shared technical hurdles, future research can foster cross-disciplinary solutions—such as transferring robust motion-suppression algorithms from wearable health devices to agricultural monitoring nodes—thereby accelerating the maturity of bioacoustic sensing networks.

8. Challenges and prospects

Despite significant developments in acoustic sensing and signal processing, the advancement of bioacoustics faces multifaceted challenges. This section outlines a roadmap for future bioacoustic research by examining key bottlenecks in hardware, signal processing, data resources, and system integration, followed by targeted research directions to address these limitations.

8.1 Systematic categorization of challenges

(1) Hardware and sensing limitations: the primary physical bottleneck across domains is acoustic impedance mismatch and signal attenuation. In medical monitoring, soft tissues act as severe low-pass filters, dampening high-frequency pathological features, while wearable sensors suffer from motion artifacts and sweat-induced degradation. In ecological and agricultural applications, sensors face extreme environmental degradation (e.g., biofouling in oceans, extreme temperatures) and struggle to capture ultraweak signals (e.g., 20–150 kHz plant emissions) without invasive deployment.

(2) Signal processing and algorithmic constraints: current algorithms lack robustness against nonstationary interference. In human–machine interfaces (HMIs), extracting weak vocal signals amidst ambient conversational noise remains difficult without distorting the target signal. Similarly, in wildlife monitoring, separating specific species from a “cocktail party” of overlapping multispecies vocalizations, wind, and anthropogenic noise requires massive computational power that far exceeds the capacity of remote nodes.

(3) Data scarcity and lack of standardization: severe “labeling crisis” plagues bioacoustics. Unlike general audio, annotating clinical murmurs or rare avian calls requires specialized expertise, leading to small, biased datasets. Furthermore, cross-domain applications suffer from a lack of standardized, open-access databases and rigorous testing frameworks. This is particularly evident in plant acoustics, where the mechanistic correlation between acoustic emissions and biological functions remains controversial due to inconsistent data collection.

(4) System integration and deployment: the transition from wired laboratory setups to wireless, continuous monitoring creates a severe power-integration dilemma. Wearable patches demand ultrasmall form factors with strictly limited battery capacity, while agricultural AIoT nodes must operate autonomously for months. Transmitting massive raw acoustic data over limited bandwidths (e.g., LoRa or underwater acoustic channels) further paralyzes real-time system integration.

8.2 Concrete future research directions

To unlock the full potential of bioacoustics and address the aforementioned bottlenecks, future research should be strategically directed across the following corresponding dimensions:

(1) Hardware innovation (Edge-AI and self-powered sensors): future developments should focus on embedding ultralow-power microcontrollers directly onto conformable acoustic patches for real-time, on-sensor anomaly detection (e.g., arrhythmia alerts). For ecological applications, integrating energy-harvesting technologies (e.g., triboelectric, solar, or bioenergy) into mobile robotic platforms (drones, gliders) is critical to achieving battery-free, long-term deployment.

(2) Signal processing (Multimodal Fusion and Explainable AI): to overcome motion artifacts and noise, algorithms must evolve toward multimodal architectures that inherently fuse acoustic data with electrophysiological (ECG/EMG), kinematic (IMU), or environmental signals. Additionally, developing explainable AI (XAI) is urgent to translate “black-box” acoustic feature representations into physiologically grounded, clinical, or ecological insights.

(3) Data strategies (self-supervised learning and federated databases): to solve the data annotation bottleneck, research must leverage self-supervised learning (SSL) paradigms that pretrain foundation models on massive unlabeled physiological and ecological audio. Simultaneously, global efforts must prioritize establishing standardized, multicenter, open-access databases that correlate acoustic spectral signatures with rigorous ground truths (e.g., multiomics data in plants or clinical imaging in humans).

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2 (4) System-level integration (AIoT and closed-loop frameworks): the ultimate trajectory is the
3 deployment of intelligent, networked systems. In precision agriculture, this means deploying low-cost
4 ultrasonic microphone arrays alongside soil and weather sensors within an AIoT framework to create
5 automated, closed-loop irrigation and pest control systems. In ecology, integrating federated learning across
6 decentralized mobile networks will enable global-scale biodiversity mapping without the transmission
7 bottleneck of raw audio.
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13 **9. Conclusion**
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15 Acoustic sensing technologies, encompassing both bioacoustics and biomedical acoustics, have evolved
16 from a niche research field to a transformative technology with far-reaching implications for healthcare,
17 ecology, and beyond. This review has charted the rapid advancements in bioacoustic sensing (from
18 piezoelectric transducers to MEMS devices) and signal processing (from traditional feature engineering to
19 deep learning), highlighting their roles in decoding human physiology, animal behavior, and plant ecology.
20 The interdisciplinary nature of bioacoustics—bridging acoustics, biology, materials science, and artificial
21 intelligence—has enabled breakthroughs in noninvasive monitoring, from intelligent stethoscopes for cardiac
22 disease diagnosis to acoustic camera systems for biodiversity tracking.
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26 Despite significant progress, challenges in noise resilience, sensor miniaturization, vocal mechanisms
27 and data standardization must be addressed to realize the full potential of bioacoustics. The integration of AI,
28 flexible electronics, and multimodal sensing frameworks offers a roadmap for future innovations. The true
29 potential of next-generation bioacoustics lies not in isolated advancements within medical or ecological silos
30 but in the translation of technologies across these boundaries. The cross-pollination of sensor designs and AI
31 models among human, animal, and plant acoustics will be the catalyst for the next paradigm shift. Looking
32 ahead, bioacoustics is poised to play a central role in solving global challenges, from early disease detection
33 in resource-constrained settings to monitoring climate change impacts on ecosystems. By fostering cross-
34 disciplinary collaboration and prioritizing technological standardization, the field can unlock new frontiers in
35 understanding life systems and driving sustainable development. The next decade promises to witness
36 bioacoustics emerge as a cornerstone of precision medicine, smart agriculture, and ecological conservation,
37 shaping a more interconnected and responsive world.
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48 **Conflict of interest**

49 The authors declare that they have no competing financial interests.
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51

52
53 **Author contributions**

54 Chengyu Li, Xiaoming Tao: conceptualization, methodology and writing-review & editing
55 Chengyu Li, Wenbo Li, Jingyang Wu, Shuo Wang, Han Liao: data curation, investigation, writing-original
56 draft, formal analysis
57 Xiang Yu, Li Cheng & Xiaoming Tao: visualization, supervision, funding acquisition
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